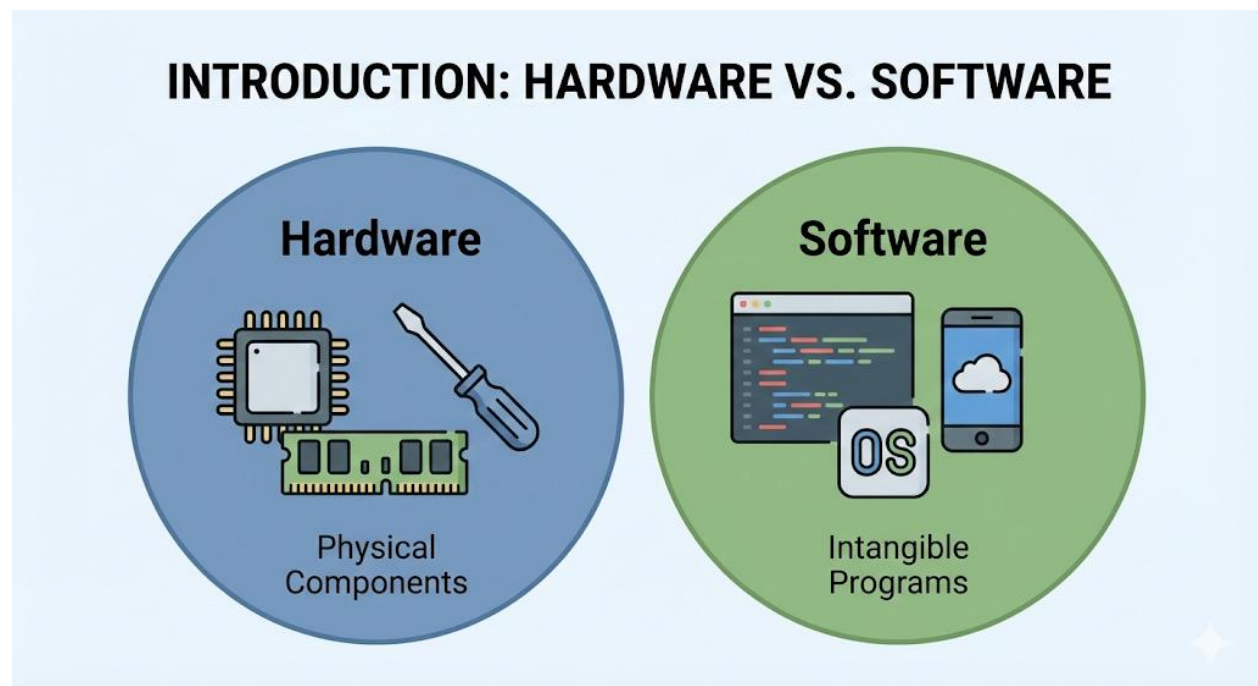


SS1 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

FIRST TERM

WEEK 1: INTRODUCTION TO COMPUTER HARDWARE AND GSM REPAIRS

MEANING OF COMPUTER HARDWARE



Computer hardware refers to all the **physical, tangible components** of a computer system.

These are the parts that you can see, touch, and pick up.

This is in direct contrast to **software**, which is the set of **intangible instructions, programs, and data** that tells the hardware what to do.

- **Analogy:** If a computer is a human body, the **hardware** is the brain, bones, and organs (the physical parts). The **software** is the person's thoughts, knowledge, and personality (the non-physical instructions and data).

Hardware includes everything from the *outside* (like the monitor, keyboard, and mouse) to the *inside* (like the motherboard, CPU, and hard drive).

MEANING OF GSM REPAIRS

GSM stands for **Global System for Mobile Communications**. It is the international standard that allows mobile phones to connect to a network.

GSM Repairs is the specialized technical field of **diagnosing, troubleshooting, and repairing mobile communication devices**, such as feature phones, smartphones (like Android and iPhone), and tablets.

SCOPE OF THE FIELD

The "scope" defines what this subject covers. It is a very broad field that includes:

- **Computer Hardware Scope:**
 - **Identification:** Naming every internal and external component.
 - **Assembly:** Knowing how to build a complete computer from individual parts.
 - **Upgrading:** Replacing older parts with newer, better parts (e.g., adding more RAM, installing a faster SSD).
 - **Maintenance:** Cleaning (e.g., removing dust from fans) and optimizing.
 - **Troubleshooting:** Diagnosing why a computer is not working (e.g., "Why won't it turn on?" "Why is it so slow?").
 - **Component Replacement:** Replacing faulty parts like a dead power supply or a broken fan.
- **GSM Repairs Scope:**
 - **Troubleshooting:** Diagnosing phone problems (e.g., "Why won't it charge?" "Why is the screen blank?").
 - **Physical Repairs:**
 - Screen and digitizer replacement (the most common repair).
 - Battery replacement.
 - Charging port repair (which often requires soldering).
 - Speaker/microphone replacement.
 - **Software Repairs:**

- "Flashing" the phone (re-installing the operating system).
- Unlocking patterns and passcodes.
- Resolving software "brick" issues.

IMPORTANCE OF HARDWARE AND GSM REPAIRS

Learning this skill is extremely important in the modern Nigerian economy.

1. **Economic Empowerment (Job Creation):** This is a *vocational skill*. It provides a direct path to entrepreneurship (opening your own repair shop) or employment (working as a technician).
2. **Cost Savings (Affordability):** Devices are expensive. It is significantly cheaper to **repair** a broken phone screen or upgrade a slow computer than it is to **replace** the entire device.
3. **Environmental Impact (Reduces E-Waste):** By repairing and extending the life of electronics, we reduce "Electronic Waste" (e-waste). This prevents toxic materials (like lead and mercury from circuit boards) from ending up in landfills.
4. **Digital Literacy & Problem Solving:** It gives you a deep, practical understanding of the technology you use every day. It teaches logical thinking and problem-solving.
5. **Data Recovery:** When a device fails, the data (photos, documents) is often more valuable than the device. Repair technicians can often recover this "lost" data.

CAREER OPPORTUNITIES

A person with these skills can pursue many careers:

- **Computer Repair Technician:** A generalist who fixes desktops and laptops.
- **IT Support Specialist (Help Desk):** Works for a company, helping employees with their computer problems.
- **GSM Repair Technician:** The "phone repairer" in a market or shop. This is a massive, high-demand industry.
- **System Builder:** Builds custom, high-performance PCs for clients (e.g., for video editing or gaming).

- **Electronics Refurbisher:** Buys broken or used computers/phones, repairs them, and resells them.
- **Data Recovery Specialist:** A specialist in retrieving data from dead hard drives.
- **IT Asset Manager:** Works for a large organization, managing the lifecycle (purchase, repair, disposal) of all IT equipment.

EVALUATION

1. In your own words, what is the main difference between computer hardware and software?
2. What does the acronym "GSM" stand for?
3. List three common tasks in *computer hardware* and three common tasks in *GSM repairs*.
4. Explain two economic reasons why learning hardware/GSM repair is important.
5. What is "E-Waste," and how does this profession help reduce it?

ASSIGNMENT

1. Write a 200-word essay on the topic: "Why I am interested in learning computer and GSM repairs."
2. Interview a local computer or phone repair technician. Ask them (a) what the most common repair they do is, and (b) what the most difficult part of their job is. Write a 1-page report on your findings.
3. List 5 different hardware components you can see on your desk or in your classroom right now.
4. Define the following terms: Troubleshooting, Entrepreneurship, Soldering, Flashing (a phone).
5. Research and list 3 "vocational skills" (like this one) and 3 "academic skills" (like history). What is the main difference?

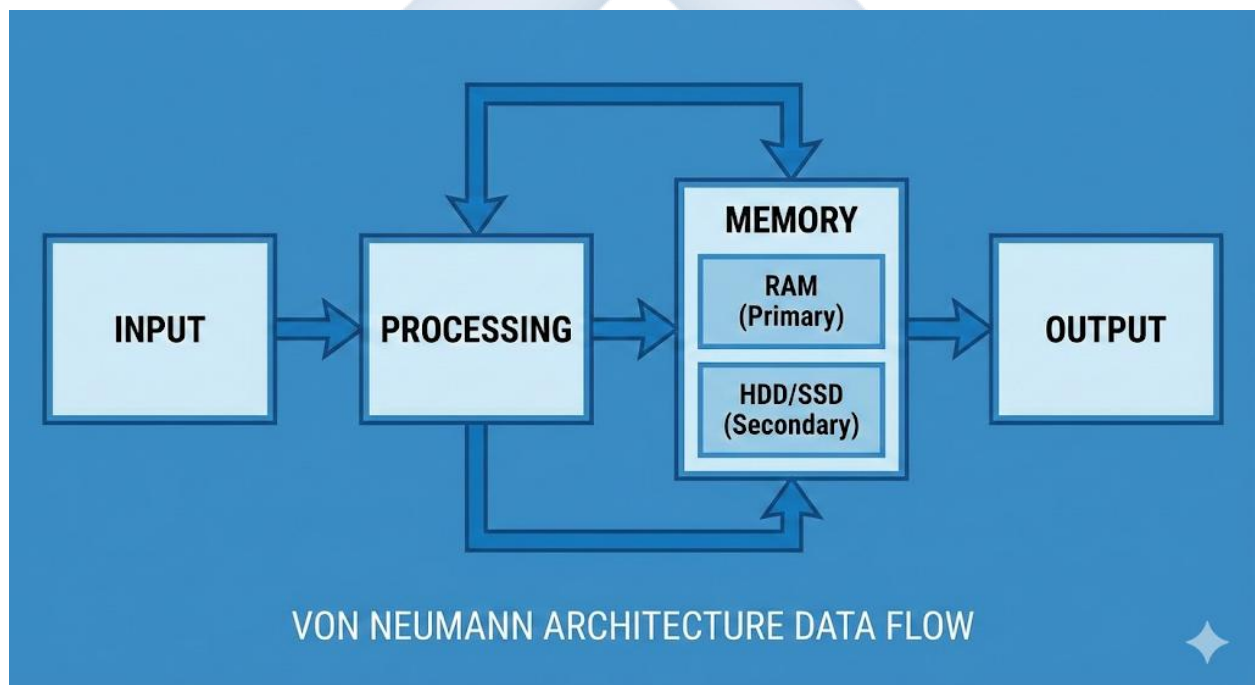
WEEK 2: CLASSIFICATION OF COMPUTER HARDWARE

INTRODUCTION

All computer hardware, no matter how complex, can be classified into one of four main categories based on its **function**. This classification model was famously described by **John von Neumann** and is the basis for all modern computers.

The four categories are:

1. **Input Devices**
2. **Output Devices**
3. **Processing Unit**
4. **Storage Devices**



1. INPUT DEVICES

- **DEFINITION:** An input device is any piece of hardware that is used to **send data into the computer**. It is the "sense organ" of the computer. It is how we, the human user, give instructions and data to the computer.
- **EXAMPLES:**

- **Keyboard:** For inputting text and commands.
- **Mouse:** For inputting "pointer" location, clicks, and scrolling.
- **Microphone:** For inputting audio (sound).
- **Webcam:** For inputting video.
- **Scanner:** For inputting a digital copy of a physical image or document.
- **Touchscreen:** A special device that is *both* an input (when you touch it) and an output (when you see it).

2. OUTPUT DEVICES

- **DEFINITION:** An output device is any piece of hardware that **presents data from the computer** in a human-understandable form. It is the "voice" or "face" of the computer. It is how the computer communicates its results back to us.
- **EXAMPLES:**
 - **Monitor (or Screen):** For visual output (images, text, video).
 - **Printer:** For "hard copy" (physical) text and image output.
 - **Speakers / Headphones:** For audio (sound) output.
 - **Projector:** For large-scale visual output (on a wall or screen).
 - **Haptic Feedback Motor (in a phone):** For vibration (touch) output.

3. PROCESSING UNIT (THE "BRAIN")

- **DEFINITION:** This is the "brain" of the computer. It is the component responsible for **executing instructions, performing calculations**, and managing all the other hardware.
- **MAIN COMPONENT:**
 - **CPU (Central Processing Unit):** A small, powerful chip that performs almost all the "thinking" (calculations and logical operations).
 - *Examples:* Intel Core i7, AMD Ryzen 5, Apple M3, Snapdragon (in phones).
- **OTHER COMPONENT:**

- **GPU (Graphics Processing Unit):** A specialized processor designed *only* for handling complex graphics and video. It is the "brain" for art and gaming.

4. STORAGE (MEMORY)

- **DEFINITION:** Hardware components that are used to **store digital data**. This is the computer's "memory." It is crucial to divide this into two types: Primary and Secondary.

A. PRIMARY STORAGE (MEMORY)

- **DEFINITION:** This is the computer's **short-term memory**. It is *very fast* and holds the data and instructions that the CPU is **actively working with *right now***.
- **Analogy:** This is your **work desk**. You pull files from the cabinet (secondary storage) and put them on your desk (primary storage) to work on them.
- **Key Property:** It is **VOLATILE** (temporary). When the power is turned off, all data in primary storage is **erased**.
- **MAIN EXAMPLE: RAM (Random Access Memory).**

B. SECONDARY STORAGE (STORAGE)

- **DEFINITION:** This is the computer's **long-term memory**. It is *slower* than RAM but can hold large amounts of data. It is where your files, operating system, and programs are kept.
- **Analogy:** This is your **filing cabinet** or **bookshelf**.
- **Key Property:** It is **NON-VOLATILE** (permanent). The data is saved even when the power is turned off.
- **MAIN EXAMPLES:**
 - **HDD (Hard Disk Drive):** An older, mechanical drive with spinning disks. (Large capacity, cheap, but slow).
 - **SSD (Solid State Drive):** A modern, chip-based drive with no moving parts. (Very fast, durable, but more expensive).
 - **USB Flash Drive, SD Card, CD/DVD.**

EVALUATION

1. What are the four main classifications of computer hardware based on the Von Neumann architecture?
2. What is the main function of an *input* device? Give 3 examples.
3. What is the main function of an *output* device? Give 3 examples.
4. What is the main difference between **Primary Storage (RAM)** and **Secondary Storage (HDD/SSD)**?
5. What does the term "**Volatile**" mean in the context of computer memory?

ASSIGNMENT

1. Create a two-column table. In Column 1, list the following devices: Printer, Keyboard, Monitor, Mouse, Webcam, Speakers, Scanner. In Column 2, classify each as "Input" or "Output."
2. Write a 200-word essay explaining the "work desk" and "filing cabinet" analogy for RAM and Hard Drives.
3. What is the "brain" of the computer called? What does the acronym stand for?
4. A smartphone's touchscreen is a "hybrid" device. Explain why.
5. Define the following terms: Peripheral, Volatile, Non-Volatile, CPU, RAM.

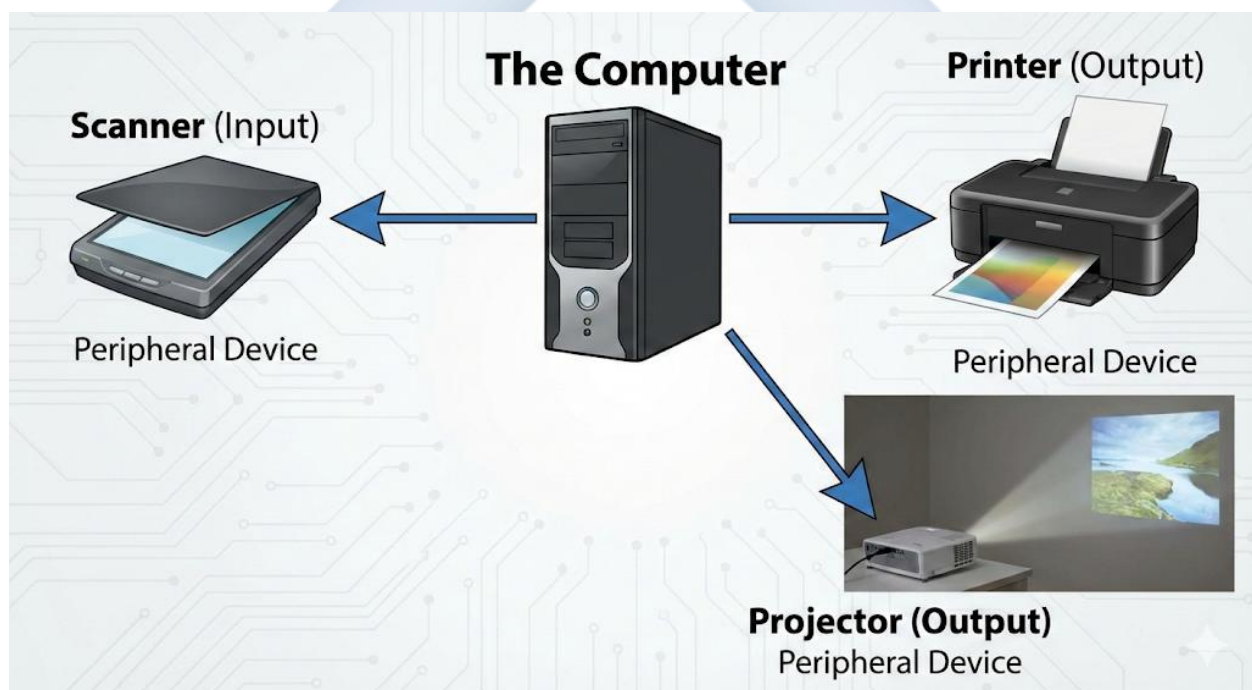
WEEK 3: PERIPHERAL DEVICES

DEFINITION OF PERIPHERAL DEVICES

A **peripheral device** is any external piece of hardware that is connected to a computer to expand its capabilities. They are "peripheral," meaning they are on the *periphery* (the edge) of the main system.

- A computer (CPU, RAM, Motherboard, PSU) can *function* without them, but it wouldn't be very *useful*.
- They are the **input** and **output** devices (like those from Week 2) and some storage devices.
- They connect via ports (like USB, HDMI) or wirelessly (like Bluetooth, Wi-Fi).

This week, we will focus on three key peripheral devices: **Printers, Scanners, and Projectors**.



1. PRINTERS

- **FUNCTION:** A **printer** is an **output device** that takes digital data from a computer (text, images) and creates a physical, hard copy on paper.
- **TYPES OF PRINTERS:**

- **A. INKJET PRINTERS:**

- **How it Works:** They work by spraying *microscopic droplets* of liquid ink from a "print head" onto the paper.
- **Pros:** Excellent at printing high-quality, vibrant **color photos**. Cheaper to buy.
- **Cons:** Liquid ink cartridges are expensive and can dry out if not used. Slower than laser.
- **Maintenance:**
 - Regularly use the "print head cleaning" utility to prevent clogs.
 - Replace ink cartridges when empty.
 - Use the correct type of paper (e.g., photo paper for photos).

- **B. LASER PRINTERS (LaserJet):**

- **How it Works:** They use a laser beam to "draw" an image onto a rotating drum. This drum attracts a fine, dry powder called **toner**. The toner is then transferred to the paper and "fused" (melted) onto it with heat.
- **Pros:** Extremely **fast**. Produces very sharp, crisp **black-and-white text**. Toner cartridges last a long time (e.g., 2,000 pages), making the cost-per-page very low.
- **Cons:** More expensive to buy. Color laser printers are very expensive.
- **Maintenance:**
 - Replace the toner cartridge when empty.
 - Gently clean up any spilled toner.
 - Periodically replace the "drum unit."

- **C. THERMAL PRINTERS:**

- **How it Works:** Uses a heated print head to burn an image onto special "thermal" paper (no ink or toner needed).
- **Use:** Receipts, POS (Point of Sale) systems, *Keke* and *Danfo* tickets.
- **Maintenance:** Replace the roll of thermal paper.

2. SCANNERS

- **FUNCTION:** A **scanner** is an **input device** that does the *opposite* of a printer. It converts a physical (hard copy) document or photograph into a digital file (e.g., a PDF or JPEG).
- **TYPES OF SCANNERS:**
 - **A. FLATBED SCANNER:**
 - **Description:** The most common type. It has a flat glass platen and a lid (like a photocopier).
 - **Use:** For scanning single pages, books, photos, and fragile documents.
 - **B. SHEET-FED SCANNER (Document Scanner):**
 - **Description:** The document is "fed" through a slot, and it passes over the scan head.
 - **Use:** For scanning a large *stack* of loose pages quickly (e.g., in an office).
- **MAINTENANCE:**
 - The *only* maintenance is to keep the **glass platen (screen) perfectly clean**.
 - Use a lint-free cloth and a gentle glass cleaner. Any dust, fingerprints, or smudges on the glass will appear on your digital scan.
 - Check the USB cable connection.

3. PROJECTORS

- **FUNCTION:** A **projector** is an **output device** that takes the visual output from a computer and "projects" it onto a large surface, like a white wall or a projector screen.
- **USE:** Classrooms (for teaching), boardrooms (for presentations), and home cinemas.
- **MAINTENANCE (CRITICAL):**
 - a. **LENS:** The lens must be kept clean. Use a special lens-cleaning cloth (like for eyeglasses).
 - b. **AIR FILTER:** Projectors generate *intense heat*. They have powerful fans and a **removable air filter** to suck in cool air. This filter *must* be cleaned regularly (e.g., monthly). If the filter gets clogged with dust, the projector will **overheat and shut down** or the expensive lamp will burn out.

- c. **LAMP (BULB):** The projector's lamp has a limited lifespan (e.g., 2,000 - 4,000 hours). Once it "blows," it must be replaced (and it is very expensive).
- d. **Cool Down:** *Never* unplug a projector immediately after turning it off. The "Off" button puts it in standby, and the fan will run for 1-2 minutes to *cool the lamp down* safely.

EVALUATION

1. What is a peripheral device? Give 3 examples.
2. What is the main difference in technology (how they work) between an Inkjet and a LaserJet printer?
3. What is "Toner," and which type of printer uses it?
4. What is the primary maintenance task for a (a) Flatbed Scanner and (b) Projector?
5. Why should you *never* unplug a projector immediately after turning it off?

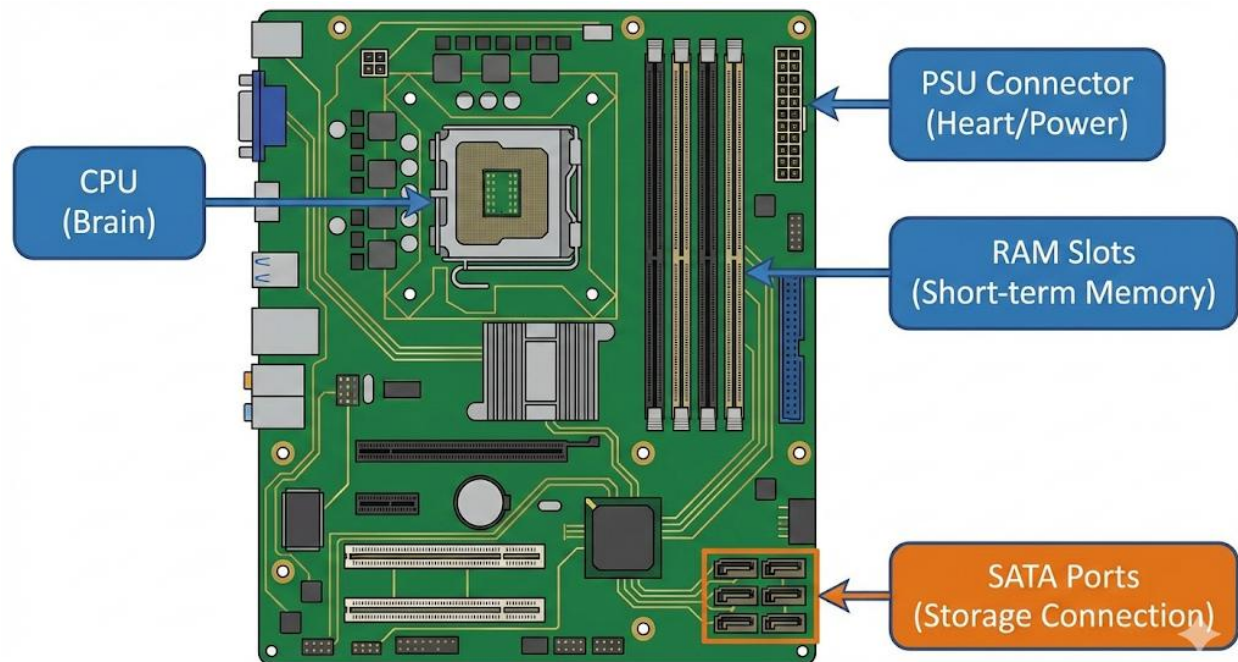
ASSIGNMENT

1. Write a 200-word essay comparing an Inkjet printer to a LaserJet printer. Which one would you recommend for a small business that prints 100s of black-and-white letters per day, and why?
2. Research "Multifunction Printer" (MFP). What are the 3 or 4 functions it combines into one device?
3. What is "Hard Copy"? What is "Soft Copy"? Which device creates which?
4. Look at a receipt from a store. What *type* of printer most likely printed it?
5. Define the following terms: Peripheral, Hard Copy, Toner, Print Head, Platen, Air Filter.

WEEK 4: INTERNAL COMPONENTS OF A COMPUTER

📖 INTRODUCTION TO THE COMPUTER "INTERNALS"

This week, we "open the box." We are moving from the external *peripheral devices* (Week 3) to the internal *core components*. These are the essential parts found inside the computer's **case (or tower)** that work together to make the computer "think."



1. MOTHERBOARD (MOBO)

- **DEFINITION:** The **motherboard** (or "Mainboard") is the "**nervous system**" or "backbone" of the computer. It is the largest printed circuit board (PCB) inside the case.
- **FUNCTION:** Its primary function is to **connect all other components**. It provides the sockets for the CPU, slots for the RAM, and connectors for the hard drive and power supply. It provides the "**bus**" (electrical pathways) that allows all the components to **communicate with each other**.
- **KEY PARTS OF A MOTHERBOARD:**
 - **CPU Socket:** The square mount where the CPU is installed.
 - **RAM Slots:** The long, narrow slots where the RAM sticks are installed.

- **PCIe Slots (Expansion Slots):** Used to install additional "cards," like a Graphics Card (GPU) or a Wi-Fi card.
- **SATA Ports:** Used to connect storage drives (HDD/SSD).
- **I/O (Input/Output) Panel:** The set of ports visible on the *back* of the computer (e.g., USB, Ethernet, Audio ports).

2. CPU (CENTRAL PROCESSING UNIT)

- **DEFINITION:** The CPU is the "**brain**" of the computer. It is a small, square chip that sits in the CPU socket on the motherboard.
- **FUNCTION:** Its job is to **fetch, decode, and execute instructions**. It performs all the calculations and logical operations that make programs run. The *speed* of the CPU (measured in **Gigahertz, GHz**) is the main factor in how "fast" or "slow" a computer feels.
- **COOLING (ESSENTIAL):** The CPU gets *extremely* hot (over 90°C). It *cannot* run without a **cooling solution**.
 - **Heat Sink:** A block of metal (usually aluminum) with fins, which sits on top of the CPU (with **thermal paste** in between) to draw the heat away.
 - **Fan:** A fan that blows cool air over the heat sink.
 - (This combined unit is called the "CPU Cooler").

3. RAM (RANDOM ACCESS MEMORY)

- **DEFINITION:** RAM is the computer's "**short-term memory**" or "workbench." It is a set of "sticks" (modules) that plug into the RAM slots on the motherboard.
- **FUNCTION:** RAM holds the data and instructions for the applications you are **currently running**.
- **ANALOGY:** Your **Hard Drive (HDD)** is the "filing cabinet" where you store all your programs. When you *run* a program (e.g., open a web browser), the computer *copies* that program from the slow HDD into the *super-fast* RAM. The CPU then works *only* with the copy in RAM.

- **PROPERTY:** RAM is **VOLATILE** (temporary). When you turn the computer off, *everything* in RAM is erased.
- **IMPORTANCE:** The *amount* of RAM (measured in **Gigabytes, GB**) determines how many programs you can run at the same time (**multitasking**). Not enough RAM is the most common cause of a "slow" computer.

4. POWER SUPPLY UNIT (PSU)

- **DEFINITION:** The PSU is the "**heart**" of the computer. It is a metal box (usually at the bottom or top of the case) with a fan and a large bundle of cables.
- **FUNCTION:** The PSU is a *power converter*. It takes the **220V AC (Alternating Current)** from the wall socket ("NEPA light") and **converts** it into the various **low-voltage DC (Direct Current)** voltages that the internal components need.
- **OUTPUT VOLTAGES:** It supplies several different voltages:
 - **+12V:** For high-power components (CPU, Graphics Card).
 - **+5V:** For components like hard drives.
 - **+3.3V:** For sensitive components on the motherboard.
- **RATING:** A PSU's capacity is measured in **Watts (W)** (e.g., 500W, 750W). You must choose a PSU that provides enough "Watts" to power all your components.

EVALUATION

1. What is the "motherboard," and what is its main function?
2. What is the "CPU," and what is its main function?
3. Why is a "heat sink" and "fan" essential for a CPU?
4. What is the "RAM," and what is its main function? Use the "workbench" analogy.
5. What is the *primary* function of a Power Supply Unit (PSU)? (Hint: It converts...).

ASSIGNMENT

1. Write a 200-word essay explaining the relationship *between* the Motherboard, CPU, and RAM.

2. A user's computer is very slow when they have 10 browser tabs and a music app open.
What internal component is *most likely* the "bottleneck" (the problem)? Explain why.
3. What is the difference between AC (Alternating Current) and DC (Direct Current)?
Which one comes from the wall, and which one do computer components use?
4. Draw a simple block diagram showing the 4 components from this lesson and how they connect.
5. Define the following terms: Gigahertz (GHz), Volatile, Multitasking, Thermal Paste, I/O Panel.



WEEK 6: SAFETY RULES AND WORKSHOP PRACTICES

↳ INTRODUCTION: THE TWO MAIN DANGERS

A computer repair workshop is a safe place *if* you follow the rules. There are two categories of danger:

1. **DANGER TO *YOU*:** Electric shock from power supplies.
2. **DANGER TO THE *COMPONENTS*:** Electrostatic Discharge (ESD), which silently kills sensitive electronics.



1. PERSONAL & ELECTRICAL SAFETY

This is about protecting *yourself*.

- **RULE 1: DISCONNECT POWER:** Before you *ever* open a computer case or phone, **unplug the power cord** from the wall socket. For a laptop or phone, **remove the battery** (if possible).
- **RULE 2: THE POWER SUPPLY UNIT (PSU):**

- A PSU has large **capacitors** inside. These can hold a *dangerous, high-voltage charge* for several minutes *even after* being unplugged.
 - **NEVER** open the metal box of the PSU itself.
- **RULE 3: AVOID LIQUIDS:** Keep all drinks (water, soda) far away from your workbench.
- **RULE 4: JEWELRY:** Remove metal rings, bracelets, or watches. If your ring touches two live circuits, it can cause a severe burn or short circuit.
- **RULE 5: HANDLE TOOLS SAFELY:**
 - Use the right screwdriver for the right screw.
 - Never use excessive force. If a part won't come out, there is probably a screw or latch you missed.

2. ELECTROSTATIC DISCHARGE (ESD) PRECAUTIONS

This is the **most important** rule for protecting the *components*.

- **WHAT IS ESD?**
 - **Electrostatic Discharge (ESD)** is the sudden flow of static electricity (a "spark") between two objects.
 - It's the *same tiny shock* you get when you walk on a carpet and then touch a metal doorknob.
- **THE DANGER:**
 - A human can't *feel* an ESD shock less than 3,000 Volts.
 - A computer component (like a CPU or RAM) can be **instantly and permanently destroyed** by as little as **10 Volts**.
 - This means you can *kill* a component *without ever feeling or seeing* the spark.
- **HOW TO PREVENT ESD (ANTI-STATIC RULES):**
 - a. **ESD WRIST STRAP (THE BEST TOOL):**
 - This is a special strap you wear on your wrist. It has a cable with an alligator clip on the end.

- You **clip the alligator clip** to an unpainted, metal part of the **computer case**.
- This "grounds" you, so your body and the computer are at the *same electrical potential*, making it impossible for a spark to jump.

b. **SELF-GROUNDING (THE "LOW-TECH" METHOD):**

- Before you touch *any* component, and *while* you are working, **always keep one hand touching an unpainted metal part of the computer case**.

c. **WORK SURFACE:**

- **NEVER** work on a **carpet**.
- Work on a clear, hard surface like a wood or laminate table.
- Professionals use a special **anti-static mat** that is also grounded.

d. **COMPONENT HANDLING:**

- Keep new components (like RAM, CPUs) inside their **anti-static bags** until the moment you are ready to install them.
- These bags are silver or pink and are designed to block ESD.

3. SAFE HANDLING OF COMPONENTS

- **GOLD CONTACTS: NEVER** touch the gold-colored "fingers" (contacts) at the bottom of a RAM stick or an expansion card. The oil from your skin can cause corrosion and a bad connection.
- **EDGES:** Always hold components (like the motherboard or RAM) by their **edges**. Do not touch the flat surfaces or the tiny chips.
- **SCREWS:** Keep screws organized. Use a magnetic parts tray or small labeled cups. Using the wrong screw in the wrong hole can damage the motherboard.

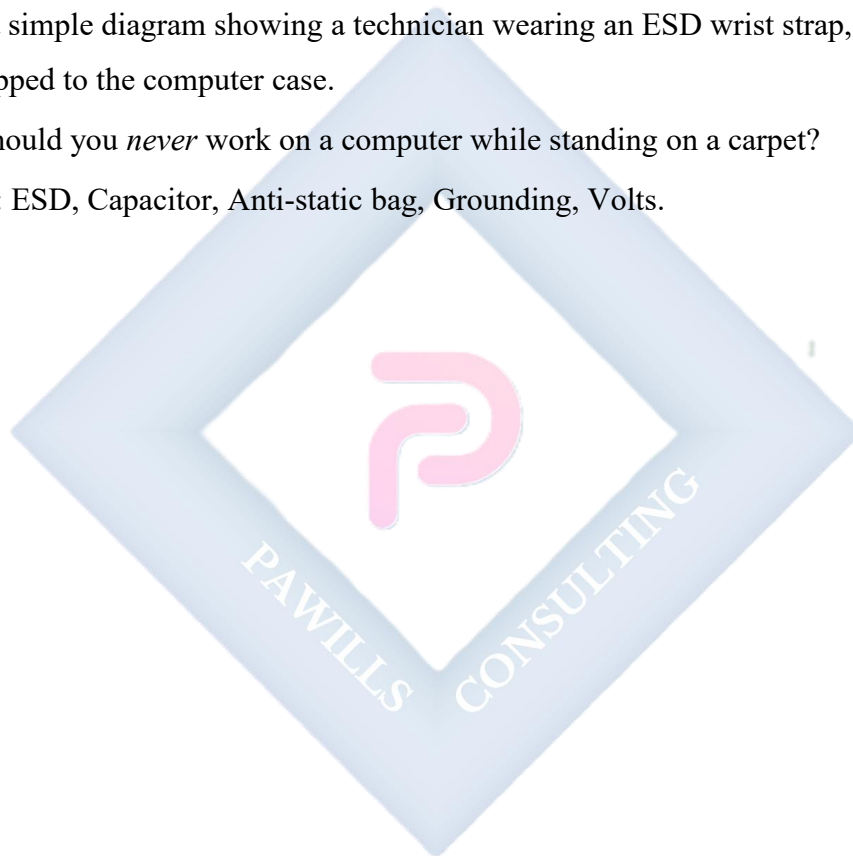
EVALUATION

1. What are the two main dangers in a repair workshop?
2. What is "ESD," and why is it so dangerous to computer components?
3. What is the *best* tool for preventing ESD? How does it work?

4. If you don't have this tool, what is the "low-tech" method for preventing ESD?
5. What part of a component (like a RAM stick) should you *never* touch?

ASSIGNMENT

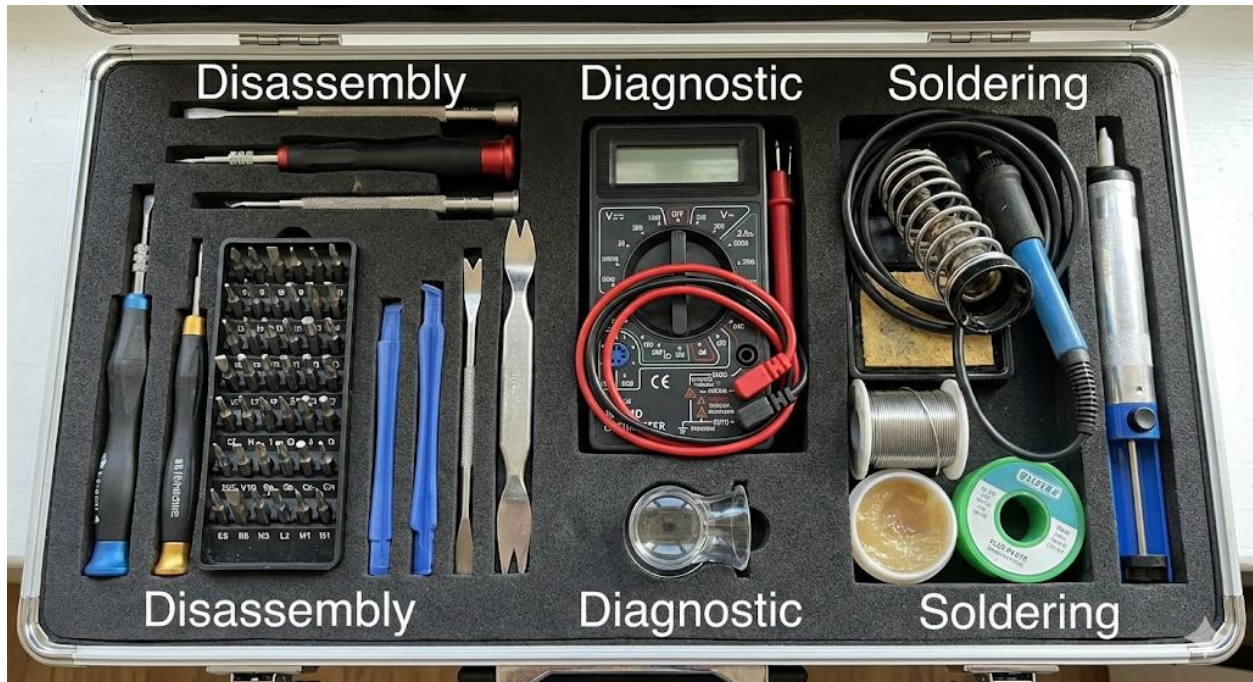
1. Write a 200-word "Safety Pledge" for a new technician, explaining the 5 most important rules they must follow.
2. Research "Capacitors" in a Power Supply Unit. Why are they still dangerous even after the PSU is unplugged?
3. Draw a simple diagram showing a technician wearing an ESD wrist strap, with the other end clipped to the computer case.
4. Why should you *never* work on a computer while standing on a carpet?
5. Define: ESD, Capacitor, Anti-static bag, Grounding, Volts.



WEEK 8: TOOLS FOR COMPUTER AND GSM REPAIRS

INTRODUCTION

A technician is only as good as their tools. Using the wrong tool can damage a device, waste time, and be dangerous. This week, we will identify the specific tools needed for both computer (desktop/laptop) and GSM (phone/tablet) repairs.



1. DISASSEMBLY TOOLS (FOR OPENING DEVICES)

These tools are used to safely open the case or housing.

- **SCREWDRIVER SET:**

- **Computer (Desktop):** You mainly need **Phillips-head** (cross-shape, sizes #1 and #2) and **Flat-head** screwdrivers.
- **GSM / Laptop (Precision Set):** This is a *specialized* set with tiny, interchangeable bits. It *must* include:
 - **Torx:** A 6-point star shape (T5, T6, T8). Used in many laptops and hard drives.
 - **Pentalobe:** A 5-point star shape. Used *only* by Apple on iPhones.
 - **Tri-Wing:** A 3-point "Y" shape. Used inside iPhones and other Apple devices.

- **Small Phillips:** (e.g., #00, #000). The most common small screw.
- **PRYING TOOLS (SPUDGERS):**
 - **Definition:** Tools used to *pry* open the plastic clips that hold laptops and phones together, *without scratching or damaging* the plastic.
 - **Plastic Spudger:** A non-marring, anti-static plastic tool. Often has one flat end and one pointed end.
 - **Metal Spudger:** Stronger, but can scratch plastic. Used for tough, glued-down components.
 - **Plastic Plectrum (Guitar Pick):** A thin, triangular tool. Very good for sliding along the *seam* of a device to "pop" it open.
- **SUCTION CUP:**
 - **Use:** A small suction cup (or a larger "pliers" version) is used to grip and *lift* the glass screen of a phone or tablet after it has been loosened.

2. DIAGNOSTIC TOOLS (FOR TESTING)

- **DIGITAL MULTIMETER (DMM):**
 - **Function:** This is the #1 diagnostic tool for *any* electronics repair.
 - **Uses:**
 - Testing the **voltage** of a laptop charger.
 - Testing the **voltage** of a phone battery.
 - Checking if a **fuse** on a circuit board is "blown" (using the **Continuity** or **Resistance** mode).
 - Testing the **DC output voltages** from a desktop PSU.
- **USB VOLTMETER/AMMETER:**
 - **Use:** A simple device that plugs between a charger and a phone to show the **Voltage (V)** and **Current (A)** being drawn. This is excellent for diagnosing charging port problems.

3. SOLDERING TOOLS (FOR BOARD-LEVEL REPAIRS)

These tools are for *advanced* repairs, like replacing a charging port, a capacitor, or a small chip on the motherboard.

- **SOLDERING IRON:**
 - **Function:** A tool that heats up to 300-400°C. It is used to *melt* solder.
 - **Soldering Station:** A professional setup with a temperature-controlled iron.
- **SOLDER (WIRE):**
 - The metal wire (an alloy) that melts to form the electrical connection.
- **FLUX:**
 - A chemical paste or liquid. It is a *cleaning agent* that removes oxidation and allows the molten solder to flow smoothly, creating a strong, shiny joint.
- **DESOLDERING TOOLS (FOR REMOVING SOLDER):**
 - **Solder Sucker (Desoldering Pump):** A small, spring-loaded vacuum pump. You melt the old solder, then "suck" it up with this tool.
 - **Solder Wick (Braid):** A copper braid that "soaks up" liquid solder (using capillary action).

4. OTHER ESSENTIAL TOOLS

- **TWEEZERS (ESD-SAFE):** For handling tiny screws, cables, and components.
- **ISOPROPYL ALCOHOL (IPA):** (90% or higher). The standard cleaning fluid for electronics. It cleans off *flux*, *thermal paste*, and *water damage* (corrosion) and evaporates quickly without leaving residue.
- **COMPRESSED AIR (BLOWER):** A can of air (or a manual rubber blower) used to safely blow dust out of fans, PSUs, and phone ports.
- **MAGNIFYING LAMP / MICROSCOPE:** Essential for GSM repairs, to see the tiny components on a phone's logic board.
- **HEAT GUN / HOT AIR REWORK STATION:** Used (carefully) to *soften* the adhesive (glue) that holds phone screens and batteries in place.

EVALUATION

1. What is the difference between a **Phillips** screwdriver and a **Pentalobe** screwdriver?
2. What is a "Spudger," and what is it used for?
3. What is the *safest* tool for measuring a high current (Amps)?
4. What is "Flux," and what is its purpose in soldering?
5. What is the primary use of **Isopropyl Alcohol (IPA)** in repairs?

ASSIGNMENT

1. Write a 200-word report on the difference between "Soldering" and "Desoldering," and name the main tool for each.
2. Why is it important to use a *plastic* spudger instead of a *metal* screwdriver to open a phone?
3. Go online and find a "repair kit" for an iPhone. List 5 *specialized* tools (e.g., Pentalobe, Suction Cup) that are included.
4. What is a "Hot Air Rework Station"? Research its use in replacing a phone's charging port.
5. Define: Torx, Pentalobe, Continuity, Flux, IPA (in this context).

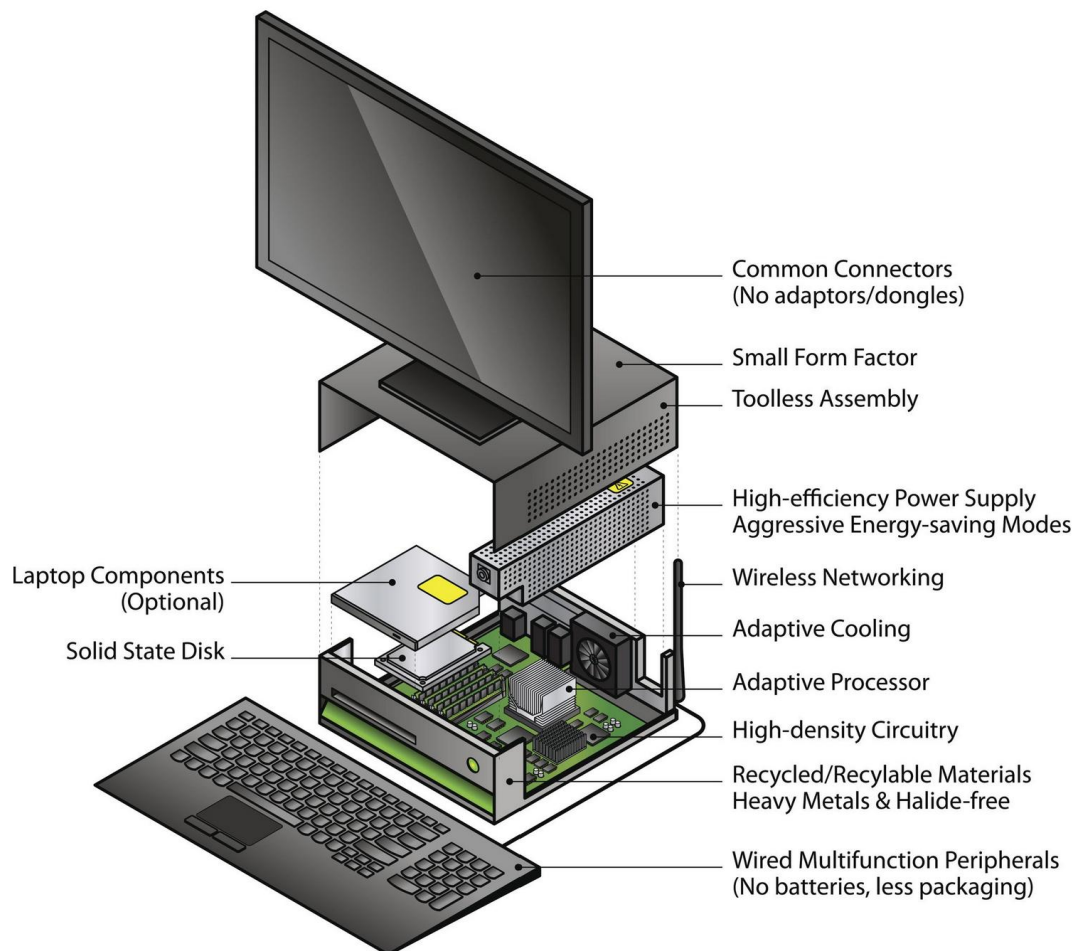
WEEK 9: COMPUTER ASSEMBLY I

TOPIC: INSTALLING RAM, HDD, AND POWER SUPPLY

INTRODUCTION

This week is a practical lesson on the **physical assembly** of a desktop computer. We will take an empty computer case (chassis) and begin installing the core components.

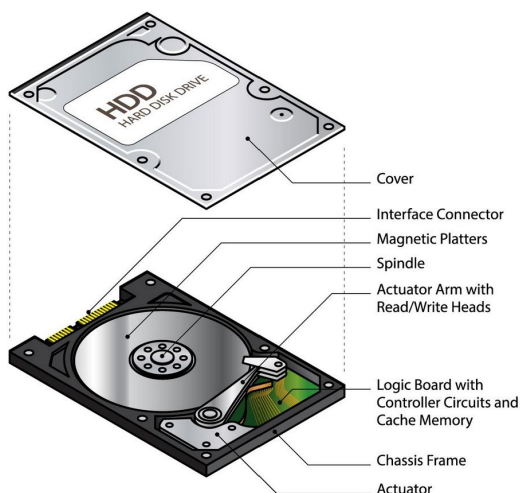
Safety Prerequisite: This *entire process* must be done with **ESD precautions** (Week 6). All components should be unplugged. The installer should be grounded using an ESD wrist strap or by constantly touching the metal case.



1. INSTALLING THE POWER SUPPLY UNIT (PSU)

The PSU is the "heart" that provides power. It is usually installed first as it is large and its cables need to be routed.

- **TOOLS:** Phillips-head screwdriver, 4 PSU screws.
- **PROCEDURE:**
 - a. **Orient the PSU:** Pick up the PSU. Locate the large **fan**.
 - b. **Position:** Align the PSU in its "bay" (the large hole at the *back* of the case, usually at the bottom).
 - c. **Fan Orientation (Critical):**
 - If the case has a *vent (grille)* on the **bottom**, the PSU fan should face **DOWN** (to pull in cool, fresh air).
 - If the case is solid at the bottom, the fan must face **UP** (to pull air from inside the case).
 - d. **Align Screw Holes:** Push the PSU flush against the back of the case until the 4 screw holes on the PSU align with the 4 holes on the case.
 - e. **Secure:** Screw in the 4 PSU screws from the *outside* of the case.
 - f. **Cables:** Tuck the large bundle of power cables into the side of the case for now.



2. INSTALLING RAM (RANDOM ACCESS MEMORY)

RAM is the "short-term memory" and is installed directly onto the motherboard.

- **TOOLS:** ESD wrist strap, RAM module(s).
- **PROCEDURE:**
 - a. **Locate Slots:** Identify the RAM slots on the motherboard (long, narrow slots, usually 2 or 4 of them).
 - b. **Open Clips:** Open the **retention clips** (levers) at *both ends* of the slot(s) you will use. (On some motherboards, only one side opens).
 - c. **Orient the RAM:** Pick up the RAM stick, holding it by the *edges*. Look at the **notch** (the small gap) in the gold contacts.
 - d. **Align the RAM:** Align that **notch** with the **key** (the small bump) in the RAM slot. *It only fits one way.* Do not force it.
 - e. **Insert the RAM:**
 - Gently place the RAM stick into the slot.
 - Place your thumbs on both ends of the stick.
 - Press down **firmly and evenly** on *both sides* at the same time.
 - You will hear a "**click**" as the retention clips snap shut and lock the RAM into place.
 - f. **Verify:** Gently pull on the stick to ensure it is locked and fully seated.

3. INSTALLING STORAGE (HDD / SSD)

This involves installing the "long-term memory" (the "filing cabinet") into the case and connecting it.

- **TOOLS:** Phillips-head screwdriver, 4 drive screws, 1 SATA data cable.
- **PROCEDURE (PART 1 - MOUNTING):**
 - a. **Locate Bay:** Find the "drive bay" or "drive cage" inside the case. This is a metal rack designed to hold 3.5-inch (HDD) or 2.5-inch (SSD) drives.
 - b. **Mount Drive:** Slide the HDD or SSD into the bay.

- c. **Align Holes:** Align the screw holes on the *side* of the drive with the holes in the drive bay.
- d. **Secure:** Screw in the 4 screws to hold the drive firmly in place. (This is important for HDDs, as they spin and vibrate).
- **PROCEDURE (PART 2 - CABLING):**
 - a. **Connect Data:** Take the **SATA Data cable** (a small, flat, L-shaped cable).
 - Plug one end into the **SATA port** on the drive.
 - Plug the other end into a **SATA port** on the **motherboard**.
 - b. **Connect Power:** Find the **SATA Power cable** coming from the **PSU** (it is a larger, L-shaped connector).
 - Plug it into the **SATA power port** on the drive.
 - The drive is now fully installed.

EVALUATION

1. What is the *first* safety step you must take before installing any component?
2. What is the "fan orientation" rule for a PSU? When should the fan face *down*?
3. How do you know which way a RAM stick fits into its slot?
4. What sound confirms that a RAM stick is correctly installed?
5. What two cables must be connected to every modern HDD or SSD?

ASSIGNMENT

1. Write a 1-page "Standard Operating Procedure" (SOP) manual, with step-by-step instructions (in your own words) for installing all three components from this lesson.
2. Research "Dual-Channel RAM." If a motherboard has 4 RAM slots (2 black, 2 grey), how must you install *two* sticks of RAM to get the best performance?
3. Draw a diagram showing a Hard Drive (HDD) and a Motherboard. Draw the two cables (SATA Data, SATA Power) that connect them, showing where the power cable comes from.

4. Why is it important to screw a Hard Drive (HDD) in securely, but less important for an SSD? (Hint: What's the difference in how they work?).
5. Define: PSU Bay, RAM Notch, Retention Clips, SATA, Drive Caddy.



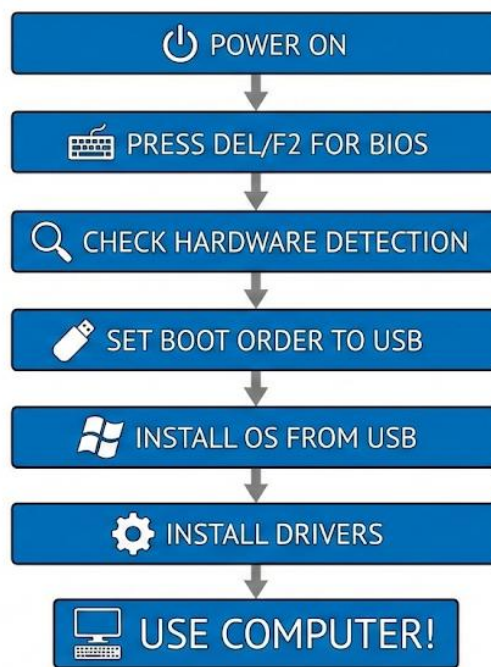
WEEK 10: COMPUTER ASSEMBLY II

TOPIC: BIOS SETUP, OS INSTALLATION BASICS

INTRODUCTION

In Week 9, we completed the **physical assembly**. All the parts are in the box. Now, we move to the **software assembly**. This is the process of "waking up" the hardware and giving it a "soul" (the Operating System) so it can become a usable computer.

This process has two main stages: **BIOS Setup** (verifying the hardware) and **OS Installation** (installing the software).



1. BIOS SETUP (BASIC INPUT/OUTPUT SYSTEM)

- **WHAT IS THE BIOS?**

- The **BIOS** (or **UEFI** on modern PCs) is a small chip on the **motherboard** that contains the *first set of instructions* the computer reads when it's turned on.
- It is the "bridge" between the physical hardware (CPU, RAM) and the main Operating System (Windows).

- **ACCESSING THE BIOS SETUP:**

- When you first power on the newly built computer, you must *immediately* press a specific key to "interrupt" the normal boot process.
- **Common Keys: DELETE, F2, F10, or F12.** (The screen will flash a message: "Press DEL to enter Setup").
- **KEY FUNCTIONS TO PERFORM IN BIOS:**
 - a. **POST (POWER-ON SELF-TEST):** The BIOS automatically runs this test. It "pings" all the hardware.
 - b. **VERIFY HARDWARE (THE "SANITY CHECK"):**
 - This is the *first and most important* thing to do in the BIOS.
 - Look at the "Main" or "System Information" screen.
 - **Check:** Does the BIOS *see* your components?
 - "CPU: Intel Core i5..."
 - "Memory: 16384 MB (16GB)"
 - "SATA Port 0: 1TB HDD..."
 - **Troubleshooting:** If your RAM or Hard Drive is *not* listed here, it means it is *not installed correctly* or is faulty. You must power off and re-install it.
 - c. **SET DATE AND TIME:** Set the system's internal clock and date.
 - d. **SET BOOT ORDER (BOOT PRIORITY):**
 - This is the *most important* step for installing an OS.
 - You must tell the BIOS *which drive to check first* for an operating system.
 - **To Install an OS:** You must change the boot order so that **#1 is the "USB Drive"** (or DVD-ROM) and **#2 is your "Hard Drive."**

2. OS INSTALLATION BASICS (OPERATING SYSTEM)

- **WHAT IS THE OS?**
 - The **Operating System** is the main software that manages the entire computer. It provides the **Graphical User Interface (GUI)** (the "desktop," icons, and mouse) that allows you to use the computer.
 - **Examples: Microsoft Windows 10/11, Apple macOS, Linux (e.g., Ubuntu).**

- **PREPARATION (INSTALLATION MEDIA):**

- You must have the OS installer on a **bootable USB flash drive** (or DVD). You usually create this on another computer (e.g., using the "Windows Media Creation Tool").

- **STEP-BY-STEP INSTALLATION PROCESS:**

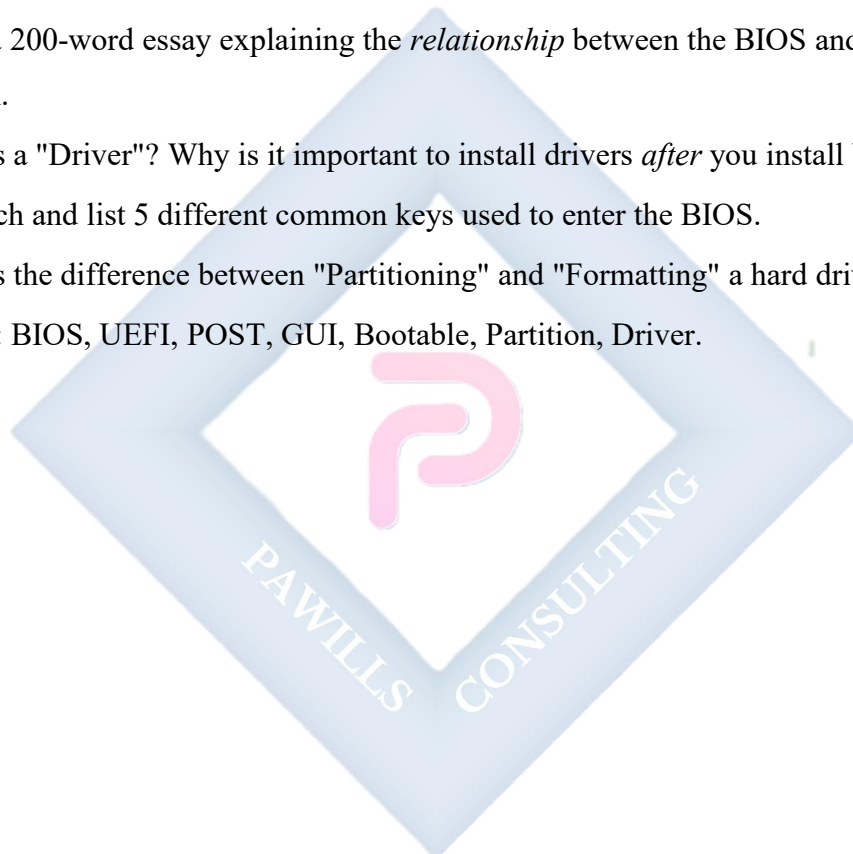
- a. **Set Boot Order:** Go into BIOS and set "Boot Priority #1" to "USB Drive."
- b. **Insert Media:** Plug the bootable USB drive into a USB port.
- c. **Save & Exit:** Save your BIOS settings and restart the computer.
- d. **Boot from Media:** The computer will now "boot" from the USB drive and launch the **Windows (or Linux) Installer**.
- e. **Follow On-Screen Prompts:** The installer will ask you for:
 - Language and keyboard layout.
 - Your "Product Key" (license).
- f. **Partitioning & Formatting (CRITICAL STEP):**
 - The installer will ask, "Where do you want to install Windows?"
 - It will show your new, blank Hard Drive (HDD) or SSD.
 - You must select the drive and "Format" it (which creates a file system, like NTFS) and create a "Partition" (a usable section). This prepares the drive to hold files.
- g. **Copying Files:** The installer will then (slowly) copy all the main OS files from the USB drive onto the hard drive.
- h. **First Reboot:** The computer will restart. **You must remove the USB drive** (or re-enter the BIOS and set Boot Priority #1 back to "Hard Drive").
- i. **Final Setup (OOBE):** The computer will now boot *from the hard drive*. You will complete the "Out of Box Experience" (OOBE), where you create a username (e.g., "SS1 Student") and password.
- j. **Install Drivers:** The *final* step. You must install **drivers**—special software that tells Windows how to use your specific hardware (like your graphics card, Wi-Fi, and motherboard).

EVALUATION

1. What does "BIOS" stand for?
2. What is the *first* thing you should check in the BIOS after building a new computer?
3. What is "Boot Order," and what must you set it to in order to install Windows?
4. What is an "Operating System (OS)"? Give two examples.
5. What is "Installation Media"? What is the most common form today?

ASSIGNMENT

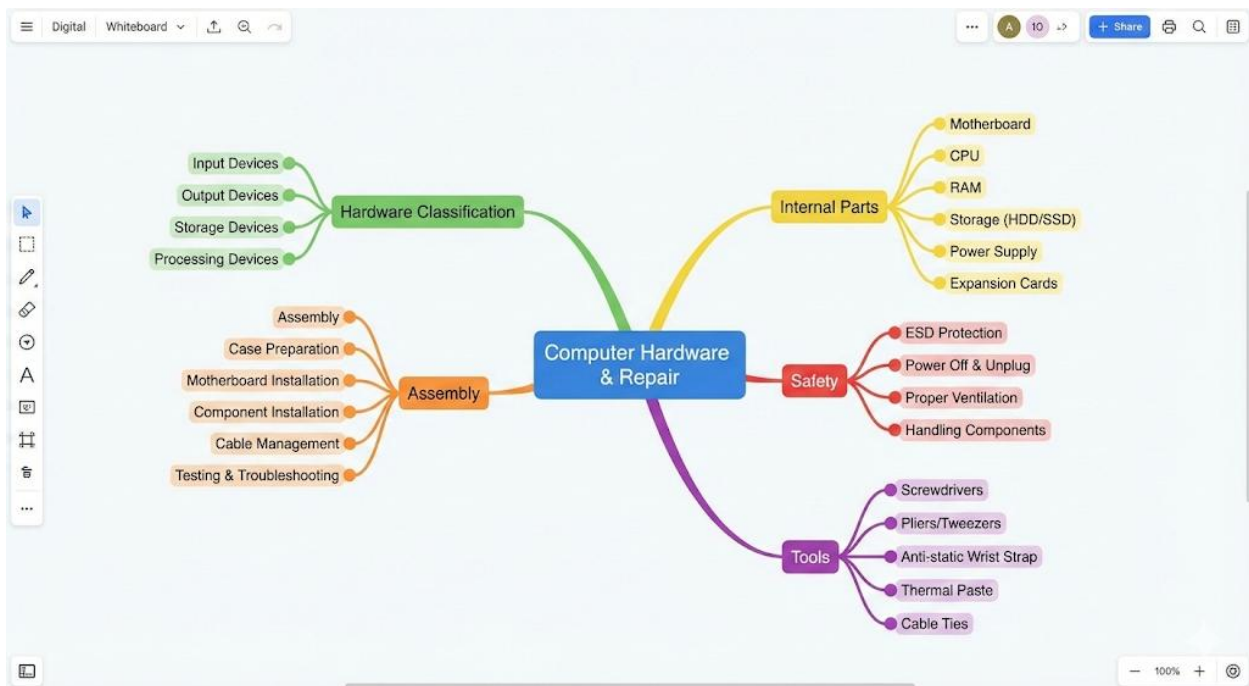
1. Write a 200-word essay explaining the *relationship* between the BIOS and the Operating System.
2. What is a "Driver"? Why is it important to install drivers *after* you install Windows?
3. Research and list 5 different common keys used to enter the BIOS.
4. What is the difference between "Partitioning" and "Formatting" a hard drive?
5. Define: BIOS, UEFI, POST, GUI, Bootable, Partition, Driver.



WEEK 11: REVISION

GENERAL REVIEW OF ALL FIRST TERM TOPICS

This week is for a comprehensive review of all concepts, components, and procedures covered this term, in preparation for the final examination.



• WEEK 1: INTRODUCTION

- **Hardware:** The *physical* parts (keyboard, CPU).
- **Software:** The *intangible* instructions (Windows, apps).
- **GSM Repairs:** Fixing mobile devices (screens, batteries, charging ports).
- **Importance:** Job creation, cost-saving, reduces E-waste.

• WEEK 2: CLASSIFICATION OF HARDWARE

- **Input:** Sends data *IN* (Mouse, Keyboard, Scanner, Mic).
- **Output:** Presents data *OUT* (Monitor, Printer, Speakers).
- **Processing:** The "brain" (CPU, GPU).
- **Storage (Primary):** **RAM.** Fast, **volatile** (temporary), "workbench."

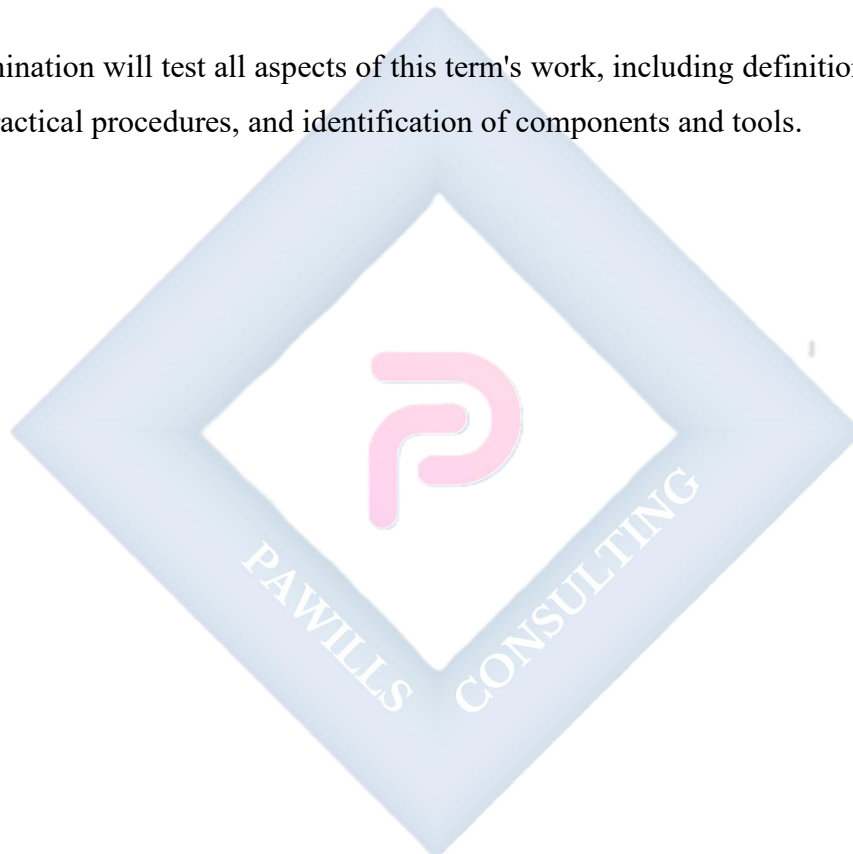
- **Storage (Secondary): HDD/SSD.** Slower, **non-volatile** (permanent), "filing cabinet."
- **WEEK 3: PERIPHERAL DEVICES**
 - **Printer:** Output. **Inkjet** (liquid ink, photos) vs. **LaserJet** (toner, text, fast).
 - **Scanner:** Input. Converts physical to digital. (Flatbed, Sheet-fed).
 - **Projector:** Output. Needs **filter cleaning** and **cool-down time**.
- **WEEK 4: INTERNAL COMPONENTS**
 - **Motherboard:** The "backbone." Connects everything.
 - **CPU:** The "brain." Executes all calculations. Needs a **heat sink/fan**.
 - **RAM:** "Short-term memory." Volatile. Key to **multitasking**.
 - **PSU:** The "heart." Converts **AC** (wall) to **DC** (components).
- **WEEK 6: SAFETY & WORKSHOP PRACTICES**
 - **Personal:** Unplug first! Beware of PSU capacitors.
 - **ESD (Electrostatic Discharge):** The *silent killer* of components.
 - **Prevention:** Use an **ESD wrist strap** or **touch the metal case** (self-grounding).
Never work on a carpet.
 - **Handling:** Hold components by the *edges*, never touch the *gold contacts*.
- **WEEK 8: TOOLS**
 - **Disassembly:** Screwdriver set (Phillips, Torx, Pentalobe), **Spudgers** (for prying plastic).
 - **Diagnostic: Multimeter** (V, I, R), Clamp Meter (for current).
 - **Soldering:** Soldering Iron, Solder, Flux, Desoldering Pump.
- **WEEK 9: ASSEMBLY I (PHYSICAL)**
 - **PSU:** Install (fan orientation is key).
 - **RAM:** Open clips, align **notch**, press *firmly* until it "clicks."
 - **HDD/SSD:** Mount in bay, connect **SATA Data** (to motherboard) and **SATA Power** (to PSU).

- **WEEK 10: ASSEMBLY II (SOFTWARE)**

- **BIOS:** Access by pressing DEL/F2.
- **Function 1:** *Verify* that RAM and HDD are detected.
- **Function 2:** Set **Boot Order** to "USB Drive."
- **OS Install:** Boot from USB, **partition/format** the drive, copy files, install **drivers**.

FINAL EXAMINATION

The final examination will test all aspects of this term's work, including definitions, comparisons, safety rules, practical procedures, and identification of components and tools.



SS1 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

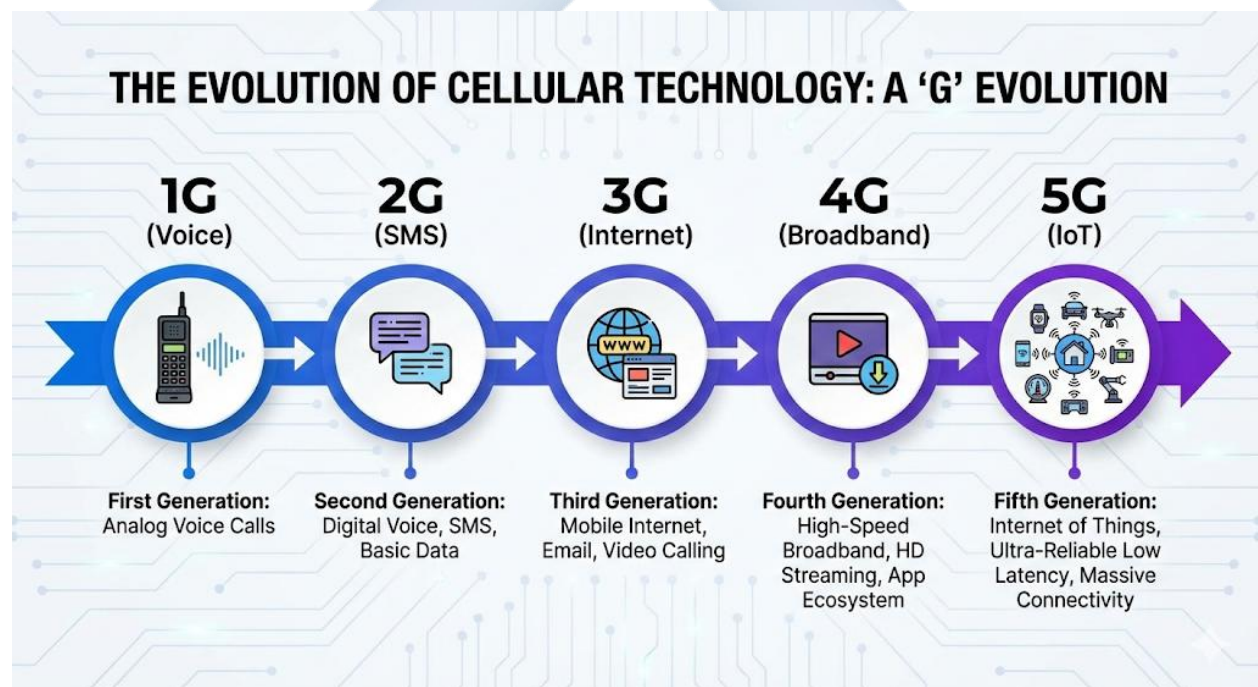
SECOND TERM

WEEK 1: INTRODUCTION TO GSM TECHNOLOGY

MEANING AND EVOLUTION OF MOBILE PHONES

GSM stands for **Global System for Mobile Communications**. It is the digital standard that was developed to replace the first generation of analog mobile phone technology. GSM is the technology that made modern mobile phones, as we know them, possible.

The "G" in 1G, 2G, 3G, 4G, and 5G stands for "**Generation**." Each generation represents a significant leap in technology, speed, and capability.



- **1G (First Generation - The "Brick Phone" Era, c. 1980s):**
 - **Technology:** Analog.
 - **Capability:** Voice calls only.

- **Characteristics:** These were the large "brick phones" (like the Motorola DynaTAC). They had poor battery life, no security (calls could be easily "listened in" on), and were very expensive.
- **2G (Second Generation - The "GSM" Era, c. 1990s):**
 - **Technology: Digital.** This is the generation where **GSM** became the standard.
 - **Capability: Voice calls + SMS (Short Message Service),** or "texting."
 - **Characteristics:** This was a revolution. Digital signals were secure (encrypted), clear, and efficient. It introduced the **SIM Card**. This is the era of iconic phones like the **Nokia 3310**. It also introduced basic mobile data called **GPRS** and **EDGE** (often shown as "E" on your phone), which was very slow.
- **3G (Third Generation - The "Mobile Internet" Era, c. 2000s):**
 - **Technology:** Digital, with a focus on data.
 - **Capability:** Voice calls + SMS + functional **Internet access**.
 - **Characteristics:** This generation brought speeds fast enough for web browsing, email, and basic video calling. This is the era that fueled the rise of the **BlackBerry** and the first **smartphones**.
- **4G (Fourth Generation - The "Smartphone" Era, c. 2010s):**
 - **Technology: IP-based (Internet Protocol).** A complete shift to data.
 - **Capability:** High-speed, "broadband" internet.
 - **Characteristics:** This is the generation we use today (often as **4G LTE**). It is fast enough for high-definition (HD) video streaming (like YouTube), online gaming, and high-quality video calls. It is the "true" smartphone era.
- **5G (Fifth Generation - The "IoT" Era, c. 2020s):**
 - **Technology:** New, high-frequency radio waves.
 - **Capability: Ultra-fast speeds** (gigabit-level) and, more importantly, **ultra-low latency** (almost zero delay).
 - **Characteristics:** This is designed not just for phones, but for the "**Internet of Things**" (**IoT**)—connecting self-driving cars, remote surgery, and smart cities.

GSM ARCHITECTURE (HOW IT WORKS)

The GSM network is a complex system that connects your phone to the rest of the world. For an SS1 level, we can simplify its architecture into three main parts:

1. THE MOBILE STATION (MS) - YOUR PHONE:

- This is the device you hold. It consists of two parts:
- **The Handset (or ME - Mobile Equipment):** This is the physical phone itself (e.g., the Tecno, Samsung, or iPhone). It is identified by a unique 15-digit serial number called the **IMEI** (International Mobile Equipment Identity).
- **The SIM Card (Subscriber Identity Module):** This is the "brain" of your subscription. It is a small, removable chip that **stores your identity** (your phone number, your network, your account details). The phone (IMEI) is just a piece of hardware; the SIM card is *you*.

2. THE BASE STATION SUBSYSTEM (BSS) - THE "CELL TOWER":

- This is the "local" part of the network that your phone talks to.
- **BTS (Base Transceiver Station):** This is the **radio tower and antenna** you see in your neighborhood. It sends and receives the radio signals to and from your phone.
- **BSC (Base Station Controller):** This is the "mini-brain" that manages a *group* of BTS towers. Its job is to handle the **"handoff."** When you are in a car on a call, the BSC is what seamlessly "hands you off" from one cell tower to the next without the call dropping.

3. THE NETWORK SWITCHING SUBSYSTEM (NSS) - THE "CORE NETWORK":

- This is the "central brain" of the entire network (e.g., the MTN or Glo central office).
- **MSC (Mobile Switching Center):** This is the "central switchboard." It connects your call to another mobile user or to the landline network.

- **HLR (Home Location Register):** This is the main, permanent **database**. It stores the master record for *all* subscribers on the network (your phone number, your services, your location).
- **VLR (Visitor Location Register):** A *temporary* database in the MSC. When you travel from Lagos to Abuja, the Abuja VLR "borrows" your details from the Lagos HLR to manage your calls locally.

EVALUATION

1. What does the acronym "GSM" stand for?
2. What two major capabilities did **2G (GSM)** introduce that **1G** did not have?
3. What is the main difference in capability between 3G and 4G?
4. What are the two parts of the **Mobile Station (MS)**?
5. What is the function of the **BSC (Base Station Controller)**? (Hint: "handoff").

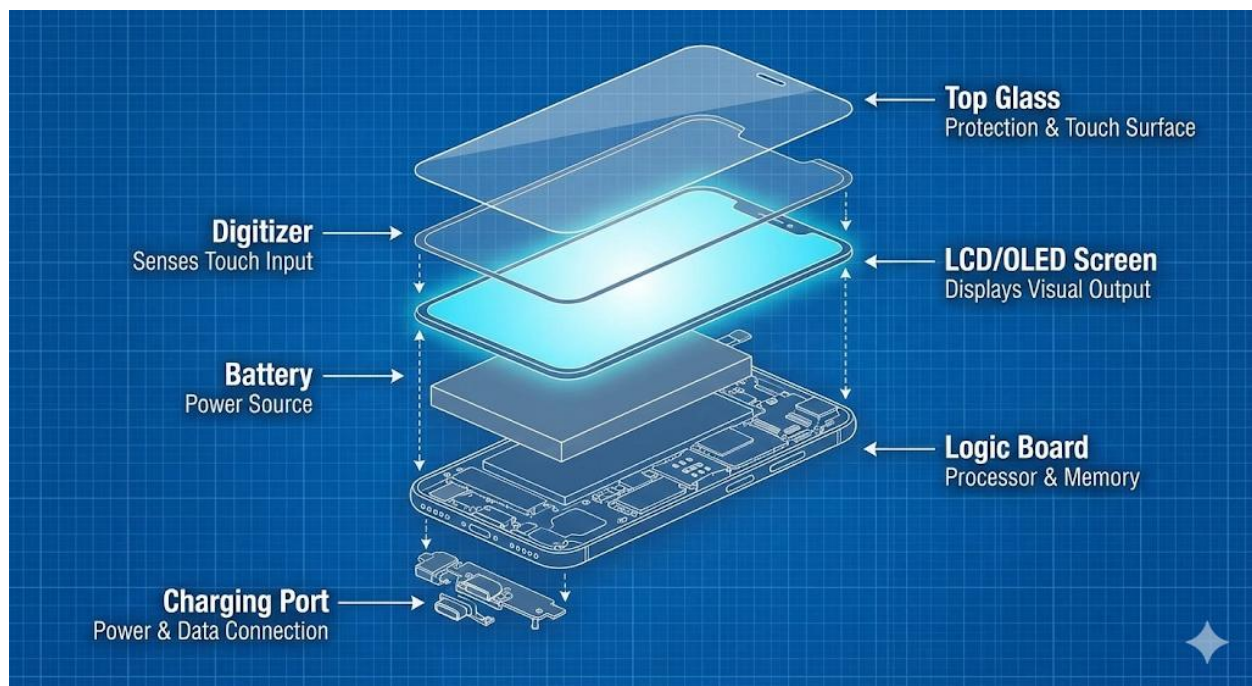
ASSIGNMENT

1. Write a 200-word essay on the evolution of mobile phones, describing the biggest change brought by each "G" (1G to 4G).
2. What is an **IMEI** number? Find the IMEI number of a mobile phone (dial ***#06#**). Write down the first 8 digits (do not post the full number).
3. What is a **SIM card**? What information is stored on it?
4. Draw a simple block diagram showing the 3 main parts of the GSM architecture (MS, BSS, NSS) and how they connect.
5. Define the following terms: 5G, SMS, Latency, BTS, MSC.

WEEK 2: GSM HARDWARE COMPONENTS

INTRODUCTION

Every mobile phone, from the oldest Nokia to the newest iPhone, is a miniature computer. It is built from the same categories of hardware we learned in the First Term (Input, Output, Processing, Storage, Power). This week, we identify the key physical components specific to a GSM phone.



1. SCREEN (DISPLAY) - OUTPUT

This is the primary *output* device.

- **LCD (Liquid Crystal Display):**
 - **Technology:** This is an older, "transmissive" technology. The pixels in the LCD screen do *not* create their own light. They act as "shutters," blocking or allowing light to pass.
 - **Backlight:** Requires a separate light source (the "backlight") to shine *through* the LCD layer to make the image visible.
 - **Characteristics:** Cheaper to produce. Blacks are not "true" black (they are a dark grey, as the backlight is always on).

- **OLED (Organic Light Emitting Diode) / AMOLED:**
 - **Technology:** This is a newer, "emissive" technology. Each individual pixel is a tiny "organic" light bulb that can turn on or off by itself.
 - **Characteristics:**
 - **True Blacks:** To show "black," the pixels simply turn *off*. This creates an "infinite" contrast ratio and saves power.
 - **Vibrant Colors:** Produces brighter, more vibrant colors.
 - **Thinner:** Does not need a separate backlight layer, so the phone can be thinner.
- **THE DIGITIZER (TOUCHSCREEN) - INPUT:**
 - This is an **invisible, transparent layer** that sits *on top* of the LCD or OLED screen.
 - **Function:** It is the *input* device that "reads" the position of your finger (your touch).
 - **Common Fault:** When you "break your screen," you might break (a) just the top glass, (b) the glass and the digitizer (touch stops working), or (c) all three (glass, digitizer, and LCD/OLED).

2. KEYPAD / KEYBOARD - INPUT

- **Membrane Keypad:** Used in older "feature phones" (like the Nokia 3310). It has a rubber membrane with carbon contacts that press onto a circuit board.
- **QWERTY Keyboard:** A full physical keyboard (like on an old BlackBerry).
- **On-Screen Keyboard:** This is not a physical hardware component. It is a *software* program that uses the **digitizer** (hardware) as its input.

3. BATTERY - POWER

This is the "power supply unit" of the phone.

- **Technology:** Almost all modern phones use **Lithium-Ion (Li-Ion)** or **Lithium-Polymer (Li-Po)** batteries.

- **Li-Ion:** High energy density (a lot of power in a small size), lightweight, and has no "memory effect" (you don't have to fully drain it before charging).
- **Li-Po:** A type of Li-Ion battery in a flexible, polymer "pouch" instead of a metal can. This allows it to be molded into very thin, custom shapes to fit inside slim smartphones.
- **Safety:** Lithium batteries are generally safe, but if they are **punctured, bent, or overcharged**, they can overheat and **catch fire or explode**. This is a critical safety rule for repair.

4. SIM SLOT - INPUT

- **Function:** The **Subscriber Identity Module (SIM)** slot is the "reader" for the SIM card.
- **Hardware:** It is a small tray or cage with a set of **tiny gold-plated pins (contacts)**.
- **Common Fault:** The most common fault is when a user inserts a SIM card incorrectly (or inserts a "nano" SIM into a "micro" slot) and **bends or breaks** one of the contact pins. This is a very difficult (board-level) repair.

5. CHARGING PORT - INPUT

- **Function:** The *input* port for power to charge the battery.
- **Types:**
 - **Micro-USB:** The older standard, D-shaped, non-reversible.
 - **USB-C:** The modern standard, oval-shaped, **reversible** (can be plugged in upside down).
 - **Lightning:** The proprietary (Apple-only) connector for iPhones.
- **Common Fault:** This is the **most common hardware fault** on any phone. It is a mechanical part that endures constant physical stress (plugging and unplugging). The port becomes loose, or the internal pins break. This repair almost always requires **soldering**.

EVALUATION

1. What is the main difference in *technology* between an LCD screen and an OLED screen?
2. What is the "Digitizer"? Is it an input or an output device?
3. What are the two main types of modern phone batteries?
4. What is the critical safety rule when handling a Lithium-Ion battery during a repair?
5. What is the *most common* hardware fault in most GSM phones, and what tool is usually required to fix it?

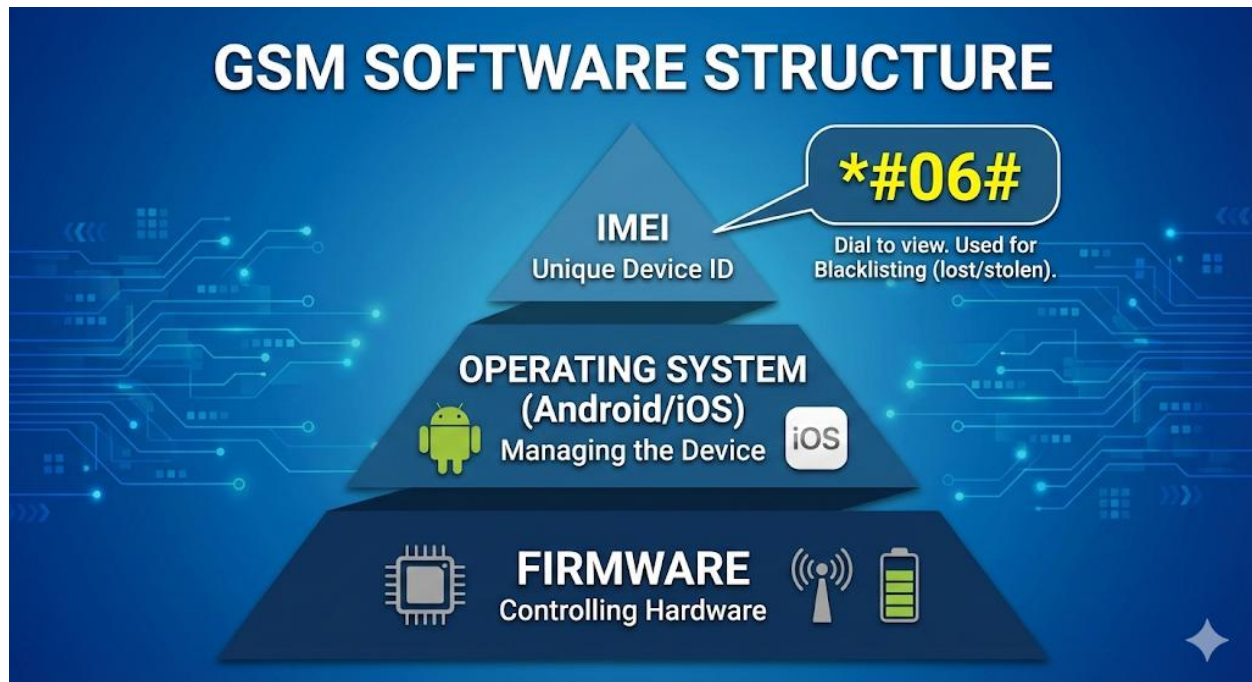
ASSIGNMENT

1. Write a 200-word essay explaining to a customer what has happened when their "screen is broken but the touch still works." (Hint: Glass vs. Digitizer).
2. Research "Memory Effect" in batteries. What was it, and why are modern Li-Ion batteries better?
3. Draw a simple diagram of a SIM card and the SIM slot, showing the "pins" making contact.
4. Find a phone at home. Identify its charging port type (Micro-USB or USB-C).
5. Define: LCD, OLED, Digitizer, Li-Po, USB-C, Soldering.

WEEK 3: GSM SOFTWARE COMPONENTS

INTRODUCTION

Last week, we covered the *hardware* (the "body"). This week, we cover the *software* (the "soul"). Software is the set of instructions that tells the hardware what to do. In GSM, there are three key software/code components to understand.



1. FIRMWARE

- **Definition:** **Firmware** is a special, low-level type of software that is **embedded** (permanently programmed) into a piece of hardware.
- **Analogy:** If the Operating System (Android) is the "President" (giving big commands), the firmware is the "civil service" that knows how to *actually* operate the machinery.
- **Function:** It is the "Basic Input/Output System" (BIOS) of the phone. It is the *first* thing that runs when you power on the device.
- **Examples:**
 - **Radio Firmware:** Controls the "radio" (transceiver) and tells it *how* to connect to a GSM (e.g., MTN, Glo) tower.
 - **Power Management Firmware:** Controls *how* the battery charges.

- **Repair Context:** When a phone is "flashed" (see Week 10), you are re-installing this core firmware.

2. OPERATING SYSTEM (OS) - IN SMARTPHONES

- **Definition:** The **Operating System** is the main, high-level software that *manages* all the phone's hardware and software. It is the "master program" that runs the phone.
- **Function:** It provides the **GUI (Graphical User Interface)**—the home screen, the icons, the app store, and the ability to run applications (apps).
- **MAJOR SMARTPHONE OPERATING SYSTEMS:**
 - **Android (Google):**
 - **Type: Open-Source.** This means Google makes the base code (AOSP) available, and other companies (like Samsung, Tecno, Infinix) can *modify* it.
 - **Characteristics:** Highly customizable, runs on thousands of different hardware models. (e.g., Samsung's "One UI," Tecno's "HiOS").
 - **iOS (Apple):**
 - **Type: Closed-Source.** Apple controls 100% of the software and hardware.
 - **Characteristics:** It is used *only* on Apple's own hardware (iPhone). This "tight integration" often leads to a very smooth and stable performance.
 - **Older (Defunct) OS:**
 - **Symbian OS:** (Nokia's original smartphone OS).
 - **BlackBerry OS:** (Used on BlackBerry phones).
 - **Windows Phone:** (Microsoft's mobile OS).

3. IMEI (INTERNATIONAL MOBILE EQUIPMENT IDENTITY)

- **Definition:** The **IMEI** is a **unique 15-digit number** that is a "fingerprint" or "serial number" for the phone's **hardware**. It is *not* software, but a unique code *stored* in the phone's firmware.
- **How to Find:** You can find the IMEI by dialing ***#06#** on *any* GSM phone.

- **PURPOSE (THE MOST IMPORTANT PART):**

- a. **Identification:** It tells the network (MTN, Glo, etc.) *what kind of device* is trying to connect (e.g., "This is a Samsung S20").
- b. **Security (Stolen Phones):** This is its main purpose. If your phone is stolen, you can report the IMEI to your network provider. They will add it to a national **"Blacklist."**
- c. **Blacklisting:** Once blacklisted, that IMEI is **barred** from all networks in the country. The phone will show "No Service" or "Emergency Calls Only," *even if the thief puts in a new SIM card.* This makes the phone useless.

- **Repair Context:**

- It is **illegal** to change or "clone" a phone's IMEI.
- A common software fault is a "corrupted" or "null" IMEI (where the phone's IMEI shows "000000..."). This is a software error, and the phone will not make calls until it is repaired (flashed).

EVALUATION

1. What is "Firmware"? Use an analogy to explain it.
2. What is the main difference between an "Open-Source" OS (like Android) and a "Closed-Source" OS (like iOS)?
3. Name two examples of modern smartphone operating systems and one example of an older, defunct OS.
4. What is the **IMEI**? How many digits does it have?
5. What is the "Blacklisting" of an IMEI, and why is it a powerful security feature?

ASSIGNMENT

1. Write a 200-word essay on the "Pros and Cons" of Android vs. iOS.
2. On any available phone, dial ***#06#** to find the IMEI. Write it down (you can 'x' out the last 4 digits for privacy).
3. What is the difference between an IMEI and a SIM card?

4. Define the following terms: Firmware, Operating System (OS), GUI, AOSP, IMEI, Blacklist.
5. Research what "rooting" an Android phone or "jailbreaking" an iPhone means. Write a 1-paragraph definition for each.



WEEK 4: GSM REPAIR TOOLS

INTRODUCTION

To repair a GSM phone, you need specialized, precision tools. Using the wrong tool (like a kitchen knife to pry open a phone) will *damage* the device. A professional technician's toolkit is divided into **Disassembly Tools**, **Diagnostic Tools**, and **Board-Level Repair Tools**.

	Disassembly Tools	Board-Level Tools
Disassembly	 Pentalobe Screwdriver  Plastic Spudger Tool  Suction Cup	 Lirrr Tool
Board-Level	 Soldering Iron	 Hot Air rework Station
Diagnostic	 Digital Multimeter	 USB ammeter

1. DISASSEMBLY & HAND TOOLS

- **Precision Screwdriver Set:** This is the *most essential* set.
 - **Phillips:** Small sizes (e.g., #000, #00) for most internal screws (Samsung, Tecno).
 - **Pentalobe:** The 5-point "star" screw. Used *only* by Apple on the *outside* of iPhones (near the charging port).
 - **Torx:** The 6-point star screw (e.g., T4, T5, T6). Used in many laptops and older phones.
 - **Tri-Wing (Y-Type):** The 3-point "Y" screw. Used by Apple *inside* the iPhone (e.g., to hold the screen and battery connectors).
- **Prying Tools (Spudgers):**

- **Plastic Spudger / Plectrum (Guitar Pick):** The *best* tool for opening plastic-body phones. They are anti-static and non-marring (they don't scratch the plastic). Used to slide into the seam and "pop" the clips.
- **Metal Spudger:** A strong, thin metal tool. Used for tough jobs where more force is needed, but *be careful* as it can scratch.
- **Suction Cup:** A small cup used to grip and *lift* the glass screen of a phone or tablet.
- **ESD-Safe Tweezers (Pointed & Angled):** Essential for picking up tiny screws, handling ribbon cables, and removing small components.
- **Magnetic Mat:** A mat with a grid. You place your screws on it as you remove them, so you (a) don't lose them and (b) remember *exactly* which hole they came from. (This prevents "Long Screw Damage" - putting a long screw in a short hole and drilling into the motherboard).

2. BOARD-LEVEL REPAIR TOOLS

These are the advanced tools for working directly on the circuit board (motherboard).

- **SOLDERING STATION:**
 - **Function:** Used to melt **solder** (a metal alloy) to create a new electrical joint.
 - **Primary Use:** Replacing a **charging port**, a **microphone**, or a **headphone jack**.
 - **Components:**
 - **Soldering Iron:** The "pen" that heats up.
 - **Solder Wire:** The "metal glue" (e.g., 60/40 Tin/Lead).
 - **Flux:** The *most important* part. A paste or liquid that *cleans* the metal and makes the solder flow smoothly. *No repair works without flux.*
- **HOT AIR REWORK STATION:**
 - **Function:** A "precision heat gun" that blows a very small, very hot (300-450°C) stream of air.
 - **Primary Uses:**

- i. **Disassembly:** Used to *melt the adhesive (glue)* that holds screens and back glass on modern, sealed phones.
 - ii. **Board Repair:** Used to "**re-flow**" or **replace** tiny **SMD (Surface-Mount Device)** components and **IC (Integrated Circuit)** chips on the motherboard.
- **DESOLDERING TOOLS:** (Tools for *removing* solder)
 - **Solder Wick (Braid):** A copper braid that "soaks up" old solder.
 - **Solder Sucker (Pump):** A small, spring-loaded vacuum.

3. DIAGNOSTIC & SOFTWARE TOOLS

- **DIGITAL MULTIMETER (DMM):**
 - **Function:** The #1 diagnostic tool.
 - **Uses:**
 - Testing **battery voltage** (e.g., 3.7V).
 - Testing the **charger output** (e.g., 5V).
 - Checking for **short circuits** on the motherboard (using **Continuity/Beep Mode**).
- **USB AMMETER / VOLTMETER:**
 - **Function:** A simple "pass-through" device that plugs between the charger and the phone's cable.
 - **Use:** It shows you the **Voltage** and **Current (Amps)** the phone is *drawing*. This is the *best* way to diagnose charging problems. (e.g., if it shows 5V but 0.0A, the phone is not charging).
- **FLASHING TOOLS (FLASHING BOXES):**
 - **Definition:** These are hardware/software tools used for *software* repairs.
 - **Function:** They interface between a PC and a phone to "flash" (re-install) the firmware, bypass security locks (FRP), or repair a corrupted IMEI.
 - **Examples:** Miracle Box, Z3X Box (for Samsung), UMT Dongle.
 - **Software-only tools:** Odin (for Samsung), SP Flash Tool (for MediaTek CPUs).

EVALUATION

1. What is the name of the 5-point "star" screwdriver used for iPhones?
2. What is a "Spudger," and why is a plastic one often better than a metal one?
3. What are the two *primary uses* of a Hot Air Rework Station?
4. What is "Flux," and why is it essential for soldering?
5. What is a "USB Ammeter" used to diagnose?

ASSIGNMENT

1. Write a 200-word report on the difference between a **Soldering Iron** and a **Hot Air Rework Station**.
2. What is "Long Screw Damage" in an iPhone repair, and what tool helps prevent this?
3. Draw a simple diagram of a technician using a **Soldering Iron**, **Solder Wire**, and **Flux** to attach a wire to a circuit board.
4. Research and list 5 safety precautions for using a Hot Air Rework Station (e.g., heat-resistant mat, not aiming at batteries).
5. Define: Pentalobe, Spudger, ESD-Safe, SMD, Flashing Box.

WEEK 6: DISASSEMBLY AND REASSEMBLY OF GSM PHONES

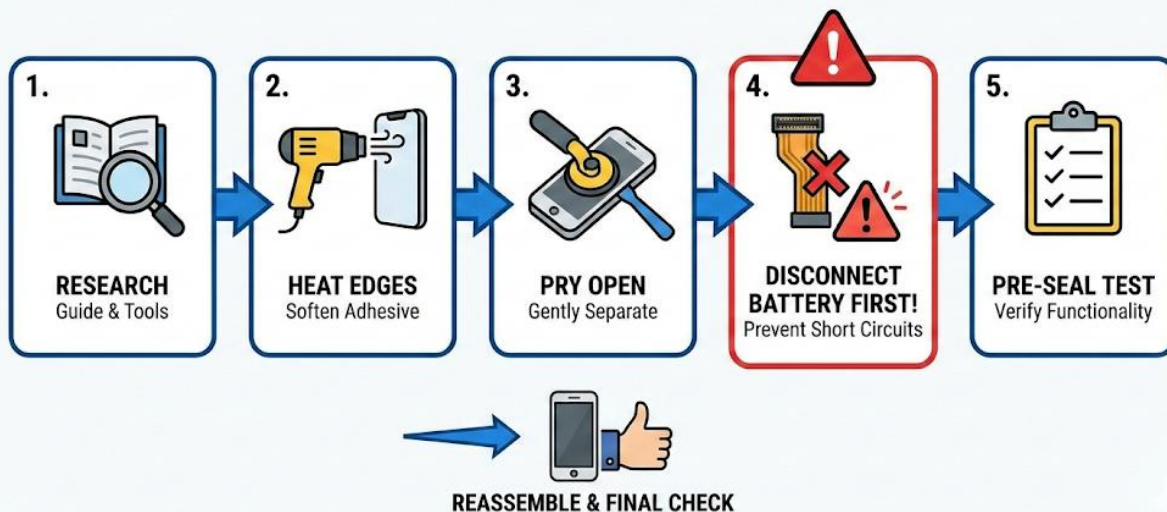
TOPIC: STEP-BY-STEP PROCESS

INTRODUCTION

This is a core practical skill. **Disassembly (Teardown)** is the process of safely taking a phone apart. **Reassembly** is the process of putting it back together.

This is a delicate operation. You are not a "carpenter"; you are a "surgeon." You must be patient, gentle, and organized. Rushing this process *will* break the phone.

PHONE DIS/ASSEMBLY: CRITICAL STEPS FLOWCHART



STEP 1: PREPARATION (MISE-EN-PLACE)

- **RESEARCH (THE #1 RULE):** Never open a phone "blind." Every single model is different.
 - Go to **YouTube** or **iFixit.com** and watch a teardown video for the *exact model* you are repairing (e.g., "Tecno Camon 19 teardown").
 - **Identify:** Does it open from the **front** (by removing the screen) or from the **back** (by removing the back cover)? This is the most important first step.
- **WORKSTATION:**

- Work on a **clean, well-lit, anti-static mat**.
- Have all your tools (from Week 4) ready: precision screwdrivers, spudgers, heat gun, suction cup.
- **SAFETY:**
 - Put on your **ESD wrist strap** (or ground yourself by touching metal).
 - Wear **safety glasses** (especially when prying at glass screens).

STEP 2: DISASSEMBLY (THE "TEARDOWN")

This is a generic guide. The exact steps *depend on your research*.

1. **POWER DOWN:** Turn the phone **completely off**.
2. **REMOVE EXTERNAL ITEMS:** Remove the SIM tray and any memory cards.
3. **REMOVE EXTERNAL SCREWS:** Remove any screws on the *outside* of the phone (e.g., the two Pentalobe screws at the bottom of an iPhone).
4. **SOFTEN ADHESIVE (FOR SEALED PHONES):**
 - Most modern phones (Samsung, iPhone) are sealed with strong glue.
 - Use a **Hot Air Gun** (on low, ~80-100°C) or a heat pad to *gently* warm the *edges* of the phone (the front or back, depending on how it opens).
 - **Warning:** Do not overheat. You are *softening* the glue, not *melting* the phone.
5. **PRY AND OPEN:**
 - Use a **suction cup** to get a small "gap" between the screen and the frame.
 - Insert a **plastic plectrum (guitar pick)** into this gap.
 - *Gently* slide the plectrum around the entire seam of the phone, cutting the adhesive.
6. **OPEN CAREFULLY (THE "RIBBON CABLE" DANGER):**
 - **DO NOT** pull the two halves apart quickly.
 - Open the phone *slowly* (like a book).
 - Look for the fragile, paper-thin **ribbon cables** (e.g., for the screen, fingerprint sensor) that are *still connecting* the two halves.

- Use a *plastic spudger* (never metal) to gently "pop" the connectors for these cables off the motherboard.
7. **DISCONNECT THE BATTERY (THE "FIRST-IN, LAST-OUT" RULE):**
- This is the **FIRST** component you disconnect *inside* the phone.
 - **Why?** This "de-energizes" the motherboard and prevents you from causing a **short circuit** if your metal screwdriver touches the wrong spot.
8. **MAINTAIN SCREW MANAGEMENT (CRITICAL):**
- As you remove internal screws, place them on a **magnetic mat** or a **labeled parts tray**.
 - Phone screws are **NOT** interchangeable. They have different lengths.
 - **Long Screw Damage:** If you put a *long* screw in a *short* hole (e.g., on an iPhone's screen shield), you will **drill through the motherboard**, cutting the circuits underneath and **killing the phone permanently**. This is an unfixable, amateur mistake.

STEP 3: REASSEMBLY

This is the *exact reverse* of disassembly, with one extra step.

1. **CONNECT COMPONENTS:** Connect all components *except* the battery.
2. **CONNECT BATTERY:** The battery is the **LAST** component you connect *before* sealing the phone.
3. **TEST (PRE-SEAL TEST):**
 - Before you glue the phone shut, power it on.
 - **Test everything:**
 - Does the new screen light up?
 - Does the touch (digitizer) work everywhere?
 - Do the cameras work? Wi-Fi? Speaker? Mic?
 - If it doesn't work, now is the time to fix it.
4. **SEAL THE PHONE:**

- Power the phone off. Disconnect the battery.
 - Clean *all* the old, gummy adhesive from the frame.
 - Apply new, pre-cut adhesive or a thin bead of B-7000 (phone glue).
 - Carefully press the two halves together. Use gentle clamps or rubber bands to hold it shut for 30 minutes while the glue cures.
5. **FINAL CLEAN:** Clean the screen and test one last time.

EVALUATION

1. What is the **#1 rule** you must follow *before* starting any disassembly?
2. What tool is used to *soften* the adhesive on a modern, sealed phone?
3. What is the **first** component you *must* disconnect inside the phone, and *why*?
4. What is "Long Screw Damage," and what tool helps prevent it?
5. What is the "Pre-Seal Test," and why is it so important?

ASSIGNMENT

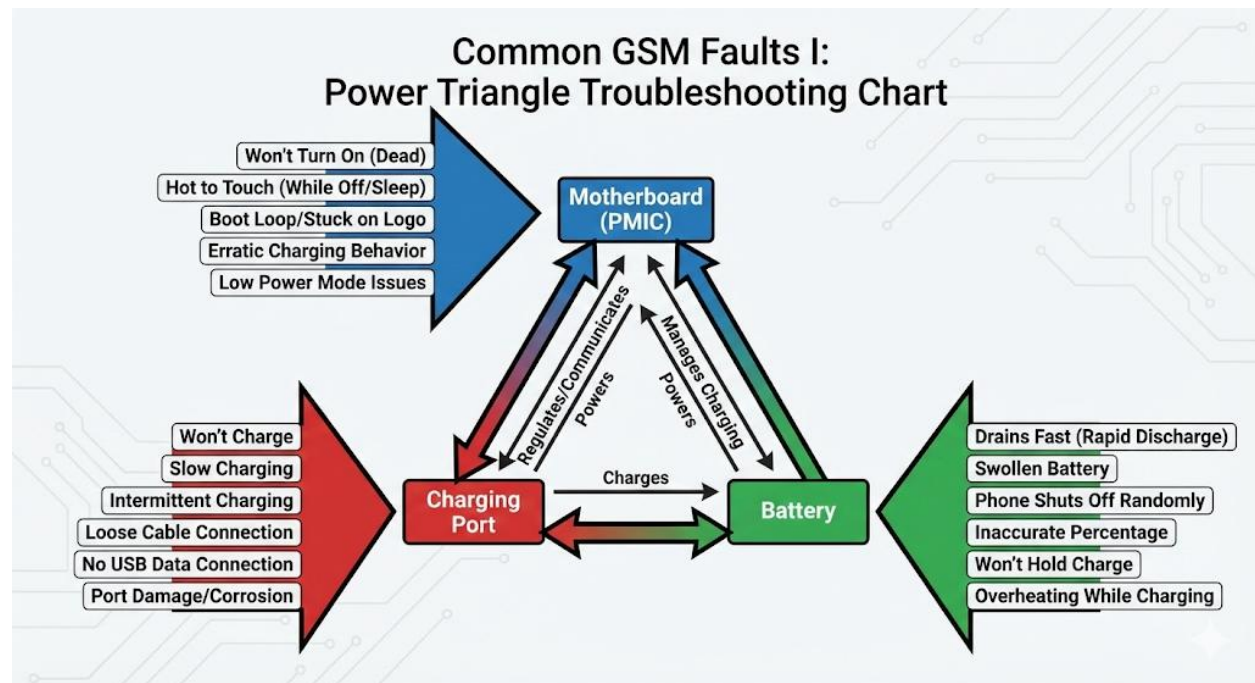
1. Go to YouTube and search for a "teardown" video of any modern smartphone (e.g., "iPhone 12 Teardown" or "Samsung S21 Teardown").
2. Watch the video and write a 1-page report on the process.
3. In your report, identify:
 - (a) Did it open from the *front* or the *back*?
 - (b) What *specialized screwdrivers* did the technician use?
 - (c) What was the *first* component disconnected after the phone was opened?
4. Draw a simple diagram illustrating "Long Screw Damage" (a long screw drilling into a motherboard).
5. Define: Teardown, Ribbon Cable, Spudger, Pentalobe, Adhesive, Short Circuit.

WEEK 8: COMMON GSM FAULTS I

🔌 TOPIC: CHARGING FAULTS, BATTERY FAULTS, POWER ISSUES

INTRODUCTION

This week, we begin diagnostics. We will focus on the "Power Triangle"—the **Charging Port**, the **Battery**, and the **Power Management (PMIC)** on the motherboard. Over 90% of "dead phone" problems are found here.



1. CHARGING FAULTS

This is the **most common hardware repair** in any shop.

- **SYMPTOMS:**

- Phone does not charge when plugged in.
- "False Charging" (shows the charging icon, but the percentage never goes up).
- Phone charges *very* slowly.
- You have to "bend" or "wiggle" the cable to make it charge.
- The cable is loose in the port.

- **DIAGNOSTIC PROCESS (Check in this order):**

- a. **RULE 1: CHECK THE CHARGER & CABLE.** *Always* test with a *known-good, original* charger and cable first. The cable is often the broken part.

- b. **RULE 2: CHECK THE PORT FOR DIRT.**

- Look inside the charging port with a bright light. It is often packed with **pocket lint** and debris, preventing the cable from seating fully.
 - **Fix:** With the phone **OFF**, gently clean the port with a *non-conductive* tool (like a sharp plastic toothpick or a dry, hard brush). *Never* use a metal pin (you can short-circuit the contacts).

- c. **RULE 3: CHECK WITH A USB AMMETER.**

- Plug the USB ammeter into the charger, and the phone into the ammeter.
 - A healthy phone will "draw" 0.5A to 2.0A.
 - If the ammeter shows 5V but **0.0A**, the phone is *not* drawing power. This confirms a hardware fault.

- d. **RULE 4: REPLACE THE PORT.**

- If the first 3 steps fail, the charging port itself is broken.
 - **Fix:** This is a **board-level repair**.
 - On some phones (e.g., Tecno, Samsung A-series), the port is on a separate, small "sub-board" that is *easy* to unplug and replace.
 - On other phones (e.g., iPhones, many others), the port is **soldered** directly to the main motherboard. This requires a **Hot Air Gun** and **Soldering Iron** to replace.

2. BATTERY FAULTS

- **SYMPTOMS:**

- The phone **drains very fast** (e.g., 100% to 0% in 1-2 hours).
 - The phone **suddenly shuts off** when it still shows 30% or 40% (a "battery crash").

- The phone is **swollen**. The screen or back cover is *lifting up* or *bulging*. This is a **fire hazard**—the battery must be replaced *immediately*.
- The phone only works when plugged into the charger.
- **DIAGNOSTICS:**
 - a. **Check Battery Health:** On an iPhone (Settings > Battery > Battery Health) or on Android (using a special app), check the "Maximum Capacity." If it is below 80%, it needs replacement.
 - b. **Visual Inspection:** Open the phone. Look for a **swollen, puffy, or "pillow-like"** battery.
- **FIX:**
 - **Replace the battery.**
 - **Old Phones:** Easy (removable back cover).
 - **Modern Phones:** Requires a full disassembly (Week 6), using a **Hot Air Gun** to soften the glue, and *gently* prying the battery out with a *plastic spudger*. **NEVER use a metal tool to pry a battery**; puncturing it can cause a fire.

3. POWER ISSUES ("DEAD PHONE")

- **SYMPTOM:** The phone is completely "dead." It shows no signs of life—no screen, no vibration, no charging light.
- **DIAGNOSTIC PROCESS (Check in this order):**
 - a. **RULE 1: "FAKE" DEAD.**
 - **Hard Reset:** The phone's *software* (OS) has crashed. Try a "hard reset" (e.g., hold Power + Volume Down for 15 seconds).
 - **Deep Discharge:** The battery is *so* dead (0.0V) that it cannot even light the charging icon. **Fix:** Leave it plugged into a known-good charger for at least 1-2 hours.
 - b. **RULE 2: TEST THE POWER BUTTON.**
 - The *power button* itself (the physical switch) might be broken or stuck.

c. **RULE 3: CHECK FOR WATER DAMAGE.**

- Open the phone. Look for **corrosion** (white or greenish "fuzz") on the motherboard. Look for the **Liquid Damage Indicator (LDI)**—a small white sticker that turns **pink or red** when wet.

d. **RULE 4: ADVANCED BOARD FAULT.**

- If the phone is clean and the battery is good, there is a **short circuit** or a failed **PMIC (Power Management IC)** on the motherboard.
- **Fix:** This requires a specialist with a multimeter to find the "short" and a hot air gun to replace the faulty chip.

EVALUATION

1. A customer's phone won't charge. What are the first *three* things you should check?
2. What tool is best for diagnosing a "false charging" problem?
3. What is the *most dangerous* symptom of a battery fault, and what should you do immediately?
4. Why must you *never* use a metal pin to clean a charging port or a metal spudger to pry out a battery?
5. What is a "Hard Reset," and when would you use it?

ASSIGNMENT

1. Write a 250-word "Troubleshooting Guide" for a customer whose phone won't charge. (Start with the simplest fix and end with the most complex).
2. What is a "PMIC" (Power Management IC)? Research its function on a motherboard.
3. Look for the "LDI" (Liquid Damage Indicator) on a phone (if you have an old one with a removable battery, it's often under it). What color is it?
4. Define: False Charging, Short Circuit, Corrosion, Hard Reset, Deep Discharge.
5. What are the safety risks of a swollen (puffy) Lithium-Ion battery?

WEEK 9: COMMON GSM FAULTS II

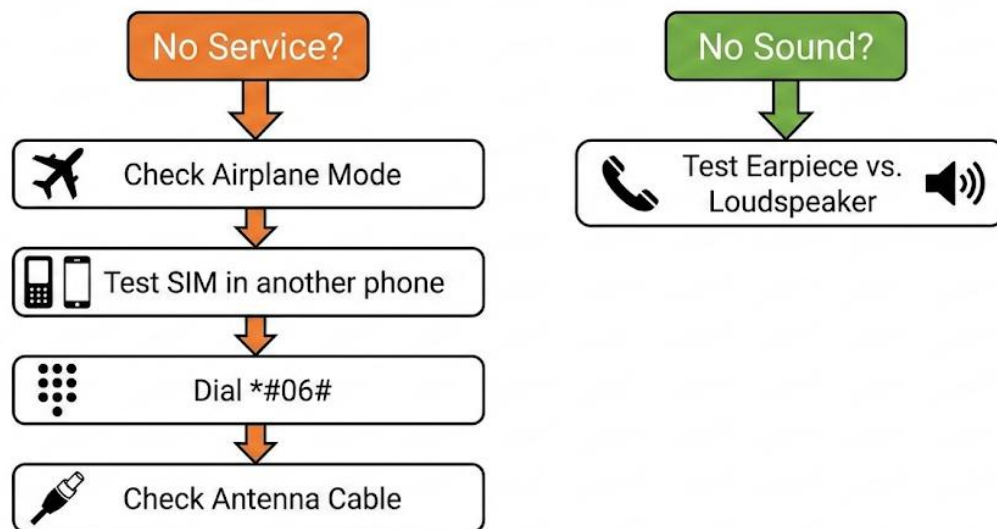
🔗 TOPIC: NETWORK ISSUES, MICROPHONE/SPEAKER PROBLEMS

INTRODUCTION

This week, we move from "power" faults to "connectivity" and "I/O" (Input/Output) faults. These are common problems that stop the user from communicating.

GSM FAULTS II: DIAGNOSTIC DECISION TREE

Visual: A Step-by-Step Guide.



1. NETWORK ISSUES

- **SYMPTOMS:**
 - "No Service"
 - "Emergency Calls Only"
 - Signal is very weak (1 bar).
 - Can't make or receive calls, even with a full signal.
 - "Invalid SIM" or "No SIM" (even when one is inserted).
- **DIAGNOSTIC PROCESS (Check in this order):**

a. **RULE 1: SOFTWARE (THE "SIMPLE FIXES").**

- **Airplane Mode:** Is it on?
- **Network Operator:** Go to Settings > Network. Try "searching" for operators and manually selecting "MTN" or "Glo."
- **Restart:** A simple reboot often fixes network glitches.

b. **RULE 2: THE SIM CARD.**

- **Test:** Put the SIM in a *different* phone. Does it work? Put a *different* (known-good) SIM in the customer's phone. Does it work?
- **Fault:** The SIM card itself may be bad, expired, or deactivated.

c. **RULE 3: THE IMEI NUMBER.**

- **Test:** Dial ***#06#**.
- **Result 1:** The IMEI is "Null" or "000000..." This is a **software fault**. The firmware is corrupted. **Fix: Flashing** (Week 10).
- **Result 2:** The IMEI is valid. **Action:** Check if the IMEI has been **Blacklisted** (e.g., the phone was reported stolen).

d. **RULE 4: PHYSICAL HARDWARE FAULTS.**

- **SIM Slot:** Open the phone. Use a magnifier to inspect the **gold pins** in the SIM slot. Are any bent, broken, or corroded? **Fix:** Advanced soldering repair.
- **Antenna Cable:** This is a *very common fault after a bad screen repair*. The **coaxial antenna cable** (a tiny wire) that connects the sub-board to the mainboard was forgotten or not "snapped" back on. **Fix:** Open the phone and re-connect it.
- **Network IC:** A chip on the motherboard is faulty. **Fix:** Advanced board-level repair.

2. MICROPHONE (MIC) PROBLEMS

- **SYMPTOM:** When you make a call, *you* can hear the other person, but they *cannot hear you*.
- **DIAGNOSTIC PROCESS:**
 - a. **Test:** Open the "Voice Recorder" app. Record your voice. Play it back.
 - If you hear *nothing*, the microphone is faulty.

- If you *can* hear your recording, the problem is with the *network* or the *other person's phone*.
 - b. **Check for Blockage:** Find the tiny **microphone hole** (usually at the bottom of the phone, near the charging port). It is often clogged with dirt or pocket lint.
- **Fix:** *Gently* clean it with a pin or a very fine brush.
 - c. **Check Permissions:** (For apps). If the mic doesn't work *only* in WhatsApp, go to Settings > Apps > WhatsApp > Permissions and make sure "Microphone" is allowed.
 - d. **Hardware Fault:** The microphone component itself is faulty.
- **Fix:** The mic is almost always **soldered** onto the charging port "sub-board." This sub-board is usually replaced as a single unit.

3. SPEAKER / EARSPIECE PROBLEMS

- **SYMPTOMS:**
 - You cannot hear the other person during a call.
 - The phone does not "ring" or play music.
- **DIAGNOSTIC PROCESS (Identify the speaker):**
 - Phones have **two** speakers:
 - **EARPIECE:** The *small, quiet* speaker at the *top* that you put to your ear.
 - **LOUDSPEAKER:** The *loud* speaker at the *bottom* for ringtones, music, and "speakerphone."
 - **Test:**
 - Make a call. If you can't hear, it's the **Earpiece**.
 - Play a YouTube video or ringtone. If you can't hear, it's the **Loudspeaker**.
 - **Causes & Fixes:**
 - i. **Software:** Volume is at 0. Or, the phone is **stuck in "Headphone Mode"** (the phone *thinks* headphones are plugged in). **Fix:** Clean the headphone jack.

- ii. **Blockage:** The speaker "grilles" (meshes) are clogged with dirt, dust, or makeup. **Fix:** Clean gently with a soft brush and Isopropyl Alcohol.
- iii. **Hardware Fault:** The speaker component is "blown" (sounds distorted) or disconnected. **Fix:** Component replacement (these are usually easy "plug-and-play" modules).

EVALUATION

1. A customer says their phone shows "No Service." List 4 *different* possible causes, from software to hardware.
2. A customer says, "When I make a call, the other person can't hear me." What is the *first* thing you should test?
3. A customer says, "When I make a call, *I* can't hear the other person, but my music plays fine." Which *specific* component is faulty?
4. What is the *most common* cause of a network antenna fault *after* a screen repair?
5. What is the "Headphone Mode" fault?

ASSIGNMENT

1. Write a 250-word "Troubleshooting Guide" for a "No Service" error.
2. Find a phone. Locate and draw a small map of its 3 key I/O ports: (a) the microphone hole(s), (b) the earpiece grille, and (c) the loudspeaker grille(s).
3. How would you *safely* clean a clogged loudspeaker grille?
4. What is a "coaxial antenna cable" in a phone? (Research and find a picture).
5. Define: Earpiece, Loudspeaker, Coaxial, IMEI, SIM Slot, Headphone Mode.

WEEK 10: GSM SOFTWARE ISSUES

TOPIC: FLASHING, UNLOCKING, VIRUS REMOVAL

INTRODUCTION

Not all "broken" phones have a hardware problem. Many are "broken" because their **software** (OS, Firmware) is corrupt. These are "soft" faults, and they are repaired using a PC and specialized software, not a soldering iron.

WHAT IS FLASHING? (GSM Software Issues)



1. "BRICKING"

- **Definition:** A "bricked" phone is a phone that cannot boot up (start) properly due to a corrupt software file. It is "as useful as a brick."
- **Symptoms:**
 - **Boot Loop:** The phone turns on, shows the logo (e.g., Samsung, Tecno), restarts, shows the logo, restarts... and never fully boots.
 - **Stuck on Logo:** The phone freezes at the logo.
 - **"Soft Brick":** The phone can still boot into "Recovery Mode" or "Download Mode." (This is *fixable*).

- **"Hard Brick":** The phone is completely dead and is not recognized by a PC.
(This is *very* difficult to fix).
- **Causes:** A failed software update, a bad "custom ROM" installation, or a virus.

2. FLASHING (RE-INSTALLING THE OS)

- **Definition:** "Flashing" is the process of completely **wiping the phone's internal storage** and **re-installing the entire system software (Firmware/ROM)** from scratch.
- **This is the "nuclear option."** It erases **ALL** user data (photos, contacts, apps). It is a last resort.
- **THE PROCESS:**
 - a. **Get the Tools:**
 - **The PC:** You need a Windows PC.
 - **The Flashing Software:** (e.g., **Odin** for Samsung, **SP Flash Tool** for MediaTek (Tecno/Infinix), **iTunes** for iPhone).
 - **The Drivers:** The PC needs the correct USB drivers to recognize the phone.
 - **The ROM (Stock ROM):** This is the most important part. You must download the *exact, correct* firmware file for the phone's *exact model number* (e.g., "Tecno-Camon-19-Pro-CI8").
 - b. **Boot the Phone to a Special Mode:**
 - **Download Mode (Samsung):** (e.g., Hold Power + Vol Down + Home).
 - **Fastboot/Bootloader Mode (Android):** (e.g., Hold Power + Vol Down).
 - c. **Connect & Flash:** Connect the phone to the PC, load the ROM file into the flashing tool, and click "Start." The tool then "flashes" (writes) the new software onto the phone.
 - d. **Reboot:** The phone reboots, fully reset to factory condition.

3. UNLOCKING

This refers to removing software-based locks.

- **SIM LOCK (NETWORK LOCK):**

- **What:** When a phone is "locked" to a specific network (e.g., a "Verizon" iPhone) and will not work with a Nigerian SIM.
- **Fix:** Requires a special **unlock code** (from the original carrier) or using a **Flashing Box** (e.g., Z3X, Miracle Box) to perform a "Direct Unlock."

- **PASSCODE / PATTERN LOCK (USER LOCK):**

- **What:** The user forgot their screen lock.
- **Fix 1 (Easy):** Use the "Find My Device" (Android) or "Find My" (iPhone) service online to remotely erase the phone.
- **Fix 2 (Hard - Wipes Data):** Boot the phone into "**Recovery Mode**" and perform a "**Wipe Data / Factory Reset**."

- **FRP (FACTORY RESET PROTECTION) LOCK:**

- **What:** A Google security feature. After a factory reset, the phone demands the *last Google account and password* that was synced to it. This is an anti-theft feature.
- **Problem:** Customers "forget" their password, or they buy a used phone that is still locked.
- **Fix:** This is a *very common* repair. It requires specialized software (Flashing Boxes or PC tools) to "**Bypass FRP**."

4. VIRUS REMOVAL

- **Symptoms:** Annoying, full-screen **pop-up ads**, apps installing themselves, phone is very slow, battery drains fast, high data usage.
- **FIXES (in order):**
 - a. **Find the App:** Go to Settings > Apps. Find and *uninstall* any strange apps you don't recognize.
 - b. **Scan:** Install a reputable Anti-Virus app (e.g., Malwarebytes) and run a full scan.
 - c. **Last Resort (Factory Reset):** The *only* 100% guaranteed way to remove a deep virus is to **Wipe Data / Factory Reset** the phone (after backing up your photos).

EVALUATION

1. What does it mean if a phone is "bricked"?
2. What is "Flashing," and what does it do to the user's data?
3. What are the 4 key things you need to flash an Android phone?
4. What is "FRP (Factory Reset Protection)"? Why was it created?
5. A customer's phone has constant pop-up ads. What are the 3 steps you would take to fix it?

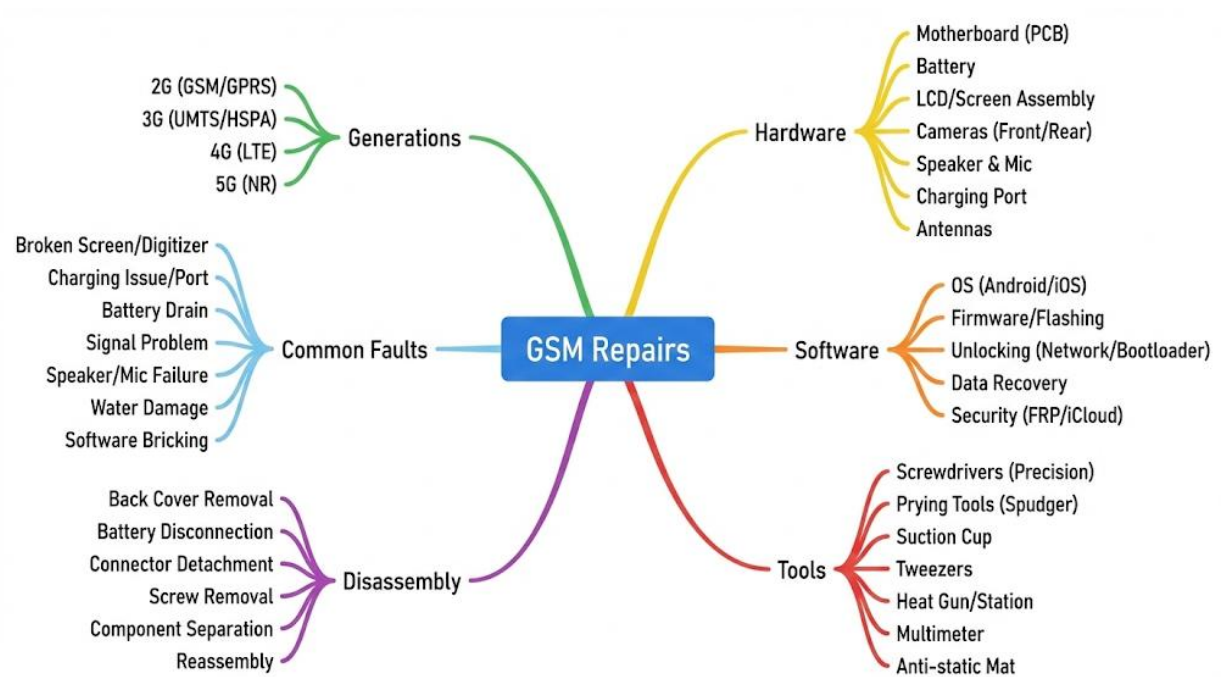
ASSIGNMENT

1. Write a 250-word "Warning Guide" for a customer, explaining what "flashing" is and why it is a "last resort" for fixing their phone.
2. Research and name two "Flashing Software" tools for PC (e.g., Odin) and one "Flashing Box" (hardware).
3. What is a "Stock ROM" vs. a "Custom ROM"?
4. Explain the *two* different ways a phone can be "locked" (Network Lock vs. User Lock).
5. Define: Boot Loop, Fastboot Mode, Recovery Mode, FRP, Adware (Virus).

WEEK 11: REVISION

📖 GENERAL REVIEW OF ALL SECOND TERM TOPICS

This week is for a comprehensive review of all topics covered this term. We have moved from basic theory to a complete understanding of GSM hardware, software, and repair diagnostics.



• WEEK 1: GSM TECHNOLOGY

- **Evolution:** 1G (Analog, Voice) → **2G/GSM** (Digital, Voice + SMS) → 3G (Internet) → 4G (Broadband/Video) → 5G (IoT/Low Latency).
- **Architecture:** MS (Phone: IMEI + SIM) → BSS (Tower: BTS + BSC) → NSS (Core: MSC + HLR).

• WEEK 2: GSM HARDWARE

- **Screen:** LCD (needs backlight) vs. **OLED/AMOLED** (emits own light).
- **Digitizer:** The invisible *input* (touch) layer.
- **Battery:** Li-Ion / Li-Po. *Safety risk* if punctured.
- **Ports:** SIM Slot (reads card) and **Charging Port** (most common fault).

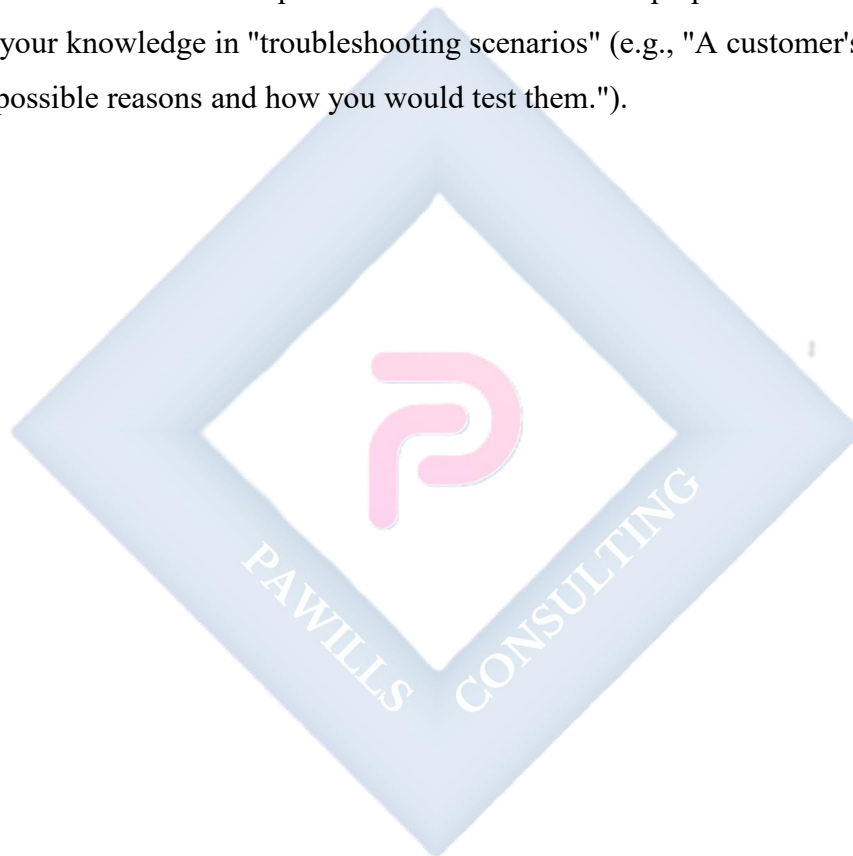
• WEEK 3: GSM SOFTWARE

- **Firmware:** Low-level "BIOS" of the phone.
- **OS:** The main user interface (e.g., **Android** (open), **iOS** (closed)).
- **IMEI:** The 15-digit "fingerprint" of the phone (find with ***#06#**). Used for **blacklisting** stolen phones.
- **WEEK 4: REPAIR TOOLS**
 - **Disassembly:** Precision Screwdrivers (**Pentalobe, Torx**), **Spudgers**, Suction Cup.
 - **Board-Level: Soldering Station** (for ports, mics), **Hot Air Gun** (for chips, glue), **Flux** (essential for soldering).
 - **Diagnostic: Multimeter** (voltage, continuity), **USB Ammeter** (charging test).
 - **Software: Flashing Boxes** (Miracle Box, Z3X).
- **WEEK 6: DISASSEMBLY / REASSEMBLY**
 - **Process:** Research → Heat → Pry → Open *carefully* (ribbon cables).
 - **#1 Rule: Disconnect Battery FIRST.**
 - **Risk: Long Screw Damage** (using the wrong screw in the wrong hole).
 - **Test:** Always do a "**Pre-Seal Test**" *before* gluing the phone shut.
- **WEEK 8: FAULTS I (POWER)**
 - **Charging:** Test cable → Clean port → Use USB Ammeter → Replace port.
 - **Battery:** Drains fast, or **swollen** (fire hazard). Replace.
 - **Dead Phone:** Check for Hard Reset, Deep Discharge, Water Damage (LDI), or Short Circuit.
- **WEEK 9: FAULTS II (I/O)**
 - **Network:** Check Airplane Mode → Test SIM → Check IMEI (***#06#**) → Check Antenna cable → Check SIM pins.
 - **Mic:** (They can't hear you). Test with Voice Recorder → Clean mic hole.
 - **Speaker:** (You can't hear them). Differentiate **Earpiece** (calls) vs. **Loudspeaker** (rings). Check for "Headphone Mode" → Clean grille.
- **WEEK 10: FAULTS III (SOFTWARE)**

- **Bricking:** Stuck on logo (boot loop).
- **Flashing:** Wiping and re-installing the OS/Firmware. *Erases all data.*
- **Unlocking:** Network Lock (SIM) or User Lock (Pattern/Passcode).
- **FRP:** Google's anti-theft lock.
- **Virus:** Pop-up ads. Fix by scanning or Factory Reset.

FINAL EXAMINATION

The final examination will test all aspects of this term's work. Be prepared to define, compare, list, and apply your knowledge in "troubleshooting scenarios" (e.g., "A customer's phone won't charge. List 3 possible reasons and how you would test them.").



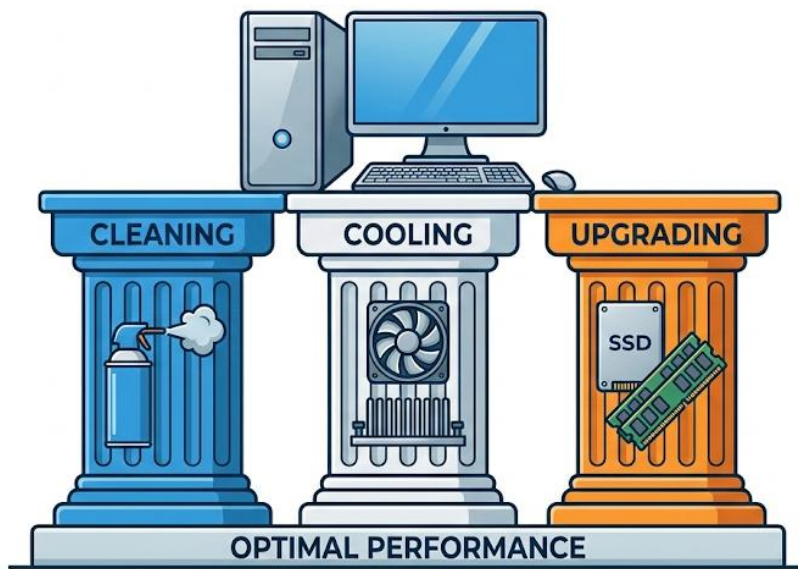
SS1 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

THIRD TERM

WEEK 1: MAINTENANCE OF COMPUTER HARDWARE

TOPIC: CLEANING, COOLING, AND UPGRADING COMPONENTS

THE PILLARS OF COMPUTER MAINTENANCE



INTRODUCTION

In the first term, we learned to identify the components of a computer. In the second term, we learned to assemble them. This term, we learn how to **maintain** them.

Maintenance is the practice of keeping equipment in good working condition. It is a critical, ongoing process. There are two types of maintenance:

1. **Corrective Maintenance:** Fixing something *after* it has broken (e.g., replacing a dead power supply). This is expensive and causes "downtime."
2. **Preventive Maintenance:** Performing routine checks, cleaning, and upgrades to *prevent* problems from happening. This is cheap, efficient, and the focus of a professional technician.

This week, we focus on the three pillars of preventive maintenance: **Cleaning, Cooling, and Upgrading.**

1. CLEANING (THE WAR ON DUST)

THE PROBLEM: DUST Dust is the **number one enemy** of computer hardware. It is not just "dirty"; it is *dangerous*.

- **How?** Dust is a **thermal insulator**. It forms a "blanket" on components.
- **The Effect:** This blanket traps heat, preventing the fans and heat sinks from doing their job. This leads to **overheating**. Overheating, in turn, causes components to run slow (see "Cooling" below) and ultimately *fail* (burn out) long before their time.

TOOLS FOR CLEANING:

- **Compressed Air / Electric Data Duster:** The primary tool. Used to blow dust out of hard-to-reach areas.
- **Soft-bristle Brushes (Anti-static):** For gently dislodging "caked-on" dust from fans and heat sinks.
- **Lint-Free Cloths:** For wiping down external cases and screens.
- **Isopropyl Alcohol (IPA):** (90% or higher). A fast-evaporating, non-conductive solvent. Used to clean thermal paste off a CPU or remove grime.

CLEANING PROCEDURES (ROUTINE CHECKS):

1. EXTERNAL (Weekly):

- Wipe down the monitor, keyboard, and mouse with a slightly damp or IPA-moistened cloth.
- Use compressed air to blow crumbs and dust out from under the keyboard keys.
- Use compressed air to clean the vents on the laptop or desktop case.

2. INTERNAL (Every 3-6 Months):

- **Safety First:** Power down the computer, unplug it from the wall, and move it to a well-ventilated area (preferably outside).

- **ESD:** Ground yourself (touch the metal case) or use an ESD wrist strap.
- **Case Fans & Filters:** Open the side panel. Most cases have removable **dust filters** (at the front, top, or bottom). Remove these, wash them with water, dry them *completely*, and re-install.
- **PSU Fan:** Use compressed air to blow air *into* the PSU fan grille. You will see a cloud of dust blow *out* the back.
- **CPU & GPU Fans:** Hold the fan blades *stationary* with your finger (so they don't over-spin) and blow the dust out of the **heat sink** fins.
- **Motherboard:** Use gentle, sweeping bursts of compressed air to clear dust from the motherboard itself.

2. COOLING (THE BATTLE AGAINST HEAT)

THE PROBLEM: HEAT Heat is the by-product of electricity. Every component in your PC generates heat. The **CPU** and **GPU** generate the *most* heat. If this heat is not removed, the components will be destroyed.

THERMAL THROTTLING: This is the computer's self-defense mechanism. When a component (like the CPU) gets too hot, its internal "thermometer" tells it to **slow down** to produce less heat.

- **Symptom:** Your computer feels "fast" when you turn it on, but after 30 minutes, it becomes very slow and "laggy." This is **thermal throttling** due to poor cooling.

COOLING SYSTEMS:

- **PASSIVE COOLING (Heat Sinks):**
 - **What:** A block of metal (usually aluminum or copper) with "fins" that increase the surface area.
 - **How:** The heat from the chip (CPU) is conducted to the heat sink. The large surface area allows this heat to *radiate* into the air.
- **ACTIVE COOLING (Fans):**
 - **What:** A fan attached to a heat sink (e.g., the "CPU Cooler") or the case.

- **How:** It *actively* blows cool air *across* the fins of the heat sink, forcing the heat away much faster.
- **THERMAL PASTE (Thermal Interface Material - TIM):**
 - **What:** A critical, silver or white paste.
 - **Function:** The surface of a CPU and a heat sink are not perfectly flat; they have microscopic gaps. Air (which is an insulator) gets trapped in these gaps.
 - **How:** A *thin layer* of thermal paste is applied between the CPU and the heat sink. This paste fills the gaps and *conducts* the heat efficiently from the chip to the metal sink.
 - **Maintenance:** This paste dries out after 2-4 years and *must be replaced*. This is a common repair for "overheating" laptops.

3. UPGRADING COMPONENTS

Definition: Upgrading is the process of replacing an existing, functional component with a newer, better, or faster one to improve the computer's overall performance.

THE THREE MOST COMMON AND EFFECTIVE UPGRADES:

3. STORAGE UPGRADE (THE #1 UPGRADE):

- **What:** Replacing an old, mechanical **Hard Disk Drive (HDD)** with a modern **Solid State Drive (SSD)**.
- **Why:** An SSD has *no moving parts*. It is 10-20 times faster at *reading and writing data* than an HDD.
- **Result:** This is the *single best* upgrade you can make. It will make an old computer feel "brand new." The Operating System (Windows) will boot in *seconds* instead of *minutes*, and programs will open instantly.

4. RAM UPGRADE (FOR MULTITASKING):

- **What:** Adding more **RAM (Random Access Memory)** (e.g., going from 4GB to 8GB, or 8GB to 16GB).

- **Why:** RAM is the "workbench" (short-term memory). If you don't have enough, the computer has to use the *slow* hard drive as "virtual memory."
- **Result:** The computer will be able to handle *more programs open at the same time* without slowing down. Essential for people who use many browser tabs, Photoshop, or video editors.

5. GRAPHICS CARD (GPU) UPGRADE:

- **What:** Adding or replacing the **Graphics Processing Unit (GPU)**.
- **Why:** This component is *only* responsible for rendering 3D graphics and video.
- **Result:** This upgrade is *only* for **gamers** or **video professionals**. It will *not* make Microsoft Word or browsing the internet any faster.

EVALUATION

1. What is "preventive maintenance," and why is it better than "corrective maintenance"?
2. Explain *why* dust is the #1 enemy of a computer. (Use the terms "insulator" and "overheating").
3. What is "Thermal Throttling"? What is the visible *symptom* to the user?
4. What is "Thermal Paste," and what is its *exact* function?
5. What is the *single most effective* upgrade for making an old computer feel "fast," and why?

ASSIGNMENT

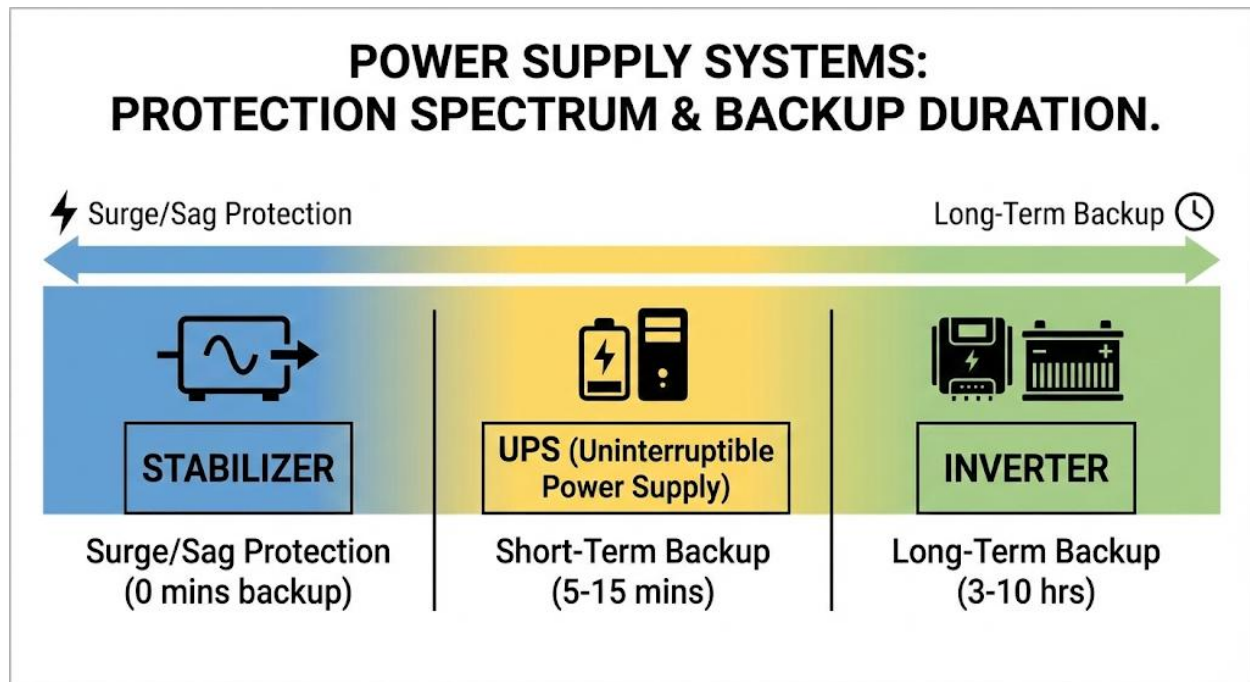
1. Write a 250-word "Maintenance Guide" for a new computer owner. Explain the steps for *Internal Cleaning* and *External Cleaning* and how often they should do it.
2. What is the difference between "Passive Cooling" and "Active Cooling"? Draw a simple diagram of a CPU, Thermal Paste, Heat Sink, and Fan, showing the path of heat.
3. Research the difference between an **HDD** and an **SSD**. Write a 3-paragraph report comparing them in terms of (a) Speed, (b) Cost, and (c) Durability.
4. A customer complains their laptop is "very slow," especially when they have 15 Google Chrome tabs open. What is the *most likely* component that needs to be upgraded? Explain your answer.
5. Define the following terms: E-Waste, Downtime, Thermal Paste, SSD, GPU, RAM.



WEEK 2: POWER SUPPLY SYSTEMS

TOPIC: UPS, STABILIZERS, INVERTERS

INTRODUCTION



In the first term (Week 4), we identified the **Power Supply Unit (PSU)** as the internal component that converts AC to DC. That PSU, however, is designed to receive *clean, stable* power.

In Nigeria, the main electrical grid (NEPA/PHCN) is notoriously "**dirty**" and "**unstable**." We face four main power problems:

6. **Spike (or Surge):** A sudden, massive *increase* in voltage (e.g., 300V) for a split second. This "fries" and *destroys* electronics.
7. **Sag (or Brownout):** A sustained *drop* in voltage (e.g., 170V). This stresses and overheats components, causing damage over time.
8. **Blackout (Outage):** A *total loss* of power. This causes data loss (unsaved work).
9. **Noise:** Fluctuations caused by other large appliances.

This week, we study the **external power systems** designed to protect our hardware from these four problems.

1. STABILIZERS (AUTOMATIC VOLTAGE REGULATORS - AVR)

- **DEFINITION:** A **Stabilizer** (or AVR) is a device that **regulates voltage** but has **NO battery**.
- **FUNCTION:** Its *only* job is to fix **sags** and **spikes**.
 - It constantly monitors the incoming AC voltage.
 - If the voltage *sags* (e.g., to 180V), an internal "buck/boost" transformer will *boost* it back up to a safe ~220V.
 - If the voltage *spikes* (e.g., to 250V), it will *trim* it back down to a safe ~220V.
- **LIMITATION:** A stabilizer has **no battery**. When NEPA cuts the power, your computer will **still turn off immediately**. It offers *no protection* against blackouts.
- **USE:** Essential for "motorized" appliances that *don't* need a battery, such as **refrigerators, freezers, and air conditioners**, which are very sensitive to sags.

2. UPS (UNINTERRUPTIBLE POWER SUPPLY)

- **DEFINITION:** A **UPS** is a device that provides **emergency, short-term power** from an internal, sealed battery.
- **PRIMARY PURPOSE:** The purpose of a UPS is **NOT** to let you *keep working* for hours. Its purpose is to give you **5-15 minutes** of battery time, which is just enough time to **SAVE YOUR WORK and SHUT DOWN YOUR COMPUTER SAFELY** during a blackout.
- It also provides excellent surge and sag protection.
- **TYPES OF UPS:**
 - a. **STANDBY (OFFLINE) UPS:**

- **How it Works:** This is the most basic and common type. The computer runs on *normal NEPA power*. The UPS just "watches" the power line.
- When it *detects* a blackout, an internal switch *transfers* (in 2-8 milliseconds) to the battery.
- **Use:** Good for home PCs and small offices.

b. **LINE-INTERACTIVE UPS:**

- **How it Works:** This is the *best* choice for most users. It is a Standby UPS but *also contains a stabilizer (AVR)*.
- **Advantage:** It can fix sags and spikes *without* switching to the battery. This *saves* the battery's life, as it only uses it for a *true* blackout.

c. **ONLINE (DOUBLE-CONVERSION) UPS:**

- **How it Works:** This is the most advanced and expensive type. The computer *always* runs on the battery. NEPA power is used *only* to charge the battery.
- **Analogy:** The computer is *never* "drinking" from the NEPA tap. It is *always* drinking from the clean "battery tank," and NEPA is just refilling the tank.
- **AdvNTAGE:** There is **zero (0 ms) transfer time** in a blackout. It produces *perfectly clean* power at all times.
- **Use:** Critical, "mission-control" equipment like hospital life-support, bank servers, and data centers.

3. INVERTERS

- **DEFINITION:** An inverter is a device that **converts DC (Direct Current) power into AC (Alternating Current) power**.
- **ROLE IN A BACKUP SYSTEM:** An "Inverter System" (as sold in Nigeria) is *different* from a UPS. It is a large-scale system designed for **long-term backup power** for *household loads* (lights, fans, TV).
- **THE SYSTEM:** It is *not* a single box. It is a system of two main components:
 - d. **The Inverter Unit:** The large (often wall-mounted) "brain."

- e. **The Battery Bank:** A set of large, **deep-cycle** batteries (e.g., 12V 200Ah batteries).
- **FUNCTION (DUAL-PURPOSE):**
 - f. **When NEPA is ON:** The inverter unit acts as a powerful **battery charger**, "filling up" the large battery bank.
 - g. **When NEPA is OFF:** The inverter "inverts" the 12V, 24V, or 48V **DC** power from the battery bank into 220V **AC** power to run the house for **3 to 10 hours**.
- **WAVEFORM (Recall from 1st Term):**
 - **Modified Sine Wave:** Cheaper. OK for lights and fans, but *dangerous* for fridges, electronics, and computers.
 - **Pure Sine Wave:** More expensive. Produces clean, grid-quality power. This is *essential* for running computers, TVs, and all modern electronics.

COMPARISON: UPS vs. INVERTER SYSTEM

Feature	UPS (Line-Interactive)	INVERTER SYSTEM
Primary Goal	Safe Shutdown (5-15 minutes)	Long-Term Backup (3-10 hours)
Loads	A single PC or small office.	An entire house (lights, fans, TV).
Battery	Small, internal, sealed.	Large, external, deep-cycle.
Form	Small, all-in-one box (plug-in).	Large, component system (inverter + batteries).
Transfer Time	4-8 milliseconds.	10-20 milliseconds.

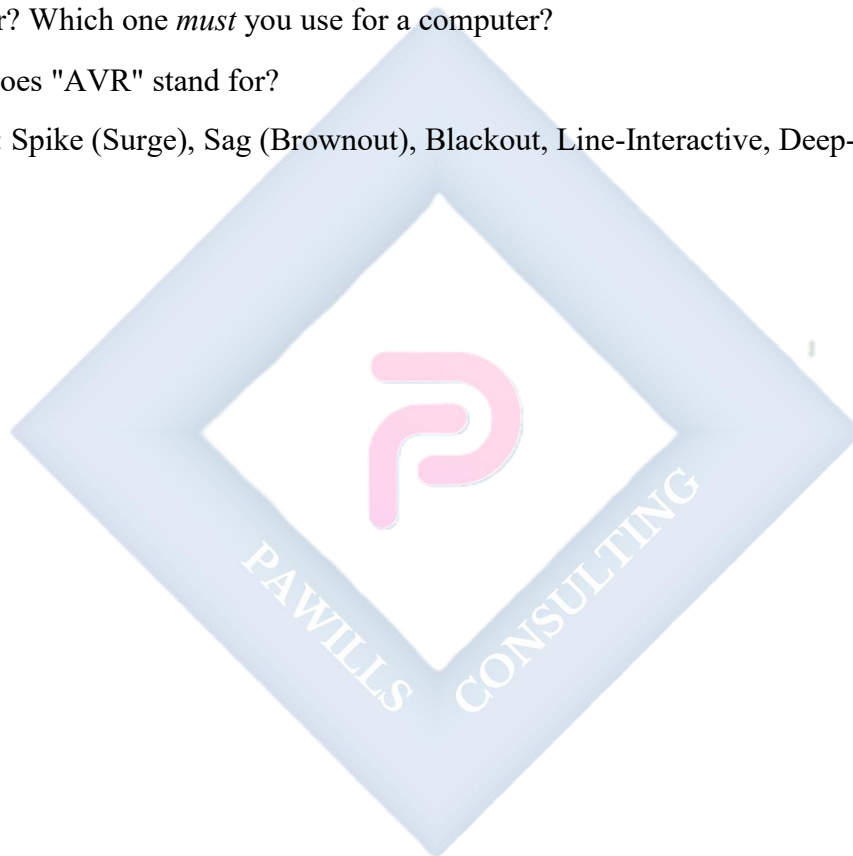
EVALUATION

1. What are the 4 main "power problems" common in Nigeria?
2. What is the *only* function of a Stabilizer?
3. What is the *primary* purpose of a UPS?
4. What is the main *advantage* of a "Line-Interactive" UPS over a "Standby" UPS?

5. Explain the two main functions of a home "Inverter System" (when NEPA is on vs. off).

ASSIGNMENT

1. Write a 250-word essay for a small business owner explaining why they need *both* a Stabilizer (for their freezer) and a UPS (for their computer).
2. Research "Online Double-Conversion UPS." Draw a simple block diagram showing how the power *always* flows from NEPA → Charger → Battery → Inverter → PC.
3. What is the difference between a "Modified Sine Wave" inverter and a "Pure Sine Wave" inverter? Which one *must* you use for a computer?
4. What does "AVR" stand for?
5. Define: Spike (Surge), Sag (Brownout), Blackout, Line-Interactive, Deep-Cycle Battery.



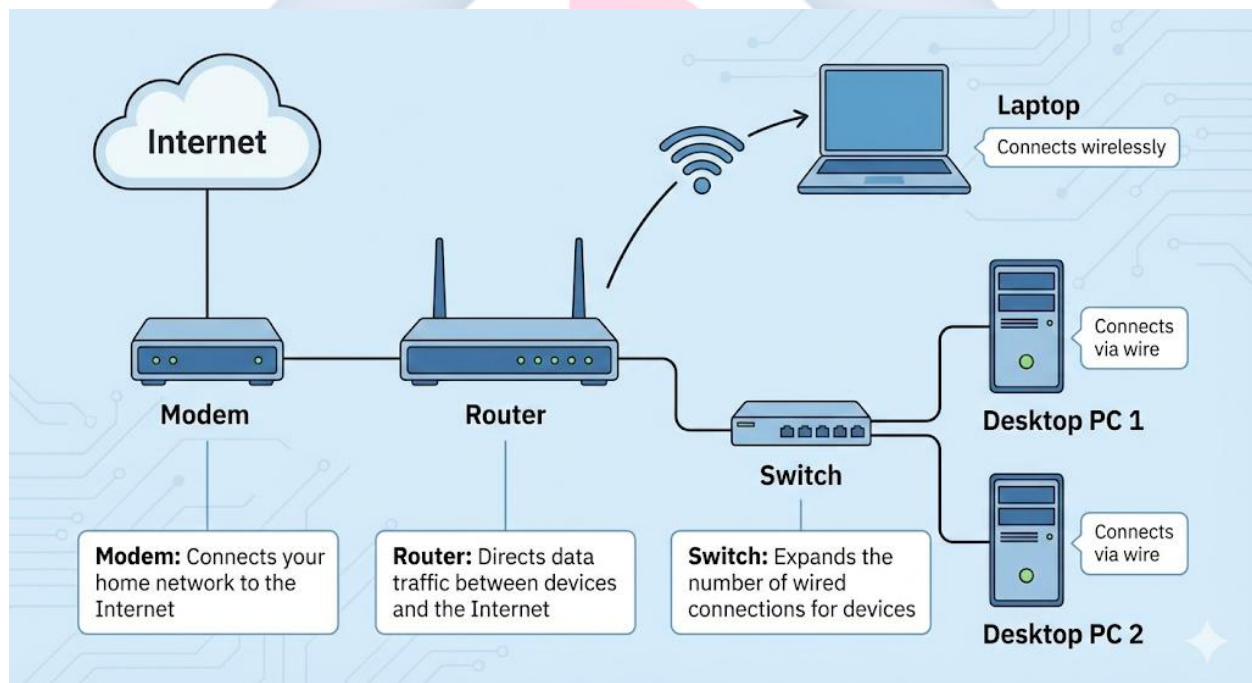
WEEK 3: INTRODUCTION TO NETWORKING HARDWARE

INTRODUCTION: LAN vs. WAN

To understand networking tools, you must first understand the two types of networks.

- **LAN (Local Area Network):** This is your **private** network. It is "local."
 - *Examples:* Your home network, your school's computer lab, your office network.
- **WAN (Wide Area Network):** This is a **public** network that covers a huge geographical area.
 - *Example:* The **Internet** is the world's largest WAN. Your Internet Service Provider (ISP), like MTN, Glo, or Spectranet, provides you with *access* to this WAN.

The hardware components we study this week are the "bridge" and "traffic cops" that connect your **LAN** to the **WAN**.



1. MODEMS

- **DEFINITION:** A **Modem** is a "translator" device. The name stands for **MOD**ulator / **DE**Modulator.

- **FUNCTION:** A modem's *only* job is to **connect your LAN to the WAN (the Internet)**. It is the "gateway" to the outside world.
- **THE "TRANSLATION" (HOW IT WORKS):**
 - a. Your computer and router speak in **Digital** signals (1s and 0s).
 - b. The internet (WAN) signal from your ISP travels over a different medium, such as **Analog** radio waves (4G/5G) or fiber-optic **light pulses**.
 - c. **MOdulation (Uploading):** The Modem *translates* the outgoing **digital** data from your computer into an **analog/light** signal to be sent over the WAN.
 - d. **DEModulation (Downloading):** The Modem *translates* the incoming **analog/light** signal from the WAN back into a **digital** signal that your computer can understand.
- **TYPES:** 4G/5G LTE Modems (like the little "Mi-Fi" devices), Fiber Optic Modems (ONTs), DSL Modems.

2. ROUTERS

- **DEFINITION:** A **Router** is the "brain" or "traffic cop" of your **Local Area Network (LAN)**.
- **PRIMARY FUNCTIONS (IT HAS 2 KEY JOBS):**
 - e. **Creates the LAN (Assigns IPs):**
 - A router's *first* job is to create your private network. It acts as a **DHCP (Dynamic Host Configuration Protocol)** server.
 - This means it *automatically assigns* a unique *private IP address* (e.g., 192.168.1.100, 192.168.1.101) to every single device that joins your LAN (your phone, your laptop, your smart TV).
 - f. **Routes Traffic (Acts as Gateway):**
 - A router's *second* job is to be the single "gateway" between your LAN and the Modem (WAN).

- It takes all the "mail" (data) from all your devices, organizes it, and "routes" it to the correct destination—either to *another device on the LAN* (e.g., sending a file to your printer) or *to the Modem* to go out to the Internet.
- **ROUTERS VS. "MODEM-ROUTERS":**
 - In many modern homes, your ISP (e.g., Spectranet) gives you *one box*. This is a **combo device** that has *both* a Modem (to connect to the WAN) and a Router (to create your LAN/Wi-Fi) built into one unit.
- **WI-FI (A ROUTER'S FEATURE):**
 - Most modern "routers" are **Wireless Routers**. This means they also contain a **WAP (Wireless Access Point)**.
 - A WAP is the hardware that *broadcasts* the Wi-Fi signal (radio waves), allowing your wireless devices to join the LAN.

3. SWITCHES (NETWORK SWITCHES)

- **DEFINITION:** A **Switch** is a simple device used to **expand a wired LAN**.
- **FUNCTION:** It is a "power strip for the network."
- **ANALOGY / USE CASE:**
 - Your router (from Job #1) *creates* the network.
 - But your router only has **4** LAN ports (the sockets in the back).
 - **Problem:** You are setting up a school computer lab with **30** desktop PCs that all need a *wired* connection.
 - **Solution:** You run *one* cable from the router to a **Network Switch**. The Switch (which has 8, 16, 24, or 48 ports) then provides enough ports for all 30 PCs.
- **ROUTER VS. SWITCH (CRITICAL DIFFERENCE):**
 - A **Router** is "smart." It *creates* a network and assigns IP addresses. It's the "boss."
 - A **Switch** is "dumb" (in a basic sense). It *does not* create a network or assign IPs. It is just an "expander" or "splitter" that *extends* the network created by the router.

EVALUATION

1. What is the main difference between a LAN and a WAN?
2. What does the word "Modem" stand for, and what is its primary "translation" job?
3. What are the two *main functions* of a router? (Hint: creating IPs and routing traffic).
4. What is a "Switch" used for?
5. Explain the key difference between a Router and a Switch.

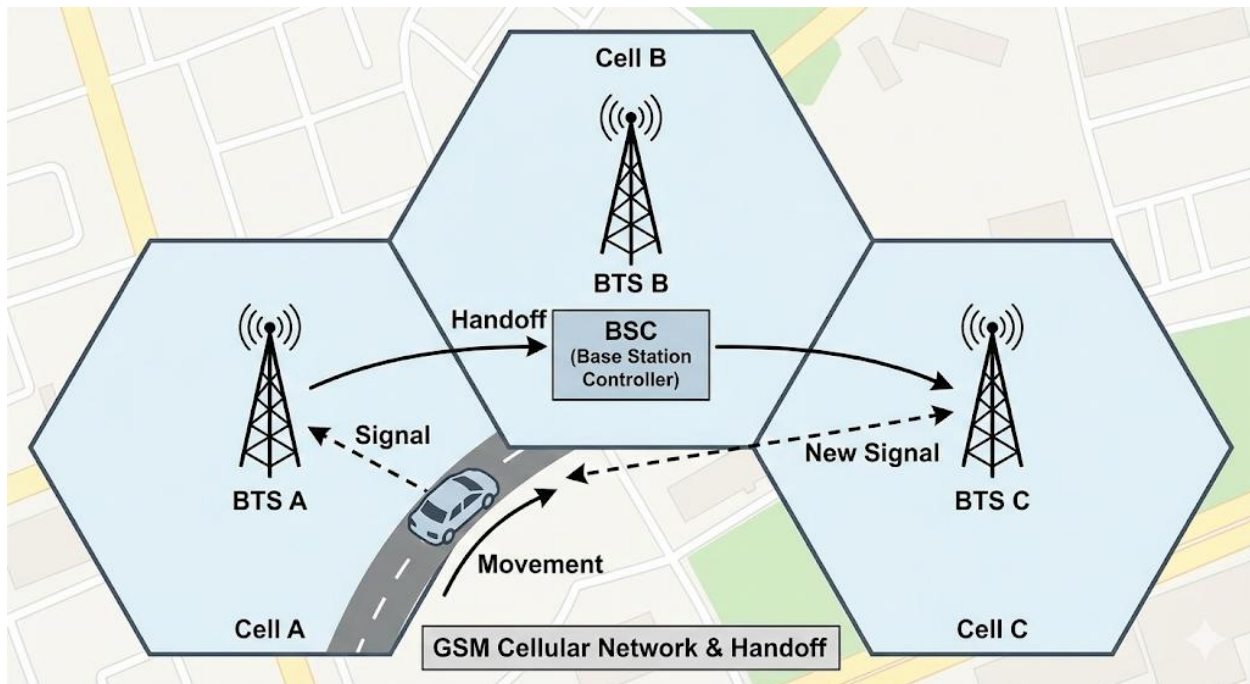
ASSIGNMENT

1. Write a 250-word essay using the analogy of a "House" to explain these 3 devices. (e.g., "The Modem is the front door to the main street (Internet). The Router is the main hallway... The Switch is a power strip in the office...").
2. What does "DHCP" stand for, and what does it do?
3. What is a "WAP" (Wireless Access Point)? Is it its own device, or a *feature* of a router?
4. Draw a simple network diagram for a small office. It must include: The Internet (a cloud), 1 Modem, 1 Router, 1 Switch, and 3 PCs (wired).
5. Define: LAN, WAN, ISP, IP Address, Wi-Fi, Ethernet Port.

WEEK 4: GSM COMMUNICATION PRINCIPLES

INTRODUCTION

In the second term, we learned *about* the GSM network (the "architecture"). This week, we dive deeper into the *physics and technology* that make it work. How does your voice travel from your phone to a tower? It all happens on invisible radio waves.



1. FREQUENCY AND FREQUENCY BANDS

- **DEFINITION:** All wireless communication (radio, TV, Wi-Fi, GSM) travels on **radio waves**. **Frequency** is the *number of waves* that pass a point in one second, measured in **Hertz (Hz)**.
 - 1 KiloHertz (KHz) = 1,000 waves per second.
 - 1 MegaHertz (MHz) = 1,000,000 waves per second.
 - 1 GigaHertz (GHz) = 1,000,000,000 waves per second.
- **FREQUENCY BANDS (THE "AIRWAVE HIGHWAYS"):**
 - The "air" (the spectrum) is a finite resource. To prevent chaos, the government (e.g., the NCC in Nigeria) *regulates* the airwaves.

- It *auctions* specific "highways" (or **frequency bands**) to specific companies for their exclusive use.
- **Example (Nigeria):**
 - MTN might be assigned the **900 MHz** band for 2G/3G.
 - Glo might be assigned the **1800 MHz** band for 2G/3G.
 - Airtel might be assigned the **2100 MHz** band for 3G/4G.
 - **4G (LTE)** and **5G** use *higher* frequency bands (e.g., 2.6 GHz, 3.5 GHz) because a higher frequency can carry *more data* (faster internet).
- **HOW THE PHONE USES THIS:** Your phone is a "multi-band" radio. When you put in an MTN SIM, your phone's *firmware* (from 2nd Term) *knows* to "tune in" and "listen" on the 900 MHz highway.

2. SIGNAL TRANSMISSION (THE CELLULAR CONCEPT)

- **THE PROBLEM WITH "ONE BIG TOWER":** In the old days (1G), they used *one* giant, high-power tower to cover a whole city.
 - *Problem 1:* It could only handle a few hundred calls at a time.
 - *Problem 2:* It required a massive, high-power transmitter (like a 50,000-Watt TV station).
- **THE GSM SOLUTION (CELLULAR NETWORK):**
 - **Divide and Conquer:** Instead of one big tower, the GSM standard divides a city into a honeycomb of small, low-power "**cells**" (from 1km to 10km wide).
 - **BTS (Base Transceiver Station):** At the center of each cell is a **BTS** (the "cell tower" or "mast"). This is a *low-power* radio that only needs to cover its small cell.
 - **Frequency Reuse:** Because the towers are low-power, two cells that are *far apart* can **reuse the same frequency** without interfering. This allows the network to handle *millions* of users.
- **THE "HANDOFF" (OR "HANDOVER"):**
 - **The Problem:** What happens when you are on a call and you drive from one "cell" to another?

- **The Solution:** This is the job of the **BSC (Base Station Controller)**.
 - i. You are in your car, connected to Tower A.
 - ii. As you drive away from Tower A, the signal gets weak.
 - iii. The BSC (which manages both towers) sees that Tower B is now getting a *stronger* signal from you.
 - iv. The BSC *seamlessly* "hands over" your call from Tower A to Tower B.
 - v. This "handoff" happens in milliseconds, so your call *does not drop*.

3. SIM TECHNOLOGY

- **SIM:** Stands for **Subscriber Identity Module**.
- **FUNCTION:** The SIM is *not* the phone; it is the "**passport**" of the *user* (the subscriber). It is a tiny, secure smart-card (a micro-computer).
- **PURPOSE:** Its one and only job is to **authenticate** you to the network (i.e., "prove you are a paying customer").
- **WHAT IS STORED ON THE SIM? (THE "BRAIN")**
 - b. **IMSI (International Mobile Subscriber Identity):** This is the *most important* number. It is a unique 15-digit number that is *your* ID in the network's main database (the HLR). This is how MTN *knows who you are* and what services (e.g., data plan) you have paid for. **This is NOT your phone number.**
 - c. **Authentication Key (Ki):** A 128-bit *secret key* (a password) that is "burned" into the chip.
 - d. **Authentication Process (Simplified):**
 - Your phone connects to the tower.
 - The network "sees" your IMSI and looks up your *secret key* (Ki) in its database.
 - The network sends your phone a "challenge" (a random number).
 - Your SIM card uses its *secret key* (Ki) to "solve" the challenge (perform an encryption) and sends the answer back.
 - The network checks if the answer is correct. If yes, you are allowed on the network.
 - This *prevents* "cloning" and unauthorized use.

- e. **Phone Number (MSISDN):** Your phone number is also stored here.
- f. **(Old) Storage:** Older SIMs had small memory for SMS and Contacts.
- **EVOLUTION OF SIM (PHYSICAL FORM):**
 - **Full-Size (Credit Card):** The original 1FF.
 - **Mini-SIM (Standard):** The large "classic" SIM (2FF).
 - **Micro-SIM:** 3FF (used in iPhone 4, BlackBerry).
 - **Nano-SIM:** 4FF (used in almost all modern smartphones).
 - **eSIM (Embedded SIM):** This is the *future*. There is **no physical card**. The "SIM" is a *chip* that is **soldered directly onto the phone's motherboard**. The network (e.g., MTN) downloads your "SIM profile" (your IMSI, key, etc.) to the chip over the air. This is what allows a phone to have multiple numbers without multiple SIM slots.

EVALUATION

1. What is a "Frequency Band," and why does the government (NCC) regulate them?
2. What is the "Cellular Concept"? Why is it better than the old "single high-power tower" method?
3. What is a "Handoff" (Handover), and what component manages it?
4. What does "SIM" stand for, and what is its *primary* purpose?
5. What is the "IMSI," and how is it different from your phone number?

ASSIGNMENT

1. Write a 250-word essay explaining the "Handoff" process to a non-technical person. (Use the analogy of passing a ball from one person to another as you run by).
2. What is the difference between the 900MHz band and the 2.6GHz (2600MHz) band in terms of *capability*? (Hint: Which one is better for 4G/data?).
3. Draw a simple diagram of the "Cellular" concept. Show 3 hexagonal "cells" with a BTS (tower) in the middle of each.
4. What is an "eSIM"? What are two major *advantages* of an eSIM over a physical Nano-SIM?

5. Define: Frequency, Hertz, BTS, BSC, IMSI, Authentication Key (Ki).



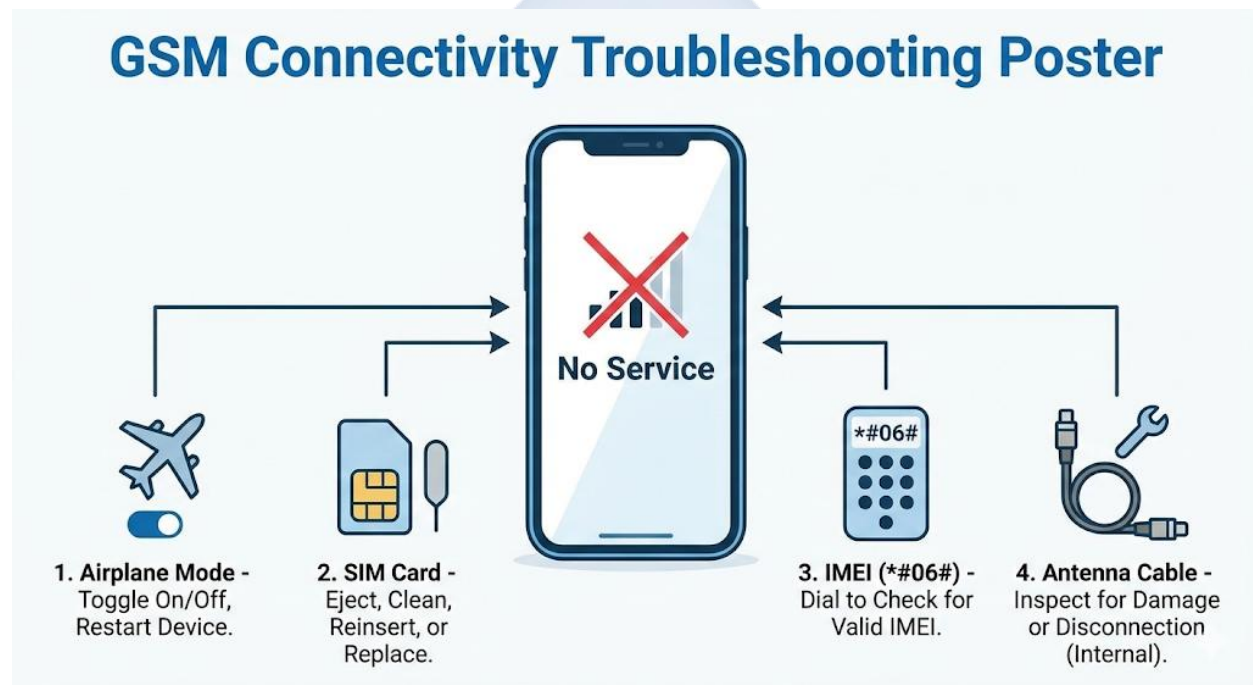
WEEK 6: GSM CONNECTIVITY ISSUES

TOPIC: NO SERVICE, SIM ERRORS, NETWORK CONFIGURATION

INTRODUCTION: THE DIAGNOSTIC MINDSET

This is a core diagnostic lesson that combines knowledge from several previous topics (GSM Principles, SIM Technology, IMEI). When a customer complains "my phone is not connecting," it can be a simple 10-second software fix or a complex, unrepairable hardware failure.

The job of a technician is to **troubleshoot logically, from simplest to hardest**. Never assume the most complex problem first.



FAULT 1: "NO SERVICE" / "EMERGENCY CALLS ONLY" / WEAK SIGNAL

This is the most common complaint. The phone is not connecting to the network.

- **DIAGNOSTIC FLOWCHART:**

STEP 1: USER/SOFTWARE (THE SIMPLE FIXES)

- **Check Airplane Mode:** Is the airplane icon on? This is the #1 "user error."

- **Reboot the Phone:** This is the #2 fix. It forces the phone's "radio" (firmware) to restart and re-scan for the network.
- **Network Operator:** Go to Settings > Mobile Network > Network Operators. Turn *off* "Automatic" and perform a *manual* search.
 - *Result A:* It *sees* the networks (MTN, Glo, etc.). This means the *radio hardware is working*. The problem is with the *SIM or account*.
 - *Result B:* It *sees no networks at all*. This means the *radio hardware is likely faulty*.

STEP 2: SIM CARD ISSUES

- **Test the SIM:**
 - (a) Remove the SIM. Wipe the gold contacts gently with a dry cloth (to remove oils) and re-insert.
 - (b) Put *this* SIM into a *known-good* phone. Does it work?
 - (c) Put a *known-good* SIM (from your own phone) into the *customer's* phone. Does it work?
- **Conclusion:** This 2-way test *proves* if the fault is with the SIM card or the phone.

STEP 3: SOFTWARE CORRUPTION (THE IMEI)

- **Test:** Dial **#06#* on the phone.
- **Result 1 (The Fault):** The IMEI is "Null," "000000000," or "Invalid."
- **Cause:** The phone's firmware (specifically the EFS partition) is corrupt. The phone *cannot* join a network because it *has no ID*.
- **Fix:** This is a *software* repair. It requires "**Flashing**" (re-installing the OS/firmware) or using a *flashing box* to repair the corrupt IMEI.
- **Result 2 (Blacklisting):** The IMEI *is* valid, but it might be **blacklisted** (reported as stolen). The phone is barred from the network.

STEP 4: HARDWARE FAILURE (THE HARD FIXES)

- **Antenna Cable:** This is the *most common hardware fault*, especially *after a bad screen repair*. The technician forgot to re-connect the tiny **coaxial antenna cable** that runs

from the mainboard to the sub-board. **Fix:** Open the phone, locate the cable, and "snap" it back on.

- **RF Chip (Radio):** The main "radio" chip on the motherboard has failed (e.g., from a hard drop). **Fix:** This is an advanced, board-level soldering repair.

FAULT 2: SIM ERRORS

- **SYMPTOM: "NO SIM CARD INSERTED" (when one is inserted)**
 - **Cause 1: Dirty Contacts:** The SIM card's gold contacts are dirty. **Fix:** Clean the contacts *gently* with a *pencil eraser* (which is a mild abrasive) and wipe clean.
 - **Cause 2: Bad SIM Cut:** The user "cut" their own SIM from Micro to Nano and damaged the chip. **Fix:** Tell the customer to get a new SIM from their provider.
 - **Cause 3: Broken SIM Slot Pins:** This is a hardware fault. The tiny gold pins *inside* the phone's SIM slot are bent or broken (often from forcing a SIM in). **Fix:** This is a very difficult soldering repair (board-level).
- **SYMPTOM: "INVALID SIM" / "SIM LOCK ACTIVE"**
 - **Cause:** This is *not* a fault. The phone is **Network Locked** to a foreign carrier (e.g., an "AT&T" iPhone). It is *programmed* to reject all other SIMs.
 - **Fix:** This is a *software* fix. It requires an "Unlock Code" from the original carrier or a software unlock using a flashing box (like a Z3X box).

FAULT 3: NETWORK CONFIGURATION (NO DATA)

- **SYMPTOM:** The phone *can* make calls and send SMS, but the **mobile data (internet) does not work** (e.g., no 'E', 'H+', or '4G' icon).
- **CAUSE:** The phone is missing its **APN (Access Point Name)** settings. The APN is the "gateway" or "address" that tells your phone *how* to access your provider's internet. It's the "username and password" for the data network.
- **FIX (Manual Configuration):**
 - a. The user can often get the settings sent to their phone (e.g., by texting "Settings" to a code).

b. Manual Fix:

- Go to Settings > Mobile Network > Access Point Names (APN).
- Click "Add" and manually type in the correct settings for the carrier.
- **Example (MTN Nigeria):**
 - Name: MTN NG
 - APN: web.gprs.mtnnigeria.net
 - Username: web
 - Password: web
- c. Save the APN, select it, and reboot the phone.

EVALUATION

1. A customer says their phone shows "No Service." List 4 *different* possible causes, starting from the *simplest* (user error) to the *most complex* (hardware).
2. What is the *first* test you should do if a customer brings you a "No Service" phone?
3. How do you check a phone's IMEI number, and what does it mean if the IMEI is "Null"?
4. What is the most common cause of an "Invalid SIM" error?
5. A phone can make calls but cannot browse the internet. What is the most likely problem?

ASSIGNMENT

1. Write a 250-word "Troubleshooting Guide" for a "No Service" error, in the form of a logical, step-by-step flowchart.
2. Find a phone. Go into its settings and *find* the Access Point Name (APN) settings. Write down the APN for your mobile carrier.
3. What is the *most common* hardware fault that causes "No Service" *after* a phone has been repaired? Explain why.
4. Research "IMEI Blacklisting." Is it permanent? Can it be reversed?
5. Define: Airplane Mode, Coaxial Antenna Cable, APN, Network Lock, SIM Slot.

WEEK 8: COMPUTER AND GSM ACCESSORIES

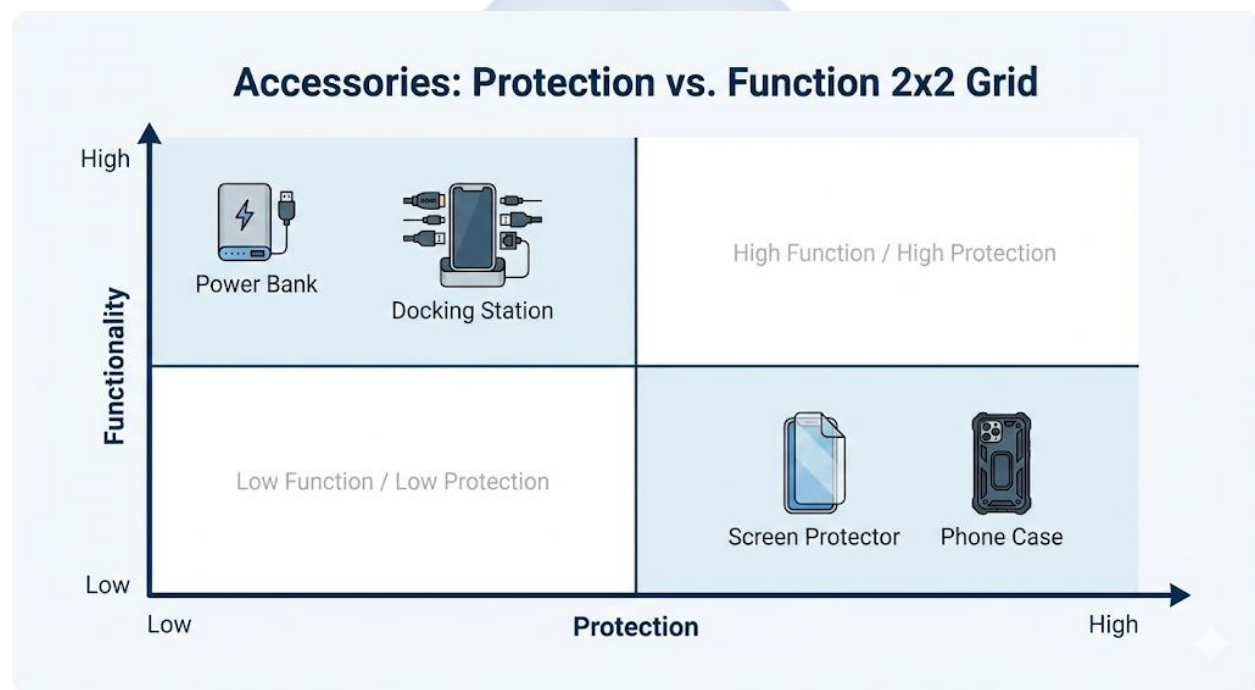
INTRODUCTION

Accessories are hardware devices that are *not* part of the core computer system. They are "add-ons" (peripherals) that enhance functionality, provide convenience, or protect the device.

For a technician, accessories are important for two reasons:

1. They are a major source of "faults" (e.g., a broken charging cable).
2. They are a major source of *revenue* (selling them to customers).

Most accessories are **consumables**—they are designed to be *replaced*, not repaired.



1. COMPUTER ACCESSORIES (DESKTOP/LAPTOP)

These are add-ons that expand the capability of a PC.

- **POWER ACCESSORIES:**
 - **Surge Protector (Power Strip):** Expands one wall socket into many, and its *primary function* is to *protect* all connected devices from a **power spike** (surge).
 - **UPS / Stabilizer:** (As covered in Week 2).

- **DATA & STORAGE ACCESSORIES:**

- **USB Flash Drive:** Small, portable, solid-state storage for moving files.
- **External Hard Drive (HDD/SSD):** High-capacity (e.g., 1TB, 2TB) storage, primarily used for **backups** or for storing large files (like movies or project files).

- **INPUT/OUTPUT (I/O) ACCESSORIES:**

- **Webcam:** An *input* device for video (for Zoom/Teams meetings, streaming).
- **External Microphone:** An *input* device for high-quality audio (for podcasting, recording).
- **Headphones/Headset:** An *output* device for private audio. A "headset" includes a microphone (input).
- **Speakers:** External speakers for high-quality audio output.

- **CONNECTIVITY ACCESSORIES:**

- **USB Hub:** A "power strip for USB." Turns one USB port into 4 or 7.
- **Docking Station:** A "master hub" for laptops. You plug your laptop (with one cable) into the dock, and the dock is *already* connected to your full-sized monitor, keyboard, mouse, printer, and network. This turns a laptop into a full desktop.

- **REPLACEMENT/MAINTENANCE:**

- These devices are **consumables**. If a mouse, keyboard, or webcam breaks, it is *replaced*, not repaired.
- A UPS is an exception; its internal battery can (and must) be *replaced* every 2-3 years.

2. GSM ACCESSORIES (MOBILE PHONES)

These are add-ons for mobile phones and tablets.

- **POWER ACCESSORIES (THE MOST COMMON):**

- **Power Bank:** A portable lithium-ion battery used to recharge a phone "on the go."

- **Car Charger:** An adapter that plugs into the car's 12V socket (cigarette lighter) to charge the phone.
- **Wireless Charger (Qi):** A "pad" that charges the phone using *magnetic induction*, without a cable (the phone must support this).
- **AUDIO ACCESSORIES:**
 - **Bluetooth Earbuds (e.g., AirPods, TWS):** Wireless *output* (speakers) and *input* (mic).
 - **3.5mm Headphone "Dongle":** A small adapter for modern phones that *don't* have a headphone jack. It converts the **USB-C** or **Lightning** port to a 3.5mm audio port.
- **PROTECTION ACCESSORIES (ESSENTIAL):**
 - **Phone Case (Jacket/Pouch):**
 - **Function:** Protects the *body, corners, and frame* of the phone from **impact** (drops) and scratches.
 - **Replacement:** A case is a consumable.
 - **Screen Protector (Tempered Glass):**
 - **Function:** This is the *most important* accessory. It is a thin, hard sheet of glass that is applied to the phone's *actual* screen.
 - **It is a SACRIFICIAL accessory.** It is *designed* to **shatter** on impact from a drop. This *absorbs* the force, protecting the *actual, expensive phone screen* (which costs N30,000+) underneath.
 - **Replacement:** When the *screen protector* cracks, it is simply *replaced* for a small cost (e.g., N1,000). This is a key service and sales item for a technician.
- **STORAGE ACCESSORIES:**
 - **MicroSD Card:** A tiny, removable storage card used to *expand* the storage of many *Android* phones (iPhones do not support this).

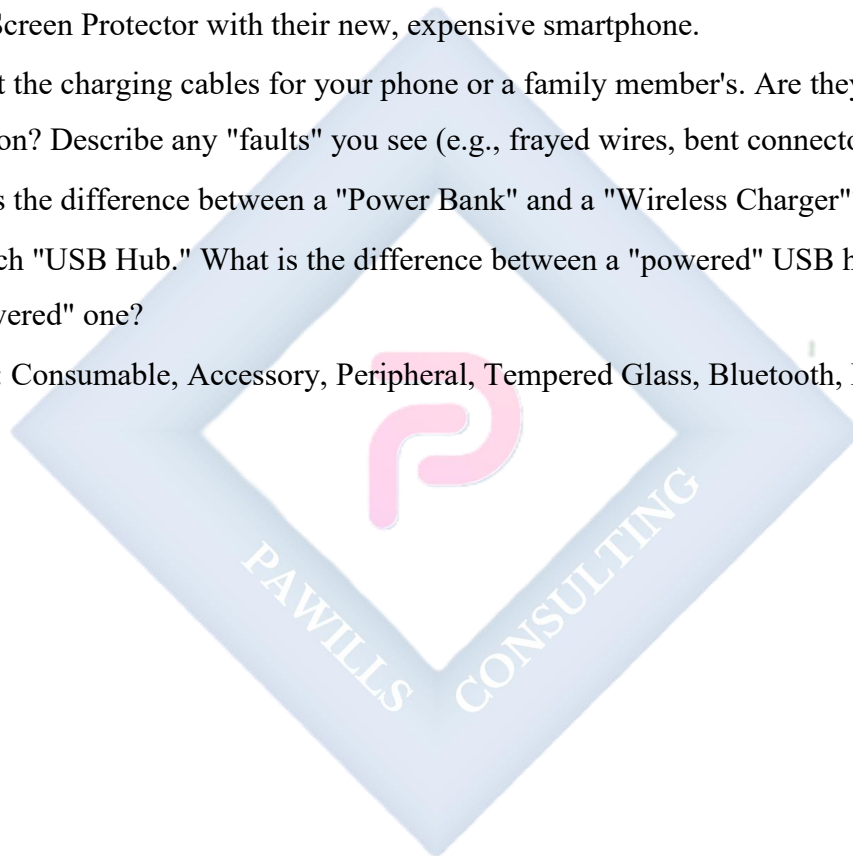
EVALUATION

1. What is an "accessory"? Are they typically repaired or replaced?

2. What is the primary *protective* function of a Surge Protector (power strip)?
3. What is a "Docking Station" used for?
4. Explain the function of a **Tempered Glass Screen Protector**. Why is it called "sacrificial"?
5. What is a "dongle" in the context of mobile phone audio?

ASSIGNMENT

1. Write a 200-word "sales pitch" to a customer explaining why they *must* buy a Tempered Glass Screen Protector with their new, expensive smartphone.
2. Look at the charging cables for your phone or a family member's. Are they in good condition? Describe any "faults" you see (e.g., frayed wires, bent connector).
3. What is the difference between a "Power Bank" and a "Wireless Charger"?
4. Research "USB Hub." What is the difference between a "powered" USB hub and an "unpowered" one?
5. Define: Consumable, Accessory, Peripheral, Tempered Glass, Bluetooth, Dongle.



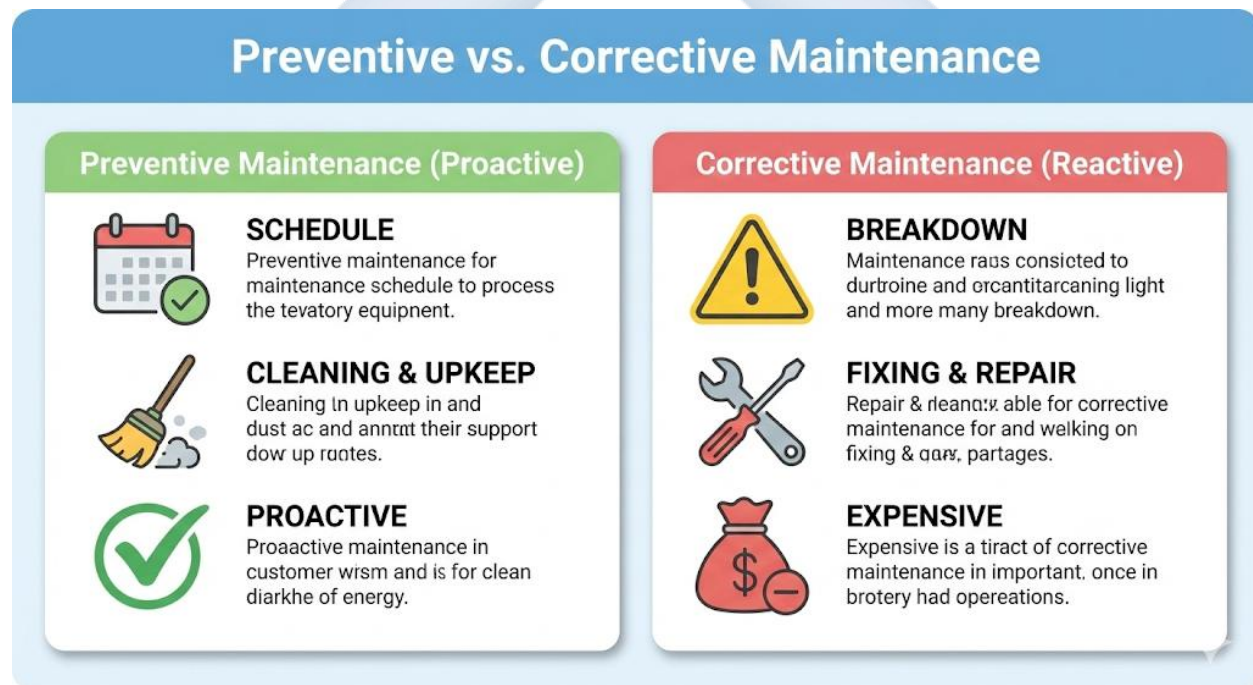
WEEK 9: PREVENTIVE MAINTENANCE PRACTICES

INTRODUCTION: PREVENTIVE VS. CORRECTIVE

As we discussed in Week 1, there are two types of maintenance. This week, we formalize the process of **Preventive Maintenance (PM)**.

- **Corrective Maintenance:** Fixing something *after* it is broken. This is *reactive*, *expensive*, and causes "downtime" (the time a device cannot be used).
- **Preventive Maintenance (PM):** A *proactive* schedule of routine checks, cleaning, and updates to *prevent* problems from ever happening. This is *cheap*, *efficient*, and *saves money*.

A professional technician's *real* job is not just to fix things, but to *stop them from breaking* in the first place.



1. ROUTINE CHECKS (PHYSICAL MAINTENANCE)

This is a scheduled checklist of physical tasks.

- **FOR COMPUTERS (Desktop/Laptop):**

a. **DUST REMOVAL (Monthly/Quarterly):**

- **Problem:** Dust is the #1 enemy. It clogs fans and heat sinks, trapping heat and causing **overheating**.
- **Action:** Use a **can of compressed air** or an **electric data duster** to blow dust *out* of all vents, fans, and heat sinks.
- **Key areas:** CPU fan/heat sink, GPU fan, PSU fan, and case fan **dust filters**.

b. **CABLE CHECK (Quarterly):**

- **Problem:** Vibrations can *loosen* internal cables over time.
- **Action:** Open the case. Gently but firmly press on all power and data (SATA) connectors to ensure they are **fully seated**. Check that fans are spinning freely.

c. **PERIPHERAL CLEANING (Weekly):**

- Wipe down the keyboard, mouse, and monitor with a soft cloth and a 70% Isopropyl Alcohol (IPA) solution to *disinfect* them.
- Use compressed air to blow debris from under keyboard keys.
- **FOR GSM PHONES (User-level):**

d. **PORT CLEANING (Monthly):**

- **Problem:** The **charging port**, **speaker grille**, and **microphone hole** get clogged with pocket lint, dust, and makeup. This is the #1 cause of charging and audio faults.
- **Action:** Power off the phone. Use a *non-conductive* tool (like a plastic toothpick or a stiff, dry brush) to *gently* clean out the debris.

e. **CASE CLEANING (Monthly):**

- Remove the phone from its case. Fine dust and grit get *trapped* between the case and the phone, scratching the body.
- **Action:** Clean the case and the phone separately with a soft cloth.

2. ROUTINE CHECKS (SOFTWARE MAINTENANCE)

A "healthy" system is also a "clean" system on the inside.

- **FOR COMPUTERS:**

- a. **ANTIVIRUS & ANTI-MALWARE (Weekly):**
 - Run a *full system scan* with a reputable tool (e.g., Windows Defender, Malwarebytes).
- b. **SYSTEM & DRIVER UPDATES (Monthly):**
 - Run **Windows Update** to get the latest security patches.
 - Update key **drivers**, especially the **GPU (Graphics) driver**, for performance and security.
- c. **DISK CLEANUP (Monthly):**
 - Run the "Disk Cleanup" utility to remove gigabytes of "junk" (e.g., temporary internet files, log files).
- d. **DEFRAGMENTATION (For HDDs ONLY - Monthly):**
 - Run the "Defragment" tool. This *organizes* the data on a *mechanical* hard drive (HDD) so it can be read faster.
 - **WARNING: NEVER** defragment an **SSD**. It is *unnecessary* (SSDs have no moving parts) and it *reduces* the lifespan of the drive.
- e. **BACKUP (Weekly/Monthly):** Create a backup of all important files to an *external* hard drive or a *cloud* service (like Google Drive).
- **FOR GSM PHONES:**
 - a. **STORAGE CLEANUP (Monthly):**
 - Go to Settings > Storage. Clear the **app cache**.
 - Go through and *uninstall* apps you no longer use.
 - Move photos/videos to a MicroSD card or the cloud.
 - b. **BACKUP (Weekly/Monthly):**
 - Ensure your contacts, photos, and WhatsApp are backing up (e.g., to your Google or Apple account).
 - c. **OS UPDATES (As they arrive):**
 - Always install the latest security updates for Android or iOS.
 - d. **REBOOT (Weekly):**
 - Restarting your phone once a week is the *easiest* preventive maintenance. It clears the RAM, kills "stuck" processes, and solves many "slowness" issues.

3. RECORD KEEPING (FOR PROFESSIONALS)

In an IT Support role, you can't just "fix and forget." You must keep records.

- **ASSET LOG:** A master spreadsheet with a list of every device (PC, phone), its **serial number/IMEI**, its user, and its specifications (RAM, CPU, HDD).
- **MAINTENANCE LOG:** A "service history" for *each* asset, showing:
 - *Example:* "PC #004 - 10-Oct-2025: Internal dust cleaning. Windows Update 22H2 installed."
- **REPAIR LOG:** A history of all *corrective* maintenance.
 - *Example:* "PC #004 - 15-Nov-2025: Replaced faulty PSU (Serial #123)."
- **WHY?** This log helps you spot *patterns* (e.g., "This model of laptop always has its fan fail") and proves you are doing your job.

EVALUATION

1. What is the difference between "Preventive" and "Corrective" maintenance?
2. What is the **#1 physical enemy** of a desktop PC?
3. What is the **#1 physical enemy** of a GSM phone's ports?
4. What is "Defragmentation," and which type of drive should you *never* defragment?
5. What is an "Asset Log," and why is it important for a business?

ASSIGNMENT

1. Write a 250-word "Preventive Maintenance Checklist" for a home PC user, dividing it into "Weekly," "Monthly," and "6-Month" tasks.
2. Why is it important to *clear the cache* on your phone? (Research this).
3. Draw a simple "Repair Log" sheet for a technician. What 5 "fields" (columns) should it have? (e.g., Date, Customer Name, Device Model, Fault, Action Taken).
4. What is "Thermal Throttling," and how does preventive cleaning *prevent* it?
5. Define: Preventive, Corrective, Defragment, SSD, App Cache.



WEEK 10: ENTREPRENEURSHIP IN ICT

TOPIC: SETTING UP A SMALL REPAIR BUSINESS

INTRODUCTION

This lesson combines all the technical skills you have learned (Hardware, GSM, Maintenance) with basic business principles. Having the *skill* is not enough; you must know how to *sell* that skill to create a profitable and sustainable business.

Entrepreneurship is the process of designing, launching, and running a new business.



1. THE BUSINESS PLAN (YOUR "MAP")

You must have a plan. This clarifies your idea and is *required* if you want to get a loan.

- **WHAT? (Mission & Vision):** What do you do? (e.g., "Fast, honest, and affordable phone and laptop repairs"). What is your goal? (e.g., "To be the most trusted repair shop in my area").
- **WHO? (Target Market):** Who are your *ideal* customers? You cannot target everyone.
 - *Example:* University students (high demand for screen/port repairs).
 - *Example:* Small businesses (who need reliable IT support).

- **WHY? (Your "USP" - Unique Selling Proposition):**
 - This is the *most important* question. The market (e.g., Computer Village) is crowded. Why will they come to *you*?
 - *Do not* compete on **Price** ("I am the cheapest"). This is a "race to the bottom."
 - Compete on **Trust, Speed, and Quality**.
- **Honesty/Trust:** "We use original parts." "We give your old, broken parts back." "No hidden fees."
- **Warranty:** "We offer a 30-day warranty on all our repairs." (This is a *powerful* USP).
- **Speed:** "Screen repair in 1 hour, guaranteed."
- **HOW? (The Financials):**
 - **Startup Capital:** How much money do you need to start?
 - **Fixed Costs:** Monthly rent for your shop.
 - **Variable Costs:** Cost of parts, electricity.
 - **Pricing:** How much will you charge? (e.g., *Cost of part* + N5,000 for labor).

2. LOCATION AND CAPITAL

- **CAPITAL:** This is the *start-up money*. You need it for:
 - a. **Rent:** For your shop/kiosk.
 - b. **Tools (Investment):** Your *initial* set of professional tools (Soldering station, screwdriver set, multimeter, etc.). *Do not buy cheap tools*.
 - c. **Inventory (Parts):** Your *initial stock* of the most common repair parts.
- **LOCATION:** This is *critical*. Your location must have high **visibility** and **foot traffic**.
 - *Good:* A "kiosk" near a university gate, a shop in a busy market (like Computer Village), a spot on a main road.
 - *Bad:* A back street, your bedroom (unprofessional).

3. TOOLS AND INVENTORY (MANAGING STOCK)

- **TOOLS:** Your tools are your "doctors' instruments." They must be professional and high-quality.
- **INVENTORY (STOCK):** This is a difficult balancing act.
 - **Problem 1 (Too much stock):** Your money is "tied down" in parts that may become obsolete (e.g., 100 iPhone 6 screens).
 - **Problem 2 (Too little stock):** A customer comes for a common repair. You say, "I have to order the part, come back in 3 days." They will *leave* and go to your competitor *today*.
 - **The 80/20 Rule (The Solution):**
 - You must *log your repairs* (record keeping).
 - You will quickly learn that **80%** of your business comes from **20%** of the repairs (e.g., Tecno/Infinix screens, iPhone batteries, charging ports).
 - You *must* keep these "Top 20%" parts **in stock at all times**.

4. SKILL, SPECIALIZATION, AND REPUTATION

- **SPECIALIZATION:** You cannot fix *everything*. It is smart to specialize.
 - e.g., "I only fix iPhones and Apple products."
 - e.g., "I only do software (flashing, unlocking)."
 - e.g., "I only do board-level (soldering) repairs *for other technicians*."
- **REPUTATION (THE #1 ASSET):**
 - In the repair business, **trust** is your *most valuable* asset. The market is full of "quacks" who steal or swap good parts from a customer's phone.
 - **How to Build Trust:**
 - i. **Honesty:** Be transparent. Show the customer the *new* part. **Give them the old broken part back.**
 - ii. **Warranty:** Offer a **warranty** (e.g., 30 days) on your repair. This shows you are confident in your work.

- iii. **Customer Service:** Be polite, communicate clearly (e.g., send an SMS when the repair is done), and finish the job *when you said you would*.

5. MARKETING

- **Word of Mouth:** The best marketing. A happy customer will tell 3 friends. An angry customer will tell 10.
- **Signage:** A clear, professional sign on your shop.
- **Social Media (Digital C.V.):**
 - Use **Instagram** and **WhatsApp**.
 - Post "before and after" videos of your repairs.
 - Post "repair-in-progress" (time-lapse) videos of you soldering.
 - This *proves* your skill, builds trust, and attracts customers.

EVALUATION

1. What is a "USP" (Unique Selling Proposition)? Give one example for a repair business.
2. What are the 3 main things a technician needs "Startup Capital" for?
3. What is the "80/20 Rule" in the context of inventory?
4. What is the *most valuable asset* for a repair technician, and what are two ways to build it?
5. What is the "digital" version of "word of mouth" marketing?

ASSIGNMENT

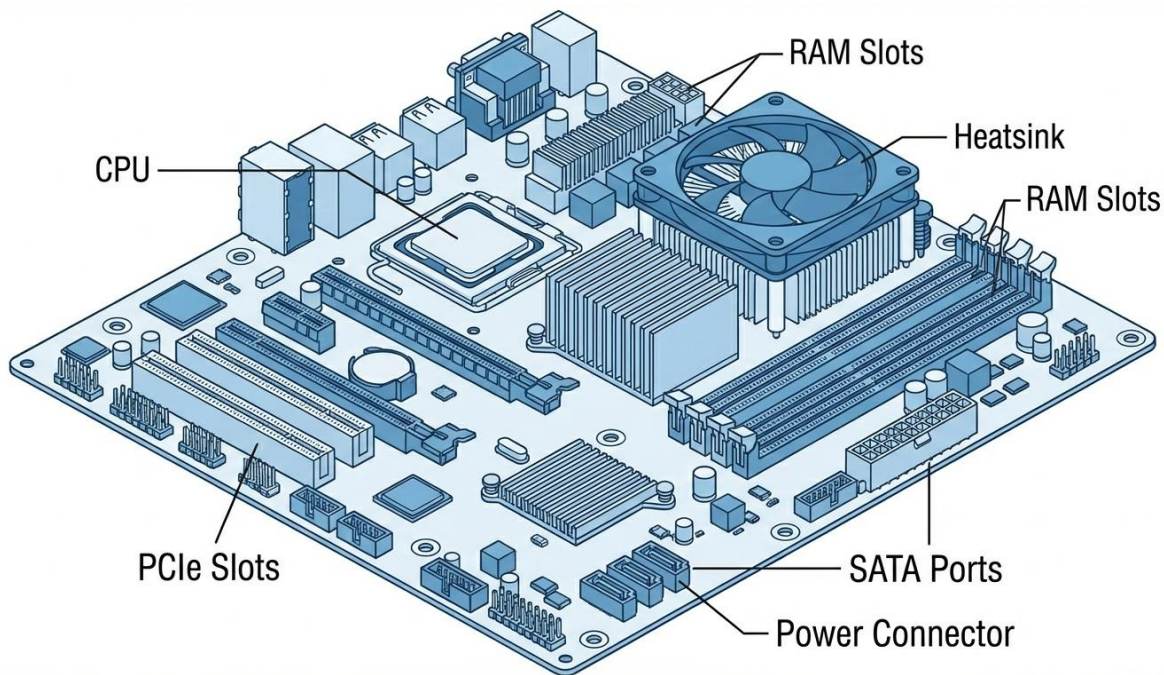
1. Write a 250-word "Business Plan" for your *own* (imaginary) repair shop. Include your **Business Name**, **Location** (e.g., "near my school"), **Target Market**, and **USP**.
2. Create a "Price List" for your new business. List 5 common repairs (e.g., iPhone X Screen, Tecno Charging Port, Laptop RAM Upgrade, PC Cleaning, Windows Install) and assign a "Labor" price (your fee, *not* including the part).
3. What is the *first* thing you would buy: N50,000 in *Tools* or N50,000 in *Parts*? Justify your choice.
4. Why is it important to *give the customer their old, broken part back*?
5. Define: Entrepreneurship, Capital, Inventory, USP, Warranty, Foot Traffic.



WEEK 11: REVISION

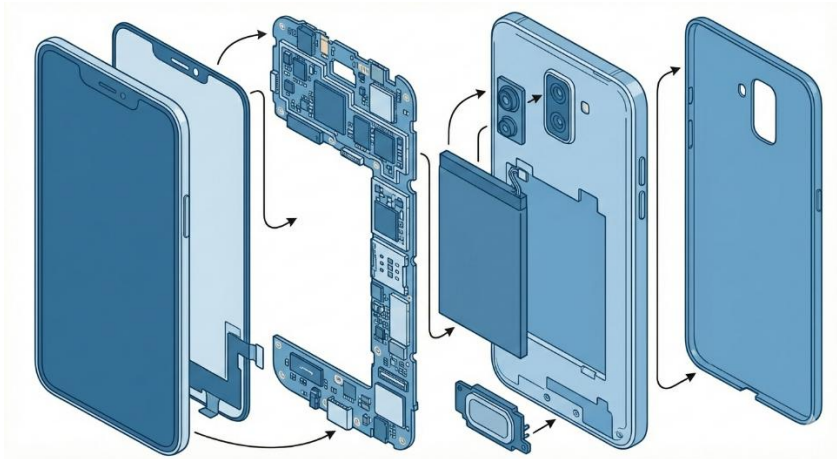
GENERAL REVIEW OF ALL THIRD TERM TOPICS

This week is for a comprehensive review of all topics covered this term. We have connected our hardware knowledge to networking, power, and the business of repairs.



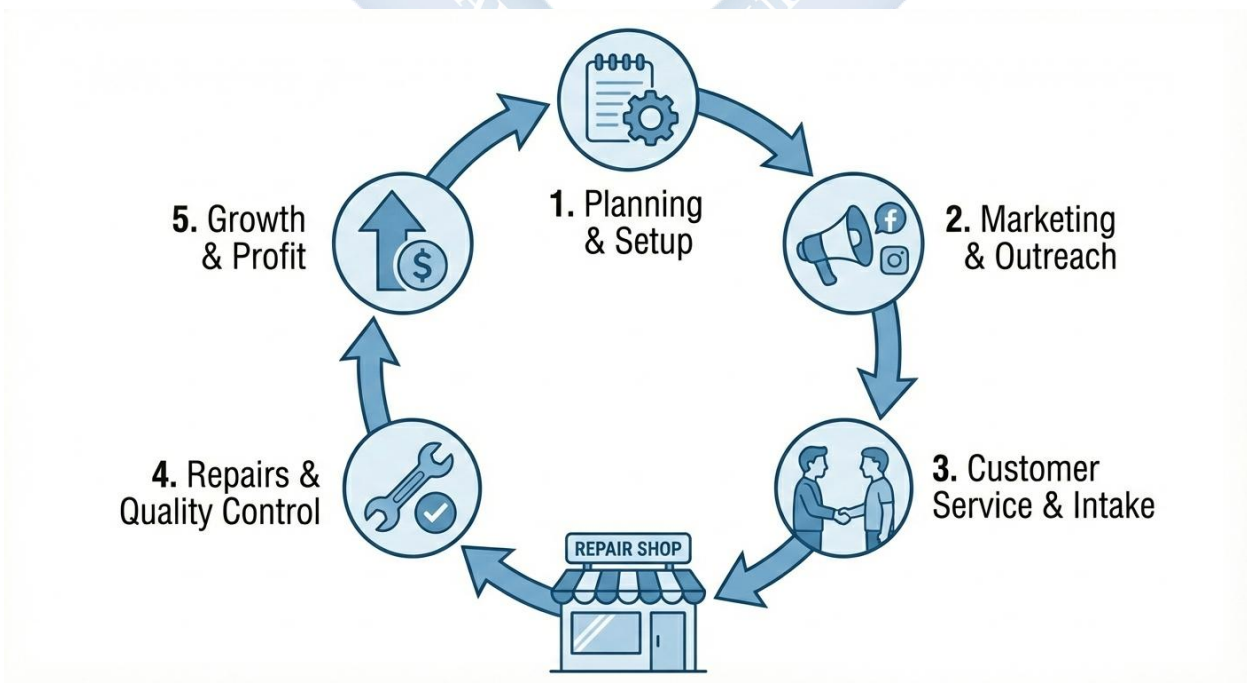
- **WEEK 1: MAINTENANCE (HARDWARE)**
 - **Cleaning:** The #1 enemy is **DUST**. Dust is an *insulator* that traps heat.
 - **Method:** Use **compressed air** and brushes to clean fans, heat sinks, and filters.
 - **Cooling: Thermal Throttling** (slowing down due to heat) is the symptom.
Caused by dust or old **thermal paste**.
 - **Upgrading:** **SSD** (for speed), **RAM** (for multitasking), **GPU** (for gaming).
- **WEEK 2: POWER SUPPLY SYSTEMS**
 - **Problem:** "Dirty" power (Spikes, Sags, Blackouts).
 - **Stabilizer:** Fixes Sags/Spikes. *No battery*.
 - **UPS:** *Short-term* battery (5-15 mins) for *safe shutdown*. (Types: Standby, Line-Interactive, Online).

- **Inverter:** *Long-term* battery system (DC to AC) for *running a house*. (Must be **Pure Sine Wave** for electronics).



- **WEEK 3: NETWORKING HARDWARE**

- **Modem:** Connects LAN to WAN (Internet). **MO**dulator/**DE**Modulator (translates signals).
- **Router:** *Creates* the LAN (assigns **DHCP/IP addresses**) and *routes* traffic.
- **Switch:** *Expands* the LAN (adds more wired ports).



- **WEEK 4: GSM COMMUNICATION PRINCIPLES**
 - **Frequency:** The "highway" (e.g., 900MHz, 1800MHz) for signals.
 - **Cellular System:** Uses many *low-power* **BTS** (towers) in "cells."
 - **Handoff:** The **BSC** *hands* your call from one tower to the next as you move.
 - **SIM:** Your "passport" (Subscriber Identity). Contains your **IMSI** (your real ID) and **Auth Key** (for security).
 - **eSIM:** An *embedded* (digital) SIM.
- **WEEK 6: GSM CONNECTIVITY ISSUES (TROUBLESHOOTING)**
 - **"No Service":** Check (1) Airplane Mode, (2) SIM card, (3) **IMEI (*#06#)** - if "null," it needs **flashing**, (4) **Antenna Cable**.
 - **"Invalid SIM":** Phone is *network locked*.
 - **"No Data":** **APN** settings are wrong.
- **WEEK 8: ACCESSORIES**
 - "Add-ons" that are *replaced*, not repaired.
 - **PC:** USB Hub, External HDD, Webcam, Headset, Docking Station.
 - **GSM:** Power Bank, Bluetooth, **Screen Protector** (sacrificial), **Case** (impact).
- **WEEK 9: PREVENTIVE MAINTENANCE (PM)**
 - **PC:** **Dust removal**, check cables, *Antivirus*, *OS/Driver Updates*, *Defragment* (HDD only), *Backup*.
 - **GSM:** **Port cleaning**, *uninstall apps*, clear cache, *OS Updates*, *Reboot weekly*.
 - **Records:** Asset Log, Maintenance Log, Repair Log.
- **WEEK 10: ENTREPRENEURSHIP**
 - **Business Plan:** (Mission, Market, **USP**).
 - **Location:** High foot traffic.
 - **Inventory:** The "80/20 Rule."
 - **Reputation:** Built on **Honesty** (return old parts) and **Warranty**.

FINAL EXAM

The final examination will test all aspects of this term's work, including definitions, comparisons, safety rules, diagnostic procedures, and business concepts.

